Detecting short gravitational-wave bursts in real LIGO noise

Frequentist, Bayesian and machine-learning analysis of simulated bursts injected into GWOSC O1 data

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Abstract

We use public LIGO strain from the first observing run (O1) as a source of *real*, non-Gaussian, non-stationary noise, inject simulated bursts at a known optimal SNR, and compare three ways of detecting them at the same false-alarm probability: a frequentist matched-filter bank (profile-likelihood ratio), a Bayesian analysis of the same likelihood (exact grid posterior, MCMC, Bayes factors) and machine-learning detectors (feature-based classifiers, convolutional networks, an autoencoder, a Transformer and a GRU). All methods share the data, the injections and a split by chunk of data. (i) The whitened noise is Gaussian in the bulk but has a glitch floor of a few 10^{-3} per second and large epoch-to-epoch variations; they dominate the cost of low false-alarm probabilities and make the calibration of a threshold epoch-dependent. (ii) With three phenomenological burst morphologies, the matched-filter bank detects sine-Gaussians with 50 % efficiency at an optimal SNR of ≈ 5 at a false-alarm probability of 10^{-2} per one-second window (an oracle that knows the parameters: 3.0), and the Bayes factor detects the same bursts, with calibrated posteriors and an explicit prior dependence. (iii) A small CNN reaches parity with the bank and is better on the shapes it was trained on; the largest effect on it is *fresh injections at every epoch*, which make a single network better than a five-member ensemble. A Transformer and a GRU detect worse than the CNN, and a tuning study — one lever at a time — shows why: the patch embedding, not the data or the optimisation, limits the Transformer, and a convolutional stem recovers most of the gap but not all. Morphology classification is as good with fresh-injection CNNs as with Bayesian model comparison (0.98 against 0.97 three-way accuracy at $\rho = 8$); a goodness-of-fit of the matched filter and the energy score of the network flag signals that belong to none of the families. (iv) With realistic binary-black-hole (IMRPhenomD) signals the banks built on bursts lose 2–3 SNR units, a network trained with such signals beats an IMR matched-filter bank on waveforms never seen in training (4.45 against 5.40), and with two detectors, coincidence and time slides push the false-alarm floor from 10^{-3} to 10^{-6} per window at a network SNR of ≈ 11 for balanced injections and ≈ 15 with the real antenna patterns. (v) A second round of studies puts error bars on these statements: the seed scatter of the CNN (0.4 in ρ_{50}) is much larger than its test-set error (0.1), the CNN advantage lives in the tails of real noise and inside its training prior (a spectral line or an out-of-prior morphology breaks it), the public data-quality flags recover most of the gap for the matched filter, and the same pipeline transfers to O2 and O3 without retraining. Parameters of binaries are recovered to ≈ 7 % in chirp mass (calibrated intervals only with exact-model MCMC), the sensitivity is expressed in h_{rss} and in Euclidean range (1.3–1.6 Gpc), and the limits of one-second windows (low masses, neutron stars, NSBH, eccentricity) are quantified.

EXECUTIVE SUMMARY AND READING GUIDE

Question. At a fixed false-alarm probability (FAP) on *real* LIGO O1 noise, how do frequentist (matched-filter bank), Bayesian (same likelihood) and machine-learning detectors compare on short unmodelled bursts and on binary-black-hole signals injected at a given optimal SNR?

Main results (SNR at 50 % efficiency, ρ_{50}, FAP 10^{-2} per one-second window, thresholds on the validation noise; see Fig. 46).

- Bank and Bayes factor are two views of one likelihood and detect the same bursts ($\rho_{50} \approx 4.9$ for sine-Gaussians; the oracle with known parameters needs 3.0: the gap is the look-elsewhere effect).
- A 1-D CNN trained with fresh injections does as well or better with no template (4.3–4.6 averaged over three morphologies, five seeds, against 5.5 for the bank); Transformers and GRUs do worse than the CNN with every lever tried.
- On binary black holes (30–100 $M_\odot$) the bank of the wrong family loses 2.0 in ρ_{50}; an IMR bank (5.4) or a CNN trained with IMR signals (4.45) recovers it.
- Bootstrap and seed studies: the error of ρ_{50} from the test set is 0.05–0.1, but the seed scatter of the CNN at FAP 10^{-2} is 0.4; the CNN advantage lives inside its training prior and in the tails of real noise (outside it, or with a 60 Hz line, it fails), and the GWOSC CAT2 flags recover most of the gap for the bank at FAP 10^{-3} (Sections 14.2, 6.6, 4.11).
- The false-alarm threshold depends on the epoch it is measured in (factor ≈ 4.5; ρ_{50} at FAP 10^{-3} is 6.8 or 10.8 depending on the background, Sec. 14.1); two detectors with time slides reach FAP 10^{-6} at a network SNR of ≈ 11.

Limits. Mostly H1 alone; SNR is optimal SNR, not distance; FAP per window is not an astrophysical rate; aligned-spin waveforms only; ML numbers hold for the training injection prior.

Reading guide. Method sections: 4 (frequentist), 5 (Bayesian), 6 and 7 (machine learning). Extensions: 8 (realistic signals), 10 (two detectors), 11 (unknown morphologies). The comparison and the threshold convention are in Sec. 14. A reader short of time can read this page, Fig. 46, Table 25 and the conclusions.

Glossary.

FAP false-alarm probability: fraction of noise-only one-second windows above the threshold.

ρ, optimal SNR $\rho^2 = \sum_k |h_k|^2$ of the whitened signal; the SNR a matched filter with exact template would get.

ρ_{50}, ρ_{90} optimal SNR at which the detection efficiency is 50 % and 90 % at a given FAP.

Matched-filter bank set of templates (here 160 sine-Gaussians, or 24 IMR templates); the statistic is the maximum of ρ^2 over templates and time shifts.

Look-elsewhere effect the maximum over unknown parameters is larger than a single-template value even in noise; the oracle–bank gap measures its price.

Fitting factor fraction of the optimal SNR recovered by the best template of a bank for a signal outside it.

Time slides shifting one detector's data by more than the light travel time to build a coincident background with no real coincident signal.

Glitch short non-Gaussian noise transient of instrumental origin; sets the tail of the background distribution.

Oracle statistic with the true parameters of the signal known; the Neyman–Pearson upper bound.

Bayes factor ratio of marginal likelihoods of signal and noise hypotheses, with a prior on the signal parameters.

Open set flagging a signal that belongs to none of the families the detector was built for.

1 Introduction

Since the first detection of a binary black-hole merger [1] gravitational-wave detectors have produced a stream of strain data that is public through the Gravitational Wave Open Science Center (GWOSC) [2]. Searches for *short, unmodelled transients* (bursts) cannot rely on a bank of astrophysically motivated waveforms and must either use phenomenological templates or make no assumption on the signal shape (excess-power and coherent-network methods [3, 4]). They all face the same statistical problem: the noise is only approximately Gaussian and stationary, the signals are weak ($\rho \sim 5$–10), the parameters of the signal are unknown nuisance parameters, and the required false-alarm probability is tiny.

This project uses that problem as a laboratory for the techniques of the course *Statistics and Data Analysis*: hypothesis testing and the Neyman–Pearson lemma, maximum likelihood, profile likelihood and Wilks' theorem, toy Monte Carlo, Bayesian inference with priors, MCMC and Bayes factors, supervised and unsupervised learning, calibration and uncertainty in machine learning. The organisation follows the earlier projects on LHCb open data (asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$) and on atomic-force-microscopy curves: one shared data set, one notebook per method, one common set of figures of merit, a report that says where each number comes from.

Research question. *At a fixed false-alarm probability on real LIGO noise, how do a frequentist matched-filter bank, a Bayesian analysis of the same likelihood and machine-learning detectors compare in detection efficiency, in the quality of their uncertainty statements, and in robustness to the non-Gaussian noise and to signal morphologies they were not designed for? And, for a signal that is clearly above the noise, which structure can tell which morphology produced it?*

Scope. Mostly a single detector (H1); three phenomenological burst morphologies, and binary-black-hole (IMRPhenomD) signals in Section 8, injected at a chosen optimal SNR; false-alarm probabilities per one-second window; a second detector with coincidence and time slides only in Section 10. This is *not* an astrophysical sensitivity estimate: SNRs are not converted into distances and the search is not a coherent network search.

Organisation. Section 2 describes the data, the whitening and the tests of the noise model. Section 3 defines the signals, the SNR and the injection chain. Sections 4, 5 and 6 are the

three analyses; Section 7 extends the machine-learning part with sequence models (Transformer, GRU), morphology classification (signal hypothesis testing) and a tuning study of the Transformer-against-CNN comparison; Section 8 repeats the detection with realistic binary-black-hole (IMRPhenomD) signals, Section 10 adds a second detector with coincidence and time slides, and Section 11 tests how signals outside the hypotheses can be flagged; Section 9 expresses the sensitivities in physical units and Section 12 estimates the parameters of the binaries; Section 13 unfolds their mass distribution; Section 14 compares the approaches (with an ablation of the CNN advantage, the threshold convention and the generalisation to O2 and O3) and Section 15 gives conclusions and limits. Most sections end with subsections added in a second round of extensions: bootstrap errors and seeds, robustness, data-quality flags, a coherent two-detector statistic, unsupervised noise models and the limits of the one-second window. Appendix A maps every number to the notebook and result file that produced it.

2 Data and noise model

2.1 Selection of the data

The data are LIGO-Hanford (H1) strain at 4096 Hz from O1 (September 2015–January 2016), downloaded with `GWpy` [5] from GWOSC. Eight chunks of 4096 s were selected, evenly spread over the run, fully inside the H1 science segments and at least 2000 s from every catalogued event (GW150914, GW151012, GW151226 and the two candidates); the closest chunk is more than 1.7×10^5 s from an event, so the chunks can be treated as noise only. Table 1 lists them with the train/validation/test assignment. The split is made *by chunk*, interleaved in time so that each part samples the whole run, never by window: neighbouring windows share noise. The test chunks (3 and 7, Nov. 2015 and Jan. 2016) therefore come from detector epochs, PSDs and glitch conditions that no threshold or model has seen.

chunk	start (UTC)	split	windows
0	2015-09-12 06:59	train	4056
1	2015-10-04 16:01	val	4056
2	2015-10-19 13:17	train	4056
3	2015-11-07 23:46	**test**	4056
4	2015-11-28 01:03	train	4056
5	2015-12-16 19:37	val	4056
6	2015-12-31 06:38	train	4056
7	2016-01-19 13:43	**test**	4056

Table 1. The eight H1 chunks (4096 s each). Windows are the non-overlapping one-second analysis windows left after removing 20 s at each end.

2.2 PSD, whitening and analysis windows

The noise power spectral density $S_n(f)$ of each chunk is estimated with Welch's method [6] (4 s Hann segments, 0.25 Hz resolution) using the *median* average, which is robust to glitches. Each chunk is whitened with its own PSD, band-passed to 30–500 Hz and down-sampled to

1024 Hz (the band is below the new Nyquist frequency, so no information is lost), then cut into non-overlapping windows of one second (1024 samples). One window is the unit of every later analysis. In the frequency domain a window has bins $V_k = \mathrm{rfft}(v)_k / \sqrt{512}$ at 1 Hz spacing; for stationary Gaussian noise $\mathrm{Re}\, V_k$ and $\mathrm{Im}\, V_k$ are independent $\mathcal{N}(0,1)$ in the band, so that the inner product of two signals is $\langle a, b \rangle = \mathrm{Re} \sum_k A_k B_k^*$ and the optimal SNR is

$$\rho^2 = \langle h, h \rangle = 4 \int \frac{|\tilde{h}(f)|^2}{S_n(f)}\, df. \tag{1}$$

Figure 1 shows the amplitude spectral density of the eight chunks: the shape is stable, the in-band sensitivity of chunks 1–7 is within 7 % of chunk 0, and the usual lines (60 Hz and harmonics) are visible.

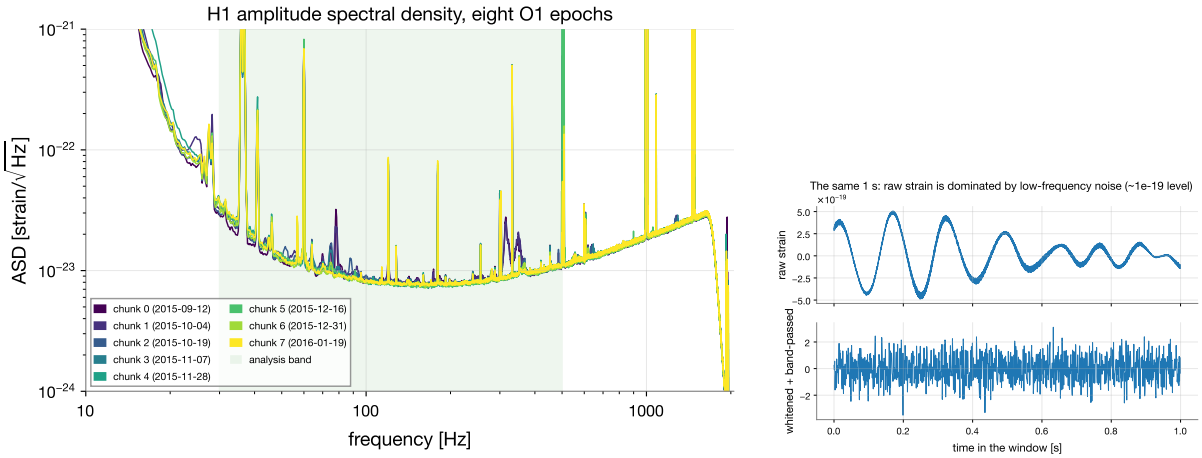


Figure 1. Left: ASD of the eight chunks. Right: the same second of data, raw and whitened (the raw strain is dominated by low-frequency noise).

2.3 How Gaussian and stationary is the noise?

The likelihood used in Sections 4 and 5 assumes independent standard-normal whitened bins. We test it against a Gaussian reference built with the same chain from ideal bins (Figure 2):

- **Bulk.** The sample distribution follows $\mathcal{N}(0,1)$ down to a probability of $\sim 10^{-4}$; the bin variance is 2 (the expected value) within a few per cent in the quiet epochs, and the correlation between adjacent bins is 0.012, at the level of the statistical floor.
- **Lines and non-stationarity.** The bin variance dips at the 60 Hz harmonics (the median-Welch PSD slightly over-estimates narrow lines at 1 Hz resolution) and shows broad excess power in the glitchier epochs (chunks 4 and 6).
- **Tails.** A fraction 10^{-4} of the samples lies beyond 5σ (Gaussian: 6×10^{-7}) and the loudest windows reach hundreds of σ; the plain standard deviation of chunk 6 is 1.8 against a robust σ of 0.95. Chunk 2 is exceptional: 3.5 % of its windows exceed the Gaussian 99.9th percentile of the window energy and the lag-1 autocorrelation of the window energy is 0.80 (long glitchy stretches).

These are the glitches and the non-stationarity of the real detector. They are the reason why *every* threshold in this report is set on real background, and why calibration is always tested on chunks that were not used to set it.

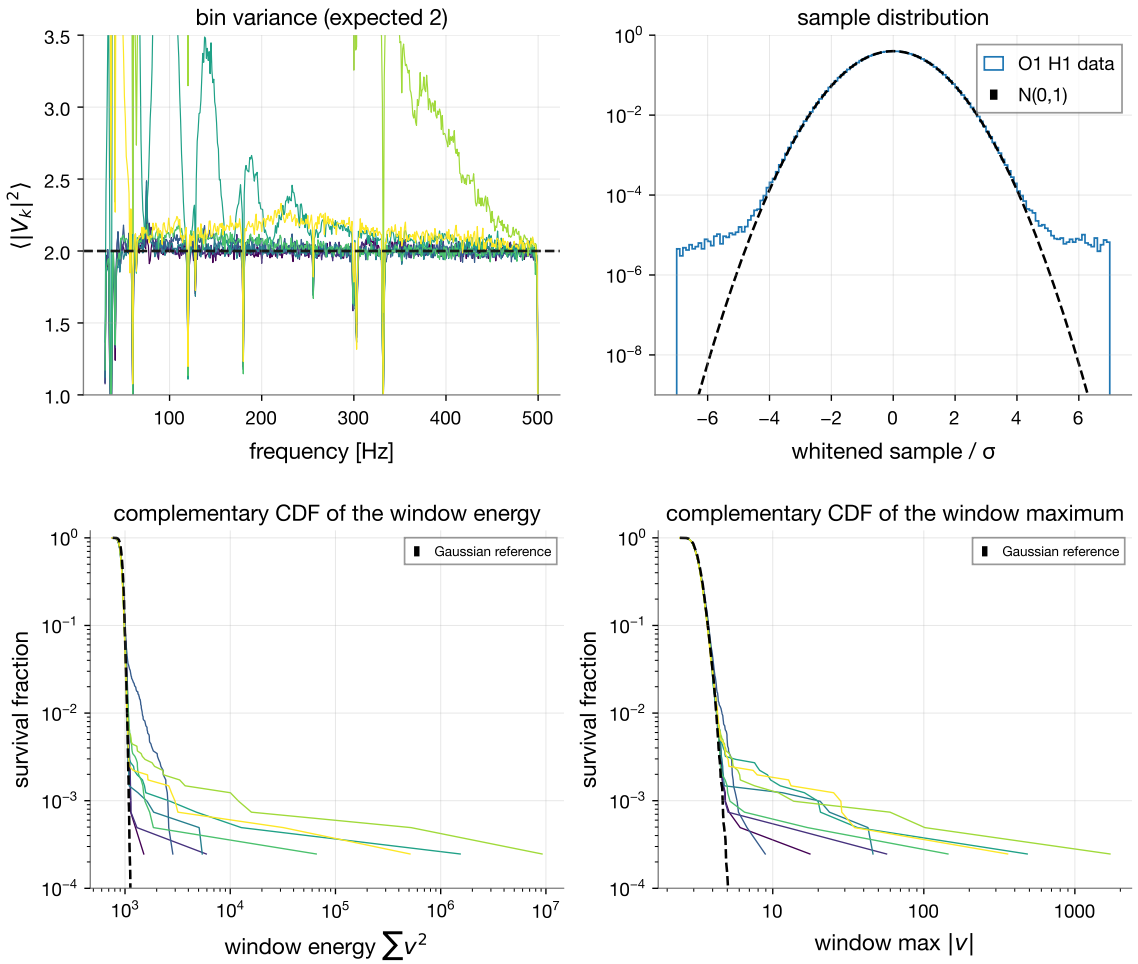


Figure 2. Top: bin variance of each chunk (left; the expected value is 2) and distribution of the whitened samples against $\mathcal{N}(0,1)$ (right). Bottom: complementary distribution of the window energy and of the window maximum for each chunk against the Gaussian reference (dashed).

3 Burst signals, SNR and injections

3.1 Morphologies

Three phenomenological morphologies are injected (Table 12); all have closed-form Fourier transforms (Gaussian integrals with complex width), which are implemented exactly and checked against a brute-force numerical transform (agreement to 10^{-4} or better for all three).

3.2 Injection chain and validation

Signals are not added in the time domain after whitening. They are built directly as whitened frequency bins $V_k = 2\tilde{h}(f_k)/\sqrt{S_n(f_k)}$ with the PSD of the chunk of the host window and added to the real whitened bins. This is equivalent to injecting in the strain (whitening is linear), gives the exact optimal SNR of Eq. (1) and has no filter transients. The exploratory notebook used the time-domain proxy "L_2 norm of the whitened waveform over the noise standard deviation";

morphology	$h(t)$	injection prior
sine-Gaussian (SG)	$e^{-t^2/2\sigma^2}\cos(2\pi f_0 t+\phi)$, $\sigma = Q/2\pi f_0$	$f_0 \sim$ log-U(50,400) Hz, $Q \sim$ U(4,15)
Gaussian pulse	$e^{-t^2/2\sigma^2}$	$\sigma \sim$ log-U(0.8,5) ms
linear chirp	$e^{-t^2/2\sigma^2}\sin(2\pi f_0 t+\pi \dot{f} t^2+\phi)$	$f_0 \sim$ U(50,150) Hz, $\dot{f} \sim$ U(150,600) Hz/s, $\sigma \sim$ U(30,80) ms

Table 2. The three burst morphologies. For all of them $t_0 \sim$ U(0.25,0.75) s inside the one-second window and $\phi \sim$ U(0,2π).

it agrees with Eq. (1) up to the constant factor $1/0.95$ (the noise standard deviation) for all morphologies, and was replaced by the analytic definition.

The chain is validated on real noise: injecting sine-Gaussians at $\rho = 0, 5, 10$ in 4000 test windows and matching with the exact template, $|z|^2$ follows a non-central $\chi^2_2(\rho^2)$ law (means 2.03, 27.1, 101.8 against 2, 27, 102; Kolmogorov–Smirnov $p = 0.64, 0.66, 0.49$). Figure 3 shows what the injections look like.

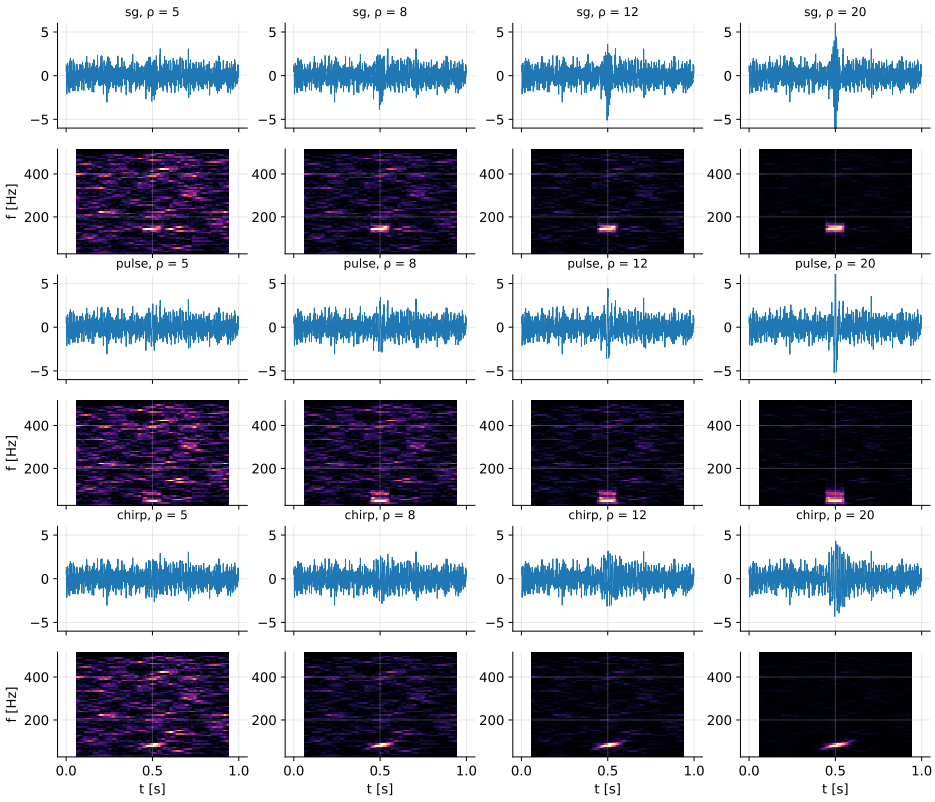


Figure 3. The three morphologies injected into the same real noise window at $\rho = 5, 8, 12, 20$: whitened series (top row of each pair) and spectrogram (bottom). At $\rho = 5$ nothing is visible by eye.

4 FREQUENTIST ANALYSIS

4.1 Hypotheses and test statistic

We test H_0 (noise only) against H_1 (noise plus a sine-Gaussian burst). For a window with whitened bins X and a unit-norm complex template $T(t_0, f_0, Q)$ the signal is $a\,T$ with $a = a_c + ia_s$ complex (amplitude and phase), and $\ln \mathcal{L} = -\frac{1}{2}\|X - aT\|^2$ up to constants. Maximising analytically over a gives the likelihood-ratio statistic

$$\lambda(t_0, f_0, Q) = -2 \ln \frac{\mathcal{L}(0)}{\mathcal{L}(\hat{a})} = \rho^2(t_0, f_0, Q) = \Big| \sum_k X_k T_k^* e^{2\pi i k t_0} \Big|^2, \qquad \lambda_{\max} = \max_{t_0, f_0, Q} \rho^2, \tag{2}$$

evaluated for a bank of 160 templates (40 log-spaced $f_0 \in [40, 450]\,\mathrm{Hz} \times Q \in \{3, 6, 10, 16\}$) and for all $t_0 \in [0.15, 0.85]$ s (the time shift by one FFT). At a *fixed* template Wilks' theorem [7] applies: $\lambda \sim \chi^2_2$ under H_0 (two degrees of freedom: the two quadratures). For $\lambda_{\max}$ it does not: (t_0, f_0, Q) exist only under H_1, which is Davies' problem [8] and the look-elsewhere effect. Two other statistics are used for comparison: the *oracle*, $|z|^2$ at the true parameters (the Neyman–Pearson optimal test of a simple hypothesis, not available in practice), and an *excess-power* statistic [3], the maximum over time and over five durations (10–200 ms) of the z-scored band energy, which assumes nothing about the morphology.

4.2 Background: Wilks, trials factor and real noise

Scanning 20 000 windows of ideal Gaussian noise, $\lambda_{\max}$ is accurately described by $N_{\mathrm{eff}} = 6.8 \times 10^3$ independent χ^2_2 trials, $P(\lambda_{\max} > x) = 1 - (1 - e^{-x/2})^{N_{\mathrm{eff}}}$: a single-template threshold of $\lambda \approx 9$ at FAP 10^{-2} becomes ≈ 27 for the bank. On real noise (Fig. 4) the distribution follows the Gaussian one up to $\lambda_{\max} \approx 22$ and then leaves it, developing a nearly flat *floor* at a few 10^{-3} per window: the glitches of Section 2. The thresholds set on the training chunks are 33.6 (FAP 10^{-2}) and 114 (FAP 10^{-3}), against ≈ 27 and ≈ 31 for Gaussian noise with the same trials factor. Epochs differ enormously: $\sim 10\,\%$ of the windows of chunk 2 exceed $\lambda_{\max} = 60$, and this chunk drives the training threshold.

4.3 Calibration on unseen epochs

The threshold at false-alarm probability α is the $(1 - \alpha)$ quantile of the training background. On the test chunks the observed FAP of the bank is 2.2×10^{-3} for $\alpha = 10^{-2}$ (68 % interval 1.7–2.9 $\times 10^{-3}$) and 1.2×10^{-3} for $\alpha = 10^{-3}$; the excess-power statistic gives 3.7×10^{-3} and 1.1×10^{-3}. The thresholds are therefore *conservative*, by a factor ≈ 4.5 at 10^{-2}, because the glitchy chunk 2 inflates the training background. The false-alarm probability is not a property of the statistic but of the (non-stationary) background it is calibrated on; a threshold transferred to another epoch can be off by a factor of several in either direction.

4.4 Efficiency and ROC

For each morphology and SNR, 500 bursts are injected in randomly chosen test windows (with the PSD of their own chunk) and the fraction above threshold is the efficiency (Fig. 5, Table 3).

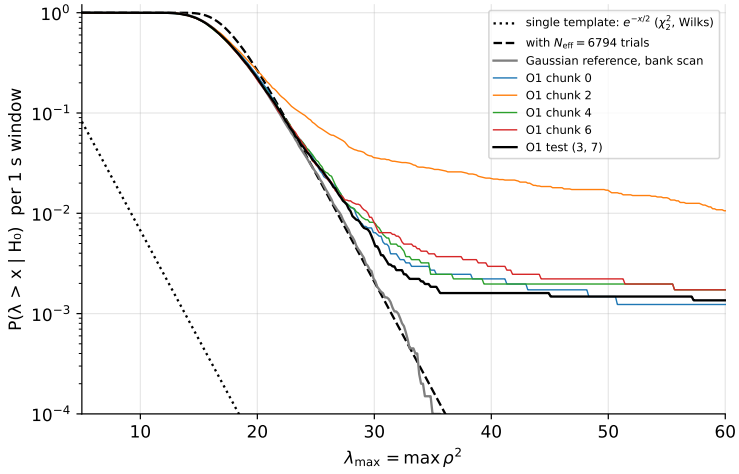


Figure 4. Survival function of $\lambda_{\max}$ under H_0: single-template χ^2_2 (dotted), χ^2_2 with $N_{\rm eff} = 6794$ trials (dashed), Gaussian simulation (grey), real chunks (colours) and test chunks (black).

- **Sine-Gaussian.** The oracle reaches 50 % efficiency at $\rho = 3.0$ and 90 % at 4.4; the bank at 5.5 and 6.9; excess power at 6.8 and 9.7. The oracle–bank gap is the price of not knowing (t_0, f_0, Q) (look-elsewhere); the bank–excess-power gap is the price of not using the signal shape.
- **Pulse and chirp.** Outside the template family the bank is mismatched but still better than the agnostic statistic: the Gaussian pulse is well covered by low-frequency, low-Q templates ($\rho_{50} = 5.9$ against 6.1), and the chirp loses only 1.7 in ρ_{50} with respect to the sine-Gaussian (7.2) while excess power needs 10.8.
- **FAP 10^{-3}.** Because of the glitch floor, lowering the FAP tenfold costs much more than in Gaussian noise: with the training-background threshold the 50 % point of the bank moves from $\rho \approx 5.5$ to ≈ 11 (dotted curves); with the validation background, a cleaner epoch set, it moves from 4.9 to 6.8 (sine-Gaussian, Section 14.1): most of the cost is the glitchy chunk 2 of the training set, i.e. a property of the epoch.

The ROC AUC at $\rho = 8$ is 0.997/0.971 (bank/excess power) for sine-Gaussians, 0.995/0.994 for pulses and 0.983/0.820 for chirps.

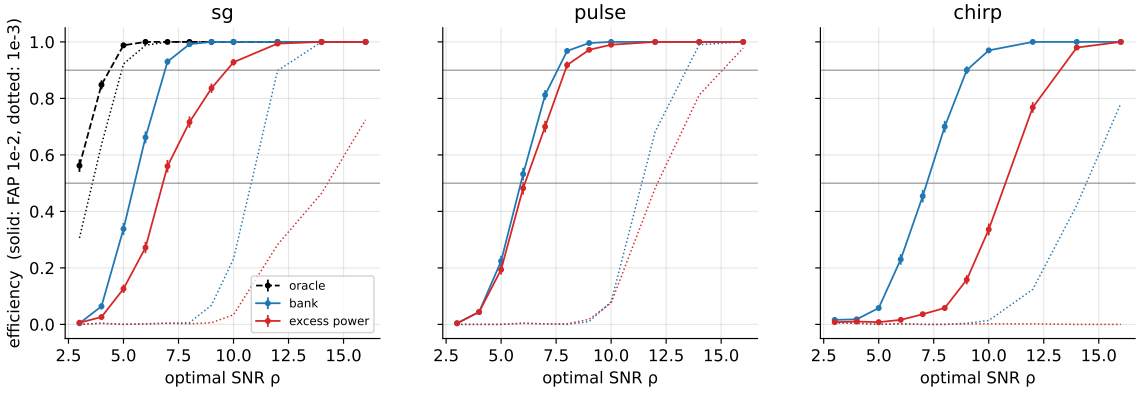


Figure 5. Efficiency against optimal SNR at FAP 10^{-2} (solid, with binomial errors) and 10^{-3} (dotted) for the three morphologies: oracle (black, sine-Gaussian only), bank (blue), excess power (red).

morphology	statistic	ρ_{50}	$\pm$	ρ_{90}	$\pm$
sine-Gaussian	oracle	3.00	0.00	4.37	0.09
	bank	5.50	0.07	6.89	0.05
	excess power	6.79	0.08	9.70	0.14
Gaussian pulse	bank	5.90	0.07	7.56	0.08
	excess power	6.08	0.10	7.92	0.06
linear chirp	bank	7.19	0.09	9.00	0.12
	excess power	10.76	0.09	13.25	0.10

Table 3. Optimal SNR at which the efficiency reaches 50 % and 90 % (FAP 10^{-2}, test chunks).

4.5 Maximum-likelihood parameter estimation

For detected sine-Gaussians the bank maximum is refined with `iminuit` on the continuous profile $\rho^2(t_0, f_0, Q)$ and the covariance of the MLE is compared with the Fisher (Cramér–Rao) matrix, obtained from the Hessian of $-\rho^2$ at the truth on noise-free data, $\mathrm{cov} = 2[\nabla^2(-\rho^2)]^{-1}$. With 300 injections per SNR (Fig. 6) the MLE saturates the bound for $\rho \gtrsim 10$ (f_0: RMS 2.8 against 2.7 Hz at $\rho = 10$; 0.90 against 0.90 Hz at $\rho = 30$) and the Hessian errors of `iminuit` agree with it. At low SNR the RMS departs upward (at $\rho = 6$ the RMS of f_0 is 34 Hz against 4.5 predicted) because the likelihood surface has secondary maxima to which the estimate jumps; the robust width stays near the bound. This is the classical threshold effect of non-linear parameters.

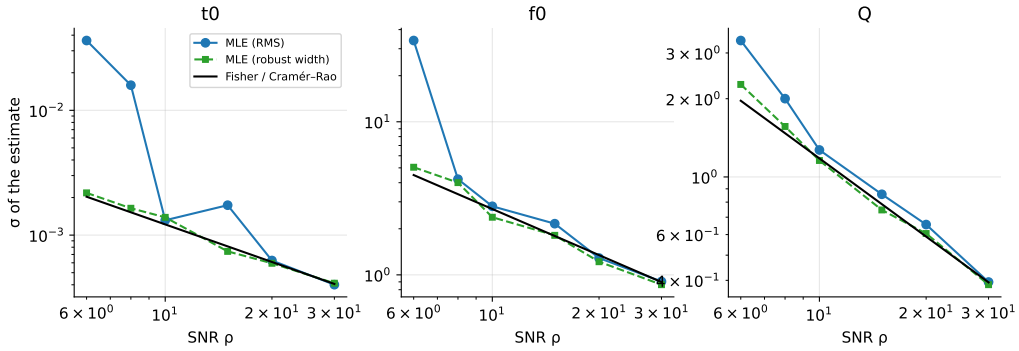


Figure 6. Standard deviation of the MLE of t_0, f_0, Q against SNR: RMS (blue), robust width (green) and Fisher/Cramér–Rao bound (black).

4.6 Toy Monte Carlo: bias, pulls and coverage

With a known template and unknown SNR $s \geq 0$ the data reduce to $z = s\,e^{i\varphi} + n$, $n \sim \mathcal{N}(0, I_2)$; the MLE is $\hat{s} = |z|$ (Rice distribution) and $-2\Delta \ln \mathcal{L}_p(s) = (\hat{s} - s)^2$, so the profile-likelihood interval is $[\hat{s} - 1, \hat{s} + 1]$ truncated at zero. With 4×10^4 toys per value of s (Fig. 7): the MLE is biased upward at low SNR ($E[\hat{s}] - s = 1.25$ at $s = 0$, $\sqrt{\pi/2}$; 0.27 at $s = 2$; 0.05 at $s = 10$); the profile-likelihood interval covers 39 % at $s = 0$ (nominal 68 %), overshoots to ≈ 74 % near $s = 1$ and is within one per cent of 68 % only for $s \gtrsim 3$ (parameter on the boundary: Wilks' regularity fails); the Neyman central and Feldman–Cousins belts [9] cover 68.3 % at every s by construction. The same experiment repeated with a fixed template injected in 5000 real windows per SNR gives the same bias, RMS and coverage within 0.03, so the single-template amplitude estimate is insensitive to the glitches (whereas the bank statistic of the previous section is not).

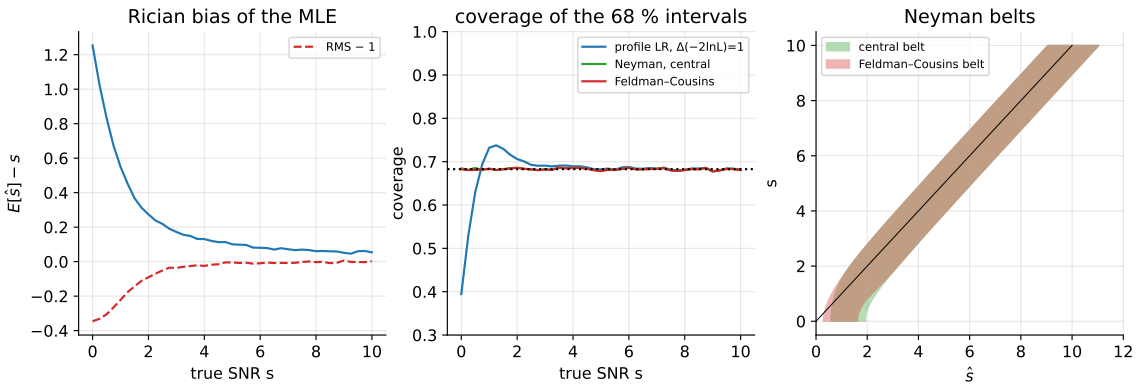


Figure 7. Toy Monte Carlo of the amplitude estimate. Left: bias and RMS -1 of $\hat{s}$. Centre: coverage of the 68 % profile-likelihood, central Neyman and Feldman–Cousins intervals. Right: the two Neyman belts.

4.7 Systematic study

The observable is the sine-Gaussian efficiency $\varepsilon(\rho = 8)$ at FAP 10^{-2}; each variant changes one analysis choice and recomputes the whole chain on the *same* windows and injections, so that differences are correlated. Barlow's criterion [10] (a variation is significant if $|\Delta|/\sigma_\Delta > 2$, $\sigma_\Delta^2 = |\sigma_{\mathrm{var}}^2 - \sigma_{\mathrm{nom}}^2|$) gives Table 4. Only the use of the *mean* instead of the median in the Welch average is large and unambiguous ($+0.025$): the mean PSD is inflated by glitches, lowers the whitened-noise level and the threshold (42.9 against 45.0). All other variants change ε by at most 0.007 (1–3σ). We quote 0.025 as the systematic uncertainty on $\varepsilon(\rho = 8)$.

variant	threshold	ε	Δ	$\lvert\Delta\rvert/\sigma_\Delta$ (Barlow)
nominal	45.0	0.928	–	–
PSD mean-average	42.9	0.953	$+0.025$	5.3
PSD 2 s segments	44.6	0.935	$+0.007$	2.8
PSD 8 s segments	44.6	0.933	$+0.005$	2.4
band 20–500 Hz	45.0	0.929	$+0.001$	1.1
band 30–400 Hz	45.0	0.925	-0.003	1.9
coarse bank (20 f_0)	44.0	0.921	-0.007	2.9

Table 4. Systematic variants (threshold from chunks 0 and 2, efficiency from 1000 injections in chunks 3 and 7).

4.8 Signal-consistency test and a combined ranking statistic

The look-elsewhere and glitch costs of the bank can be reduced without changing the template family (notebook 16). The best template is split in 8 frequency bands of equal power; $\chi^2 = \sum_i |z_i - z/8|^2 \cdot 8$ measures how far the distribution of the matched-filter output over the bands departs from that of the template (Allen's χ^2, 14 degrees of freedom for Gaussian noise and an exact template), and the re-weighted SNR is $\rho/[(1 + c^3)/2]^{1/6}$ for $c = \chi^2/\mathrm{dof} > 1$. A logistic regression and a gradient-boosted classifier on $(\rho^2, \chi^2, \text{new SNR}, \max|v|, \text{kurtosis}, f_0, Q)$ are trained on the training chunks only. At FAP 10^{-2} nothing changes: the bulk of the background is Gaussian. At 10^{-3}, where the glitch tail sets the threshold, the χ^2 veto alone recovers 1.4 of the 2.2 units that separate the bank from the CNN and the learned ranking statistics do

statistic	FAP 10^{-2}	FAP 10^{-3}
bank ρ^2	5.51	7.76
bank with χ^2 veto ($c \leq 3$)	5.44	6.35
new SNR	5.63	6.72
logistic regression (7 features)	5.54	6.46
gradient boosting (7 features)	5.38	6.34
CNN, fresh injections (notebook 07)	4.50	5.48

Table 5. Mean ρ_{50} over sine-Gaussian, pulse and chirp (validation thresholds; bootstrap error 0.09 at 10^{-2}, 0.2 at 10^{-3}).

no better than it (6.3–6.5): consistency with the template, not a richer combination, carries the gain, and a residual ≈ 0.9 remains in favour of the network. The logistic coefficients confirm that ρ^2 and the new SNR carry the weight and that a large $\max|v|$ lowers the score.

4.9 Rigour of the background: epochs, local thresholds, tails and rate limits

Notebook 23 examines the single-detector bank statistic on all eight chunks.

Epoch dependence. Setting the threshold on seven chunks and measuring the FAP in the eighth gives observed/nominal ratios of 0.30–0.86 for seven chunks and 4.3 (FAP 10^{-2}) or 4.9 (10^{-3}) for the glitchy chunk 2, which also makes the thresholds of the others conservative. The ratio is weakly related to the share of glitchy windows (Spearman 0.29) and to the PSD level (0.21): the epoch effect is carried by a few chunks of loud transients, not by a smooth drift.

Time-local background. Taking the threshold of each window from the windows within $\pm$150–600 s of the same chunk reduces the spread of the observed FAP across chunks from 0.013 (leave-one-chunk-out) to 0.0006–0.0012; the price is a slightly lower efficiency for loud signals (0.976 against 0.997 at $\rho = 8$) because loud local glitches raise the neighbours' threshold, and a gain at low SNR (0.18 against 0.13 at $\rho = 4$, which is the usual optimistic effect of a threshold lowered in quiet stretches).

Tail extrapolation. A generalised Pareto fit to the top 5 % of the pooled train+validation background (1217 exceedances) gives $\xi = 0.91$: a tail so heavy that the extrapolated threshold at FAP 10^{-5} is 5700 (68 % bootstrap 3500–8800) against 41 for Gaussian noise with $N_{\mathrm{eff}} = 6800$. It reproduces the test FAP at 10^{-3} ($1.2 \cdot 10^{-3}$ observed) but underestimates it by a factor ≈ 9 at 10^{-4}: the empirical tail of the glitches is not a clean power law, and extrapolation beyond the resolution of the background is not safe; the time slides of Section 10 are the defensible way to go lower.

Rate limits. With the 2.25 h of two-detector data of notebook 14 and 62 zero-lag triggers against 63.1 expected, the Feldman–Cousins 90 % upper limit on the number of detectable signals is 13.0 (against 2.3 for zero events and no background), i.e. rates of 5.8 per hour for signals of network $\rho = 15$ and 14 per hour at $\rho = 10$ (using the efficiencies of notebook 14 at FAR $< 10^3\,\mathrm{yr}^{-1}$): the search is too short for a physically interesting limit.

Rare-event Monte Carlo. In ideal Gaussian noise the tail of $\lambda_{\max}$ can be reached by subset simulation (multilevel splitting with preconditioned Crank–Nicolson moves that leave the Gaussian law invariant; notebook 24, four independent runs of 2000 samples per level, $p_0 = 0.1$). The thresholds at FAP $0.1, 10^{-2}, \ldots, 10^{-10}$ agree with the formula $1 - (1 - e^{-\lambda/2})^{N_{\mathrm{eff}}}$, $N_{\mathrm{eff}} = 6794$, within 0.1–0.6 at FAP $\geq 10^{-6}$ and ≤ 2 at 10^{-10}, with run-to-run scatter of 0.1 to 3 (Fig. 8): the effective-trials model of the look-elsewhere effect is correct in Gaussian noise far beyond the reach of plain Monte Carlo (20 000 windows cover only 10^{-3}), so every excess over it measured in real noise (Section 4.9) is due to the noise and not to the statistic.

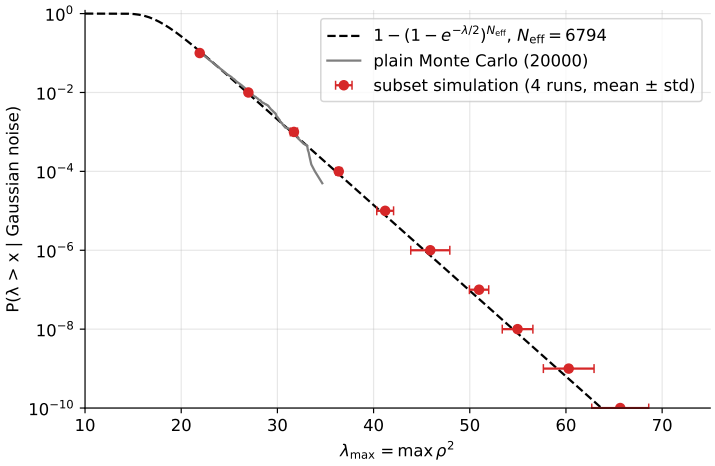


Figure 8. Gaussian-noise false-alarm probability of the bank statistic: analytic formula, plain Monte Carlo and subset simulation.

4.10 Bank design, time-local whitening and PSD uncertainty

Notebook 27 varies three choices of the frequentist pipeline (Fig. 9).

Number and placement of templates. The fitting factor on sine-Gaussians of the project prior rises from 0.78 (20 templates) to 0.91 (40), 0.97 (80), 0.985 (160, the project) and 0.996 (640); the mean ρ_{50} at FAP 10^{-2} from 6.24 to 5.80, 5.61, 5.57 and 5.57 (error 0.06): beyond ≈ 80 templates a denser bank buys nothing, because the loss from the look-elsewhere effect balances the gain in overlap. At FAP 10^{-3} the differences (8.0–8.5 for 80–640 templates) are within the error of 0.6. A random placement of 40 or 160 templates is as good as the regular grid (5.76 and 5.62 against 5.80 and 5.57; fitting factor 0.94 and 0.98). The 160-template bank is therefore near the point of diminishing returns, and a bank of 80 would give the same sensitivity at half the cost.

Time-local whitening. Whitening each 512 s block with a PSD estimated from the block and its two neighbours instead of the whole chunk does not reduce the epoch dependence (leave-one-chunk-out observed/nominal FAP from 0.37 to 4.2 against 0.30 to 4.3; ρ_{50} 5.5/7.9 against 5.5/7.7 at $10^{-2}/10^{-3}$): the epoch effect comes from glitches, not from a slowly varying PSD.

PSD uncertainty. In synthetic Gaussian noise with the PSD of a real chunk, the FAP at the ideal threshold of 10^{-2} (10^{-3}) becomes $1.46 \cdot 10^{-2}$ ($2.3 \cdot 10^{-3}$) if the PSD is estimated on 32 s of data, $1.1 \cdot 10^{-2}$ ($2.0 \cdot 10^{-3}$) on 64 s and 0.96–$0.86 \cdot 10^{-2}$ (1.6–$1.1 \cdot 10^{-3}$) on 256–4096 s: with the

4096 s chunks of the project the estimation noise is negligible, and a PSD from less than ≈ 100 s would need the Student-t correction of the likelihood.

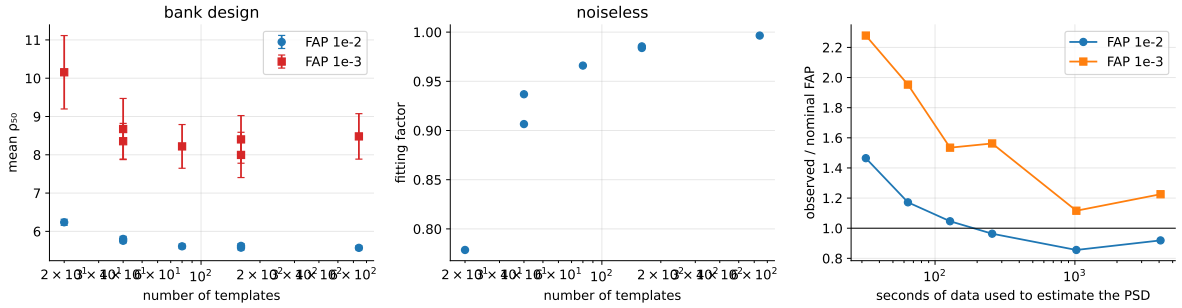


Figure 9. Left: mean ρ_{50} against the number of templates. Centre: fitting factor. Right: observed over nominal FAP when the PSD is estimated on T seconds.

4.11 Data-quality flags

Notebook 35 vetoes the windows that are not entirely inside the GWOSC segments `H1_CBC_CAT2` (the data that remain after the category-2 vetoes; `CAT1` is identical to the science mode in our chunks). The dead time is 0.9 % overall and 7.4 % in the glitchy chunk 2 (0 elsewhere except 0.1 % in chunk 6), yet the flags remove 40 % of the loudest 1 % and 39 % of the loudest 0.1 % of the bank statistic (and only 4 % and 3 % of those of the CNN, which already rejects them). The leave-one-chunk-out observed/nominal FAP of the bank, 0.30–4.3 at 10^{-2} and 0.49–4.9 at 10^{-3} without the flag, becomes 0.76–1.21 and 0.27–1.48 with it; the spread across chunks drops from 0.74 to 0.15 (log scale) at 10^{-2}, and the pooled threshold at FAP 10^{-3} from 110 to 57 (from $\sqrt{110} = 10.5$ to $\sqrt{57} = 7.5$ in the SNR of the threshold: the efficiency gain at fixed FAP, not measured here, is bounded by this). The glitch floor and the epoch dependence found in Section 4.9 are therefore mostly the glitches that GWOSC itself flags, and a flag-based veto should be part of any search with the template statistic; the CNN gains nothing because it never alarmed on them. The BURST flag removes less (it is a different veto set, 5–41 % of the top windows).

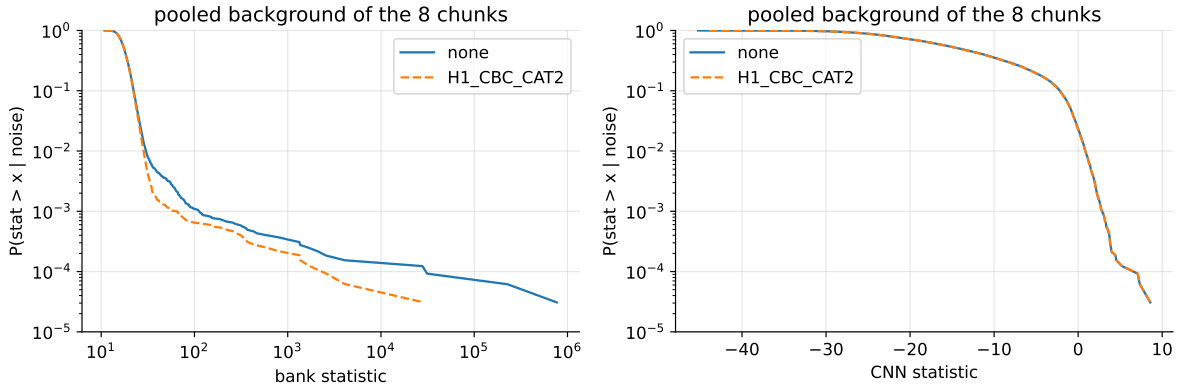


Figure 10. Pooled background of the bank statistic (left) and of the CNN (right) with and without the CAT2 veto.

5 BAYESIAN ANALYSIS

5.1 Model and exact marginal likelihood

The Bayesian analysis uses the same Gaussian likelihood as Section 4 with priors on the signal parameters (Table 6). The two linear amplitudes $a = a_c + ia_s$ get independent Gaussian priors, so that the SNR $\rho = |a|$ is Rayleigh(σ_ρ); the default is $\sigma_\rho = 10$. With $z(\theta) = \sum_k X_k T_k^* e^{2\pi i k t_0}$ and $\rho^2(\theta) = |z|^2$ (the matched-filter output of the frequentist analysis at fixed $\theta = (t_0, f_0, Q)$) the integral over a is a Gaussian integral:

$$\ln \frac{p(X|\theta,\mathcal{M}_1)}{p(X|\mathcal{M}_0)} = \frac{1}{2}\frac{\sigma_\rho^2}{1+\sigma_\rho^2}\,\rho^2(\theta) - \ln(1+\sigma_\rho^2), \qquad a \mid \theta, X \sim \mathcal{N}\Big(\tfrac{\sigma_\rho^2}{1+\sigma_\rho^2} z,\ \tfrac{\sigma_\rho^2}{1+\sigma_\rho^2}\Big). \tag{3}$$

The closed form agrees with a brute-force two-dimensional integration over a to 10^{-13} ($\ln B = 10.7017$ both ways). The posterior of θ and the Bayes factor B_{10} then need only the matched-filter cube $\rho^2(t_0, f_0, Q)$, i.e. exactly the object computed in Section 4: the two analyses differ only in how the cube is combined (maximum against prior-weighted average) and in the prior.

parameter	prior
t_0	uniform on $[0.25, 0.75]$ s (as the injections)
f_0	log-uniform on $[50, 400]$ Hz
Q	uniform on $[4, 15]$
a_c, a_s	$\mathcal{N}(0, \sigma_\rho^2)$ each, $\sigma_\rho = 10$

Table 6. Priors of the Bayesian analysis.

5.2 Exact grid posterior and MCMC cross-check

The posterior is evaluated exactly on a grid of 96 log-spaced $f_0 \times 12$ values of $Q \times 512$ time steps (5.9×10^5 cells, uniform prior per cell by construction); the SNR posterior is the mixture over cells of Rice densities. Figure 11 shows one noise window with the same sine-Gaussian injected at $\rho = 5, 8, 12, 20$: at $\rho = 5$ the (f_0, Q) posterior is broad, from $\rho = 8$ the time of arrival is determined to a millisecond and at $\rho \geq 12$ everything collapses around the truth; the noise-only window gives a nearly flat t_0 posterior and an SNR posterior piled up at small ρ (the prior speaking). An independent `emcee` [11] run (five parameters, no marginalisation, 40 walkers $\times$ 6000 steps) agrees with the grid on means and widths for $\rho = 8, 12, 20$ (e.g. $f_0 = 151.9 \pm 2.4$ against 152.0 ± 2.4 Hz at $\rho = 8$); the integrated autocorrelation time is 260–380 steps (acceptance 0.32–0.38), i.e. 420–610 effective samples. One difference is informative: at $\rho = 20$ the grid width of f_0 is 1.65 Hz against 1.14 Hz from MCMC, because the posterior is narrower than the grid cell (2.3 %, i.e. 3.4 Hz); at high SNR the grid is the limiting factor.

5.3 Bayes factor as a detection statistic

We compute $\ln B_{10}$ with the same 160-template bank as the frequentist statistic, on the same background and injections. Three facts:

- **On real noise the Bayes factor is not a false-alarm probability.** A fraction 1.2 % (0.3 %) of

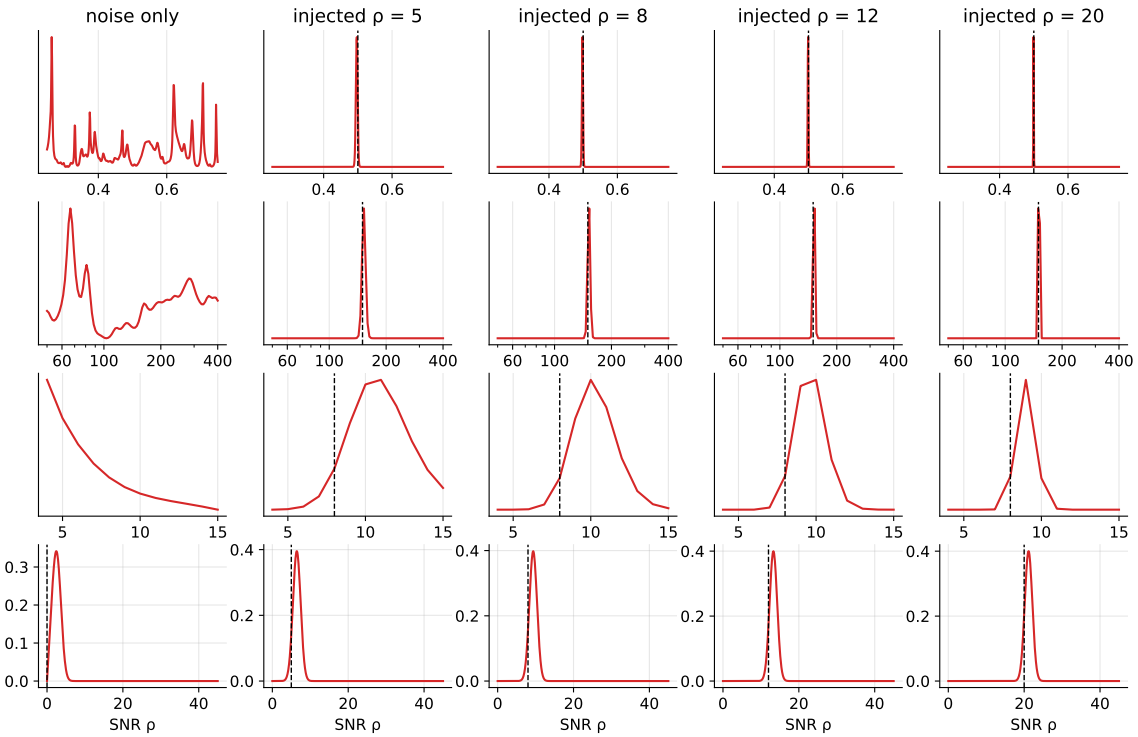


Figure 11. Posterior marginals of t_0, f_0, Q and ρ for the same noise window with no signal and with a sine-Gaussian ($t_0 = 0.5\,\mathrm{s}$, $f_0 = 150\,\mathrm{Hz}$, $Q = 8$) injected at $\rho = 5, 8, 12, 20$; dashed: truth.

the training (test) noise windows has $B_{10} > 10$ and 0.9 % (0.17 %) has $B_{10} > 100$: glitches are evidence for *something* that the Gaussian model can only call a burst. The Jeffreys scale [12] is a scale of evidence within the model; the false-alarm probability has to be calibrated on background exactly as for the frequentist statistic.

- **Same detection power.** At equal FAP (10^{-2}) the efficiencies of $\lambda_{\max}$ and $\ln B_{10}$ coincide within 0.01 at every SNR ($\varepsilon(\rho = 6) = 0.718$ against 0.714, $\varepsilon(\rho = 8) = 0.994$ for both) and so do the ROC AUCs (0.9763 against 0.9769 at $\rho = 6$; 0.9968 for both at $\rho = 8$): the ranking of the windows is almost identical.
- **The prior matters for B_{10}, not for the ranking.** At fixed data the Bayes factor falls as $-2\ln\sigma_\rho$ for large prior width (Occam penalty for a prior that wastes mass on SNRs the data exclude): for a noise window it decreases monotonically from 0 to -7 between $\sigma_\rho \to 0$ and 100, for a signal it has a maximum at σ_ρ of the order of the true SNR (≈ 5 for $\rho = 8$, ≈ 10 for $\rho = 12$); the range of the t_0 prior shifts it by the logarithm of the volume ratio (Fig. 12). Whenever a Bayes factor is quoted the prior must be quoted with it.

5.4 Calibration of the posterior

Simulation-based calibration [13] draws the truth from the prior, simulates data, and checks that the posterior CDF of the truth is uniform. With 500 simulations in ideal Gaussian noise and 500 in real test windows, the rank histograms of t_0, f_0, Q and ρ are consistent with uniformity in both (Kolmogorov–Smirnov $p = 0.24$–0.90 in Gaussian noise, 0.25–0.85 in real noise), so that waveforms, whitening convention, closed-form marginalisation and grid are correct end to end. Five hundred simulations are too few to see an effect of the glitches ($\lesssim 1$ % of windows). The coverage of the nominal 68 % credible intervals is 0.65–0.69 for Q and ρ but 0.50–0.51 for f_0 and

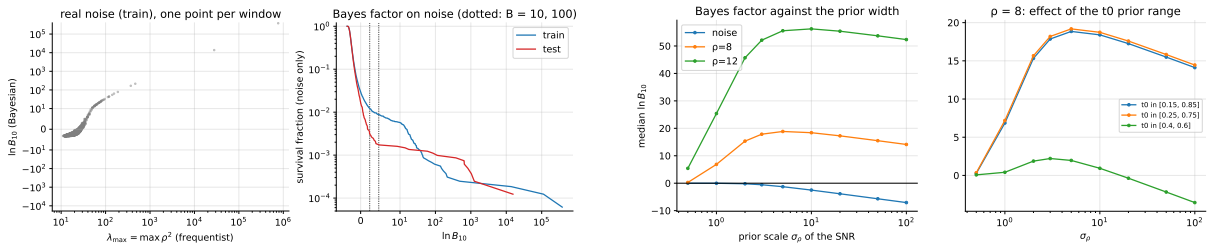


Figure 12. Left: $\ln B_{10}$ against $\lambda_{\max}$ on real noise (one point per window) and its tail. Right: median $\ln B_{10}$ against the prior width σ_ρ for noise and sine-Gaussians at $\rho = 8, 12$; and for $\rho = 8$ against the range of the t_0 prior.

0.58–0.60 for t_0, in both noises: this is a grid-resolution effect (for loud injections the posterior is narrower than a cell: 3.4 Hz in f_0, 1 ms in t_0), not a model failure, and the MCMC does not have it. A posterior-predictive check on the residual of the fit (400 injections at $\rho = 10$) gives 0.25 % of p-values below 0.01 and 2 % below 0.05 (median 0.69 instead of 0.5, because the in-band noise variance is 1.975 rather than 2 per component).

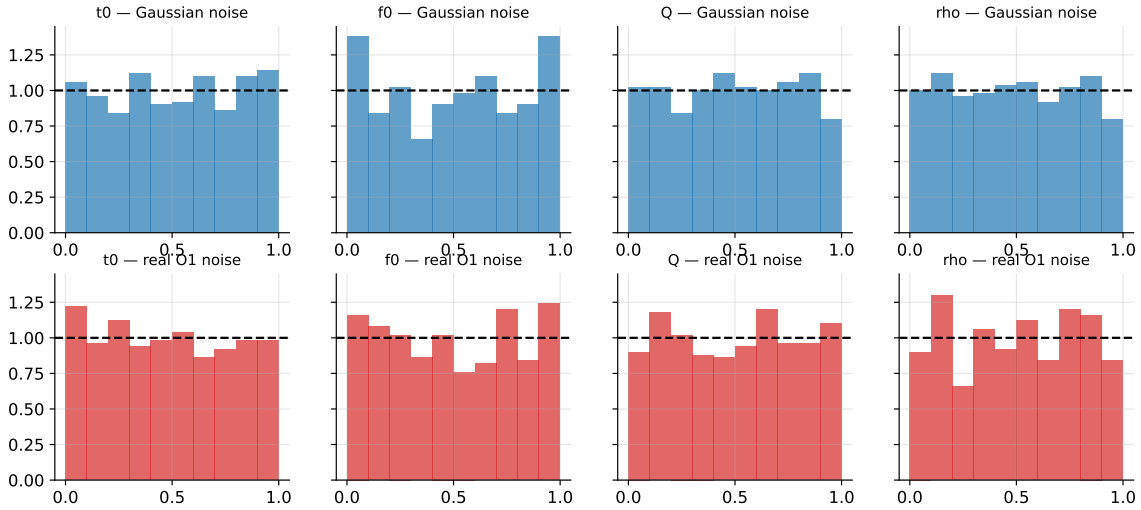


Figure 13. Simulation-based calibration: distribution of the posterior CDF of the true parameters (top: Gaussian noise; bottom: real O1 noise). Uniform = calibrated.

5.5 Credible against confidence intervals at low SNR

In the simplest setting where the two can be compared exactly (known template, $s \geq 0$ measured through $\hat{s} = |z|$) we compute the frequentist coverage of the 68 % credible intervals by integrating the Rice density (Fig. 14). For $s \gtrsim 3$ credible and confidence intervals coincide at 68–69 % (the posterior becomes Gaussian and the prior irrelevant). Below that, credible intervals — which are conditional on the observed data and make no coverage promise — behave differently: with a flat prior the central interval covers 0 at $s = 0$, the HPD interval 77 % at $s = 0$ and 85 % near $s = 1$ before dipping to 53 % at $s \approx 1.7$; with the Rayleigh(10) prior both undercover for $s \lesssim 1.5$ (54–60 % at $s = 1$) because the prior prefers larger SNRs than the truth. The Neyman/Feldman–Cousins intervals cover by construction, and the profile-likelihood interval lies in between (39 % at $s = 0$, 74 % at $s = 1$). The HPD intervals are shorter than the profile ones for $\hat{s} \lesssim 2$: informativeness is traded against guaranteed coverage.

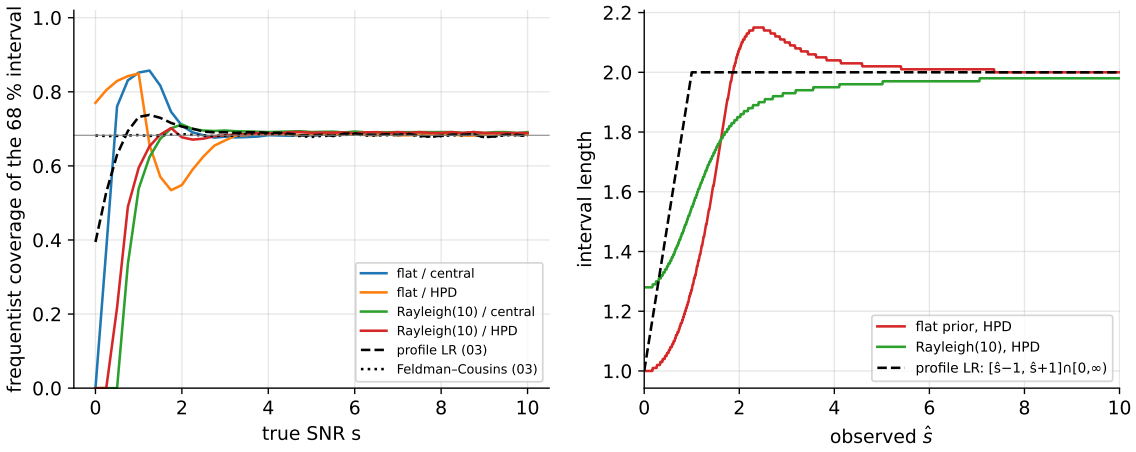


Figure 14. Left: frequentist coverage of the nominal 68 % intervals against the true SNR (credible intervals with two priors and two orderings; profile likelihood and Feldman–Cousins from Section 4). Right: interval length against the observed $\hat{s}$.

5.6 Labelled glitches and a glitch-aware Bayes factor

Real glitches have no labels, so notebook 21 injects three labelled families into real test noise: blips (one-sided decaying sinusoids), whistles (fast sweeps, 1500–5000 Hz/s) and scattered-light-like arches, scaled to a given norm ρ of the whitened bins. At the FAP 10^{-2} thresholds of the validation noise and $\rho = 8$ (Fig. 15) the bank, the Bayes factor and the CNN alarm on 97–98 % of the blips (a blip is a low-Q sine-Gaussian: a single detector cannot tell them apart) and on 38–53 % of whistles and arches, which the sine-Gaussian templates fit poorly; the mean alarm fraction on glitches grows from 0.06 ($\rho = 4$) to 0.97 ($\rho = 14$) for the bank, and the Allen χ^2 test removes only ≈ 25 % of the alarms at $\rho = 14$ while it also rejects 12 % of loud real signals. A χ^2 test is not a glitch veto against glitches that fit a template well.

A *glitch-aware* Bayes factor treats the noise hypothesis as "Gaussian noise or one of the three glitch families" with prior weight $\eta = 0.1$ for the glitch, using a 51-template glitch bank and the same analytic amplitude marginalisation ($\sigma_\rho = 10$): $\ln B_{\mathrm{rob}} = \ln B - \ln[(1-\eta) + \eta B_{\mathrm{glitch}}]$. At $\rho = 8$ it lowers the alarm rate on blips from 0.97 to 0.35, on whistles from 0.38 to 0.04 and on arches from 0.52 to 0.19, at the cost of the efficiency for real signals (0.88 against 0.96 mean over the three morphologies at $\rho = 8$; 0.78 against 0.99 for pulses, which resemble blips). The AUC of signal against glitch among the alarms rises from 0.52 to 0.87 (blips) and from 0.93 to 0.97 (whistles). On the *real* noise windows the fraction with $B > 100$ only goes from 0.17 % to 0.15 % (test) and from 0.21 % to 0.18 % (validation): the false $B > 100$ of Section 5 are not made of glitches of these three families. The gain is an upper bound for unseen glitch types, because the glitch model contains the families that were injected.

5.7 Exact-model MCMC, two-detector evidence and calibration

Exact IMRPhenomD model (notebook 32). For the dominant mode a detector sees $\mathrm{Re}[C\,h(f)]$ with one complex constant C (distance, inclination, polarisation, antenna pattern and phase), so with a Gaussian prior on C the likelihood of the aligned-spin IMRPhenomD model is analytic and only $(M, q, \chi_1, \chi_2, t_{\mathrm{merge}})$ are sampled (32 walkers $\times$1500 steps of `emcee`, ≈ 1 ms per likelihood; autocorrelation time ≈ 117 steps, i.e. ≈ 270 independent samples per chain). On 30

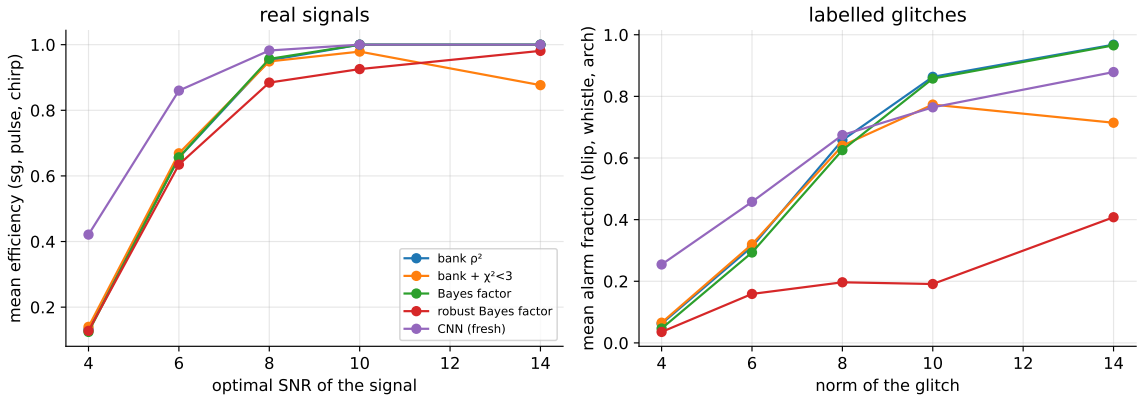


Figure 15. Alarm fraction at FAP 10^{-2} against the norm, for real signals (left, mean of sine-Gaussian, pulse, chirp) and for the labelled glitches (right).

injections per SNR the median error of M_c is 4.7, 6.3, 6.8 % at $\rho = 8, 12, 16$, of M 5.1, 7.9, 5.3 %, of q 22–28 %, of $\chi_{\rm eff}$ 0.13, 0.09, 0.06 and of the merger time 1–2 ms, with 68 % interval coverage of 0.67–0.83 for M_c, 0.60–0.70 for q, 0.67–0.73 for $\chi_{\rm eff}$ and 0.43–0.70 for M (± 0.09 from 30 injections; the merger time, whose width is 2–4 ms, has 0.5 because of the $\sim$ms offset between the template and the envelope peak that defines the injected time). The accuracy is the same as that of the non-spinning grid (≈ 7 %, Section 12): the spin and mass-ratio degeneracies, not the model, limit M_c and M; but the intervals are calibrated, while those of the grid posterior covered the truth only in 9–32 % of the cases.

Signal or glitch with two detectors. Bayes factors with amplitude marginalisation ($\sigma = 10$) of a coherent signal (one complex amplitude, real detector weights integrated over the angle), of a glitch in H1 only or in L1 only (equal prior) and of independent signals, evaluated at the peak of each template, are used to classify physical network injections against blips in one detector of the same network norm. The AUC of the odds $\ln B_C - \ln B_G$ is 0.72, 0.85, 0.91, 0.93 at network norm 6, 8, 10, 12, no better than that of $\min(\rho_H^2, \rho_L^2)$ (0.74, 0.87, 0.92, 0.94): with two detectors almost all the discrimination comes from requiring a signal in both. The odds are *not* calibrated: a $\ln B_C - \ln B_G > \ln 100$ occurs in 8 % of the noise window pairs (32 % for $\ln B_C$ alone), because the profile over 24 templates and the lag peaks is not the marginalisation of Section 5 and carries the full look-elsewhere factor; its threshold must be set on the background, as for any frequentist statistic.

Simulation-based calibration of the grid posterior. In Gaussian noise, with data drawn from the generative model (1500 simulations), the posterior of (f_0, Q, t_0) over the sine-Gaussian bank is close to calibrated but not exactly: the Kolmogorov–Smirnov distance of the rank from the uniform is 0.05–0.06 for f_0 and Q and 0.16–0.18 for t_0 (mean rank 0.61: the posterior sits $\approx$ half a lag-bin early; KS $p < 10^{-3}$ for all with this many simulations), and the central 68 % interval contains 71–74 % of the truths for f_0 and Q. A four times finer grid (640 against 160 templates) does not improve it (0.060 against 0.050 for f_0): the residual is the discretisation of the lag and of the template cells, not the density of the bank.

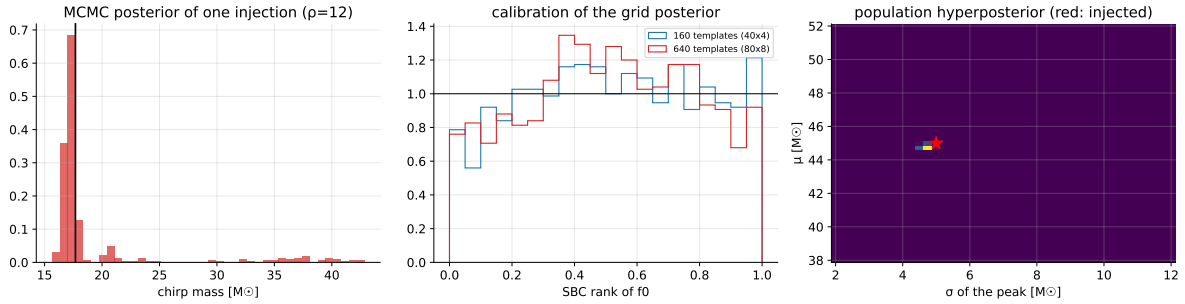


Figure 16. Left: MCMC posterior of the chirp mass of one injection. Centre: SBC rank of f_0 for the two banks. Right: hyperposterior of the mass distribution of Section 13.

6 Machine-learning detectors

6.1 Data set and models

The labelled set is built from the same chunks and the same split: for each noise window of a split we also build a signal window, the *same* window with a burst whose morphology is drawn uniformly among the three and whose optimal SNR is uniform in $[3, 15]$. Classes are balanced and the label-0 windows contain all the real glitches and non-stationarity (16 224 training and 8112 validation windows per class, 8112 test windows). A second training set contains only sine-Gaussians, to test generalisation to morphologies not seen in training. The networks see the whitened series clipped to ± 12 (glitches reach hundreds of σ); the feature pipeline uses the unclipped series.

Feature-based models. Twenty-two morphology-agnostic features: the maximum over time of the z-scored sliding energy at five durations (the excess-power statistic at each scale), the maximum and the standard deviation over time of the energy in seven frequency sub-bands (30–500 Hz, from a short-time spectrogram), and $\max |v|$, kurtosis and std/MAD of the window. No matched-filter output is used.

CNN. A one-dimensional convolutional network of 17 761 parameters: three Conv–BatchNorm–ReLU blocks (kernels 15, 9, 5; 16, 32, 64 channels; max-pooling by 4), global max-pooling (translation invariance: the burst can be anywhere), a dense layer with dropout 0.2 and one logit; binary cross-entropy, Adam [14], batch 128, early stopping on the validation loss (14 epochs). The *ensemble* is five independently initialised copies (the logits are averaged); a second network is trained on sine-Gaussians only.

Autoencoder. A convolutional autoencoder (3.6×10^5 parameters, 32 latent values for 1024 samples) trained on the real noise windows *only*; the reconstruction error is a morphology-independent anomaly score.

6.2 Regularisation, bias–variance and what the features say

Logistic regression with L_1 and L_2 penalties (Fig. 17) reaches a validation AUC of 0.927 with all features; the 5-fold cross-validated L_1 choice $C = 0.019$ keeps 18 of the 22 features and costs nothing in AUC, while for $C \lesssim 10^{-3}$ the L_1 model collapses much faster than L_2 (it sets coefficients to zero one by one). Trees (Fig. 17): the validation AUC of gradient boosting saturates at 0.938 for depth 4–6 while the training AUC keeps rising (0.936 $\rightarrow$ 0.956 from depth 1 to 12): beyond depth ~ 4 there is only variance. The learning curve (depth 4) starts at a training AUC of 1.0 against 0.925 on validation with 500 examples, and the two approach each other (0.952 / 0.938) at 3.2×10^4 examples without having converged: the feature models are data-limited, which is the practical argument for giving the network the raw series. The most important features by permutation importance are the 10 ms excess-power score and the energy in the 50–180 Hz bands; the first principal component of the features carries 46 % of the variance. Validation AUCs: ridge 0.927, random forest 0.938, gradient boosting 0.938.

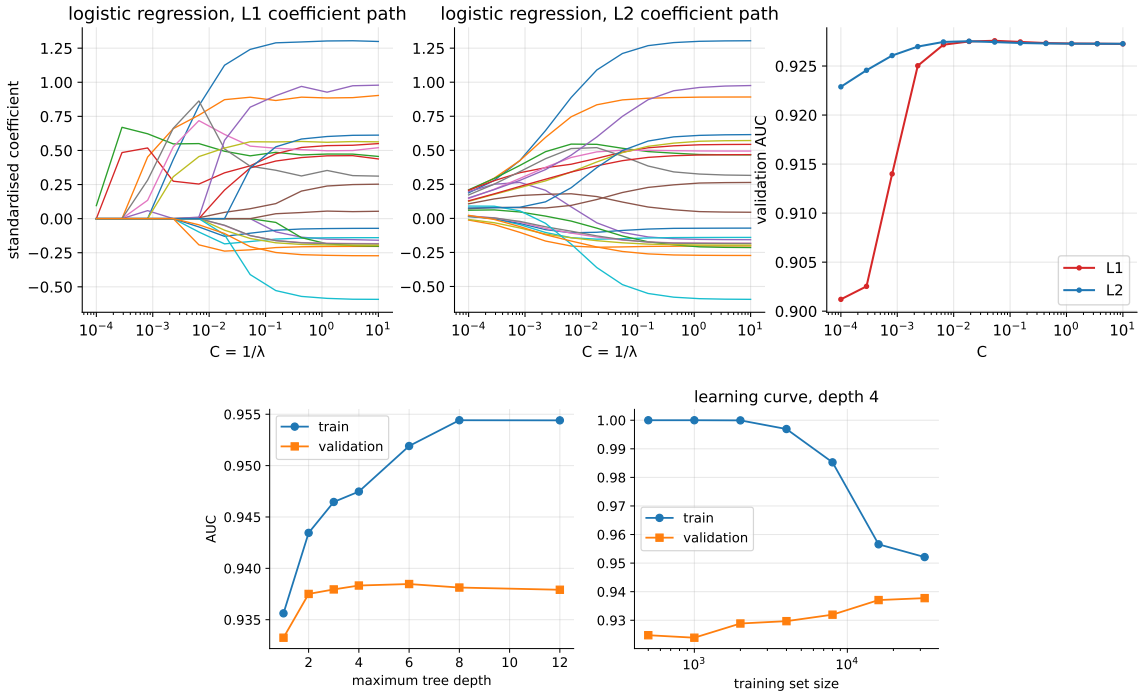


Figure 17. Top: L_1 and L_2 coefficient paths and validation AUC against C. Bottom: bias–variance (AUC against tree depth) and learning curve of gradient boosting.

6.3 Comparison with the frequentist statistics on the same injections

All statistics are evaluated on exactly the same test windows and injections, with the threshold of each set to the 99 % quantile of its own distribution on the *same* background, the validation noise (chunks 1 and 5). A common background is what makes the comparison fair: Section 4 used the training chunks, whose glitchy chunk 2 gives a more conservative bank threshold (there $\rho_{50} = 5.5$ for sine-Gaussians; here 4.9). Table 7 and Fig. 18 summarise the result; if the threshold is instead set on the test noise itself (equal observed FAP) the networks gain 0.4–0.6 in ρ_{50} (the test epochs are cleaner than the validation ones) and the bank only 0.04, so that the ensemble then also beats the bank in family (4.6 against 4.8); the conclusions for pulse and chirp and for the unseen morphology are unchanged.

statistic	sine-Gaussian		Gaussian pulse		linear chirp	
	ρ_{50}	ρ_{90}	ρ_{50}	ρ_{90}	ρ_{50}	ρ_{90}
matched-filter bank	**4.87**	**6.53**	5.34	6.82	6.28	8.21
excess power	6.54	9.10	5.77	7.54	10.10	12.21
ridge-logistic (22 features)	6.40	8.26	5.72	7.46	6.81	8.96
gradient boosting (22 features)	5.61	7.30	5.70	7.24	6.06	7.73
CNN (mixed training)	5.56	7.80	4.62	6.52	5.71	7.64
CNN ensemble (5)	5.05	7.05	**4.21**	**5.90**	**5.28**	**6.92**
CNN (sine-Gaussian only)	5.09	6.87	6.50	12.93	6.80	10.82
autoencoder, window MSE	12.09	15.14	12.44	15.06	12.32	15.14
autoencoder, local max	6.89	8.80	6.75	8.53	8.30	11.08

Table 7. Optimal SNR at 50 % and 90 % efficiency (FAP 10^{-2} on validation noise, test injections). Bold: best in each column.

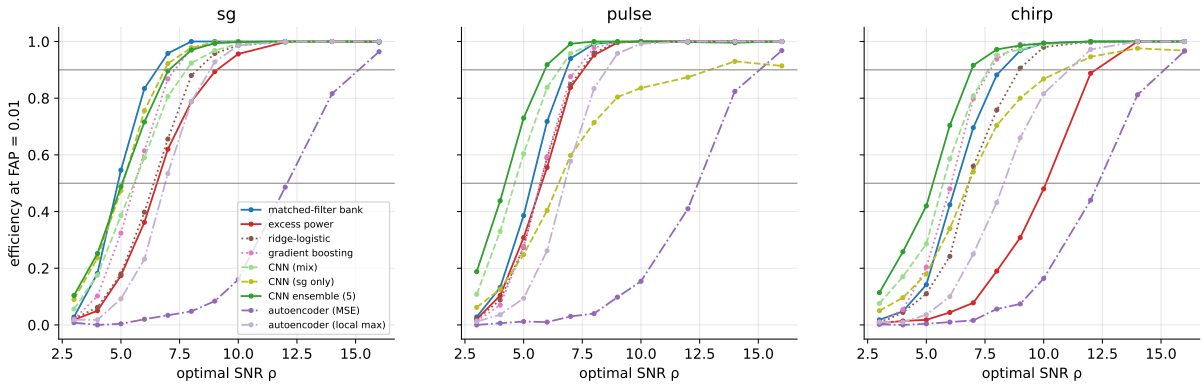


Figure 18. Efficiency against optimal SNR at FAP 10^{-2} for all statistics, on the same test injections.

- **In family (sine-Gaussian).** The matched-filter bank and the ensemble are close ($\rho_{50} = 4.9$ and 5.1); the bank is better at 90 % (6.5 against 7.1), the network wins by 0.2 when the threshold is set on the test noise. Parity is a respectable result for a network that has seen no template.
- **Morphologies in the training set (pulse, chirp).** The ensemble is about one unit of SNR better than the bank (4.2 against 5.3; 5.3 against 6.3): it has *learned* shapes that the bank covers only through its sine-Gaussian templates. Gradient boosting on 22 features is already at the bank level.
- **A morphology not seen in training.** The CNN trained on sine-Gaussians only loses efficiency exactly where its training stops: $\rho_{50} = 6.5$ (pulse) and 6.8 (chirp) against 5.3 and 6.3 for the bank, and the efficiency at $\rho = 8$ falls from 0.98 to 0.71 and 0.70 (bank: 0.99, 0.88). The advantage of a supervised detector is bought with training data; outside the training distribution the bank, which has no data to fail on, is the more robust.
- **Unsupervised.** The reconstruction error of the whole window is the total energy of a second of almost white noise — there is nothing to learn — and is inefficient ($\rho_{50} \approx 12$). The *local* maximum of the error recovers the performance of excess power (6.9, 6.8, 8.3) but not more: an autoencoder can beat an energy statistic only if the noise has structure that it can learn, which after whitening it hardly has.

- **The CNN of this section is data-limited.** Section 7.3 shows that training it with fresh injections at every epoch improves it by ≈ 0.5–0.7 in ρ_{50} (a single network then beats the five-member ensemble); the numbers of this section are those of the static-data network, kept as the reference of the comparison.
- **False alarms.** On the test chunks the observed FAP at a nominal 1 % is 0.85 % (bank), 0.92 % (excess power), 0.75 % (gradient boosting), 0.53 % (ensemble) and 0.35 % (single CNN): no statistic is immune to the epoch dependence of the background.

6.4 Calibration and uncertainty

A score is useful to a physicist if it is a probability. On balanced validation and test sets (noise plus mixed injections with SNR 3–15) all models are close to calibrated (Fig. 19): the expected calibration error is 0.7 % for gradient boosting, 2.2–2.4 % for the CNNs and 2.7 % for the ridge model on the test set. The CNNs are slightly *under*-confident at intermediate scores (observed signal fraction up to ≈ 0.15 above the prediction between 0.1 and 0.7). Temperature scaling [15] fitted on validation does not help ($T = 0.94$–1.02; ECE changes by at most 0.002): the miscalibration is not of the temperature type. For the CNNs the ECE is 0.8–1.2 % on the validation epochs they were tuned on and 2.2–2.4 % on the test epochs, so most of it is an *epoch shift*. The ensemble is better than a single network in every proper score (Brier 0.0350 against 0.0413, log-loss 0.110 against 0.128, AUC 0.9938 against 0.9909); the AUC on the test epochs is higher than on the validation ones for every model (ensemble $0.988 \to 0.994$), because the test chunks are cleaner: no sign of over-fitting to the training epochs.

The epistemic uncertainty is estimated with the ensemble [16] and with MC dropout [17] (30 stochastic passes). The ensemble spread falls with SNR (mean std 0.16 at $\rho \approx 4$, ≈ 0.001 for $\rho \geq 12$) and peaks at $p \approx 0.5$; MC dropout shows the same trend but is only moderately correlated with it (0.35): they measure different things (spread over dropout masks of one network against disagreement of independently trained networks). Rejecting the 20 % most uncertain windows raises the AUC from 0.9924 to 0.9997.

6.5 Hard-negative mining and families of real glitches

Mining. Notebook 17 fine-tunes the fresh-injection CNN for 8 epochs either as is (control) or repeating 5 times, at every epoch, the 10 % of training noise windows with the highest network score. Mean ρ_{50} over the three morphologies (validation thresholds): baseline 4.57/5.66, control 4.37/5.80, mining 4.46/5.84 at FAP $10^{-2}/10^{-3}$. The paired difference mining $-$ control is $+0.10$ (95 % interval -0.15 to $+0.33$) and $+0.04$ (-0.21 to $+0.33$): no measurable effect. With a binary cross-entropy that already sees 16 000 noise windows per epoch, the hard windows are not rare enough for re-weighting to matter at FAP $\geq 10^{-3}$; the effect might appear at lower FAP, which the available background cannot test.

Families. Notebook 18 takes the 4 % of validation and test windows with the largest bank statistic (649), describes them by their log-spectrogram and clusters them (k-means, 3 families plus an "extreme" family of the 3 % most atypical windows). The silhouette of the typical windows is only 0.08: they are noise-tail fluctuations that differ essentially by the band in which the bank fired (peak frequency 113, 218 and 280 Hz), not distinct glitch classes. The extreme family (20 windows, median $\max|v| = 4.7$) is the real-glitch population: the bank

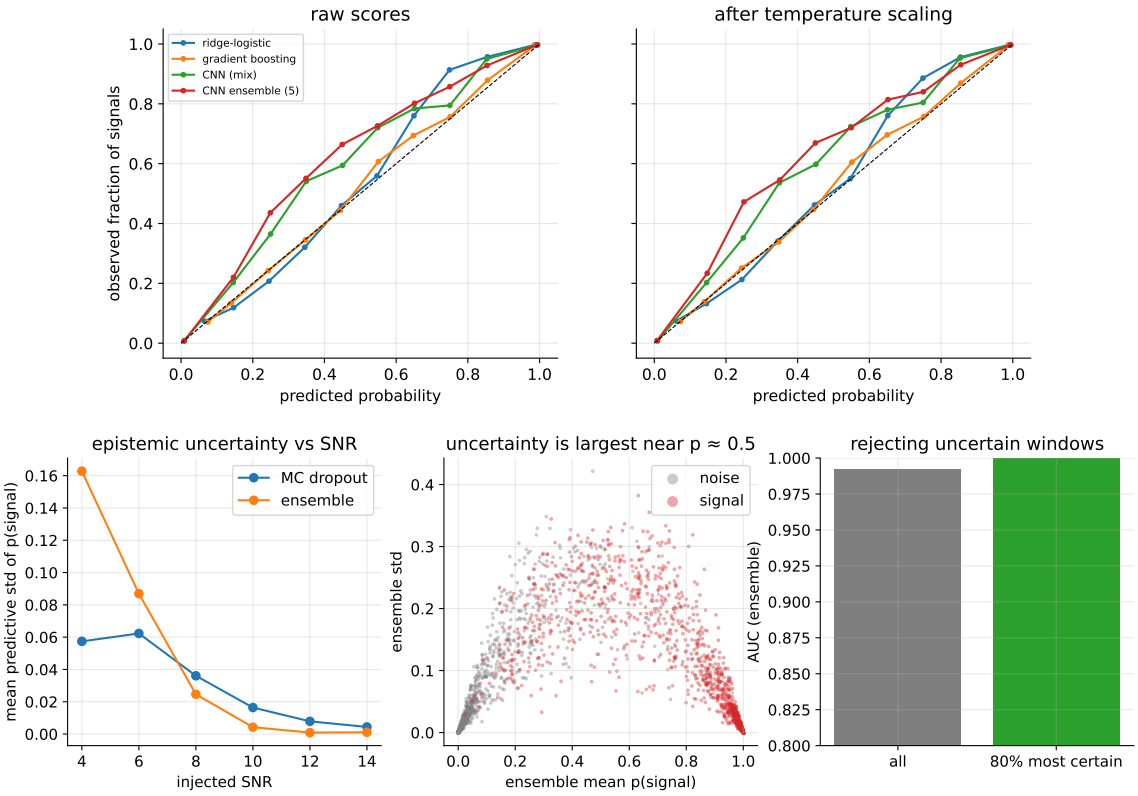


Figure 19. Top: reliability diagram on the test set, raw and after temperature scaling. Bottom: predictive standard deviation against SNR (ensemble and MC dropout), against the ensemble mean, and the effect of rejecting uncertain windows.

alarms in 95 % of them at FAP 10^{-2}, the $\chi^2 < 3$ condition removes ≈ 95 % of those alarms and the CNN alarms in 5 % (Fig. 20). Without human labels the unsupervised analysis therefore supports one conclusion: what separates the detectors is a small tail of loud glitches that the template statistic cannot reject by amplitude and that the network and the χ^2 test do.

6.6 Prior mismatch and the base rate

Prior mismatch. Notebook 20 tests the fresh-injection CNN and the 160-template bank on injections outside the training ranges (FAP 10^{-2}, validation thresholds; Fig. 21). The bank keeps its performance ($\rho_{50} \leq 7.0$ in every case, and efficiency 1.0 at $\rho = 40$ everywhere), the CNN does not:

- sine-Gaussians at 400–500 Hz: $\rho_{50} = 8.6$ for the CNN against 5.1 for the bank (efficiency at $\rho = 8$: 0.42 against 0.99); at 30–50 Hz 6.7 against 5.7;
- pulses of 5–15 ms: 9.3 against 6.7; chirps at 150–250 Hz: 7.4 against 7.0;
- loud signals: the efficiency at $\rho = 40$ of the CNN falls to 0.72 for pulses in the training range, 0.02 for pulses of 0.3–0.8 ms and 0.65 for chirps at 150–250 Hz, because the network input is clipped at ± 12 and the training SNR stops at 15: the loudest events are the ones it misses;
- mild extrapolations (broader or narrower Q, steeper chirps) are harmless: 4.6–5.3 for the CNN.

The CNN advantage of this section is therefore an advantage *inside the injection prior*; outside

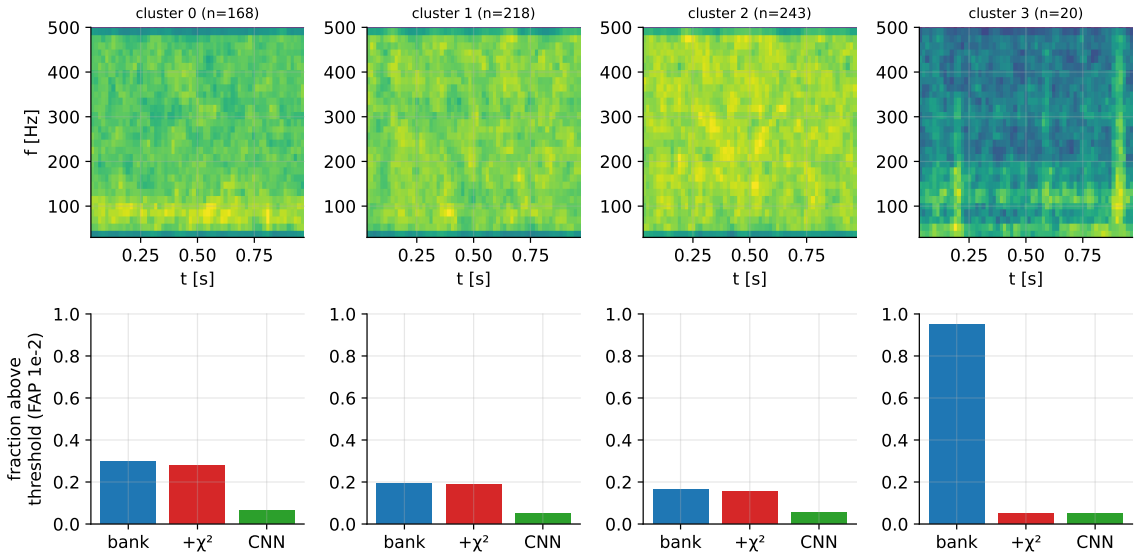


Figure 20. Mean log-spectrogram of each family of candidate windows (top) and fraction above the FAP 10^{-2} threshold for the bank, the bank with $\chi^2 < 3$ and the CNN (bottom).

it, the template statistic degrades gracefully and the network does not. The remedy (a wider prior and no clipping, or a bank veto for loud events) is not tested here.

Base rate. The efficiencies of this report describe a balanced problem. If a one-second window contains a signal with prior probability π, an alarm is real with probability $\mathrm{PPV} = \varepsilon\pi/(\varepsilon\pi + \mathrm{FAP}(1-\pi))$. With $\varepsilon \approx 0.99$ at $\rho = 8$ and for an *illustrative* rate of one detectable burst per day ($\pi \approx 1.2 \cdot 10^{-5}$) the PPV is 0.001 at a single-detector FAP of 10^{-2}, 0.01 at 10^{-3} and 0.90 at the two-detector 10^{-6} of Section 10; at one per month it is $4 \cdot 10^{-5}$, $3 \cdot 10^{-4}$ and 0.24. A PPV of 0.9 needs a FAP of 10^{-6} (one per day) or $4 \cdot 10^{-8}$ (one per month): the single-detector FAPs quoted in this report are far from sufficient for a real search, which is why the two-detector time slides are essential. The rates are assumptions, not measured burst rates.

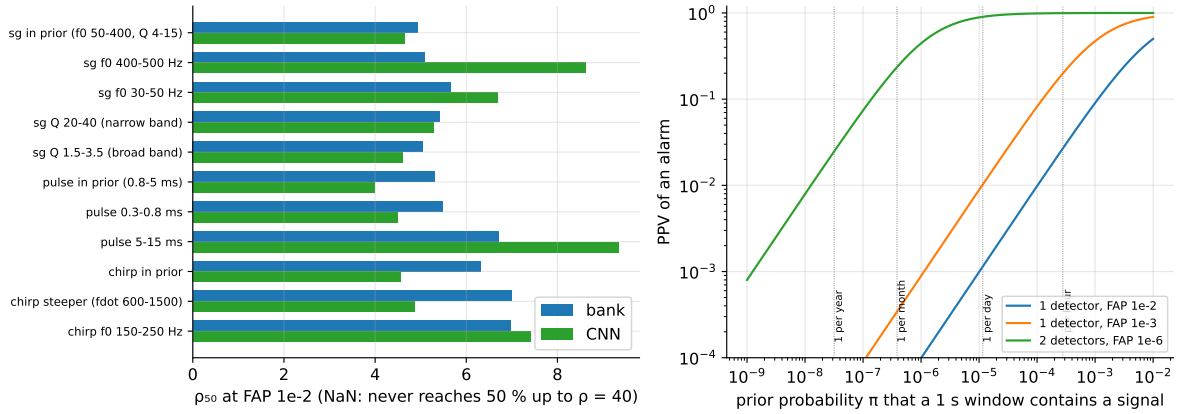


Figure 21. Left: ρ_{50} of the bank and of the CNN on injections outside the CNN training prior (a bar is missing if the detector never reaches 50 %). Right: probability that an alarm is real against the prior probability of a signal per window, with the illustrative rates marked.

6.7 Robustness, conformal thresholds, interpretability, latency and adversarial attacks

Notebook 25 stresses the fresh-injection CNN (two seeds) with the bank as reference; thresholds are those of the validation noise at FAP 10^{-2} and the efficiency is the mean at $\rho = 6$ over the three morphologies.

perturbation	FAP bank	FAP CNN	eff. bank	eff. CNN
baseline	0.0085	0.0038	0.66	0.86
signal amplitude $\times 0.9$ / $\times 1.1$	–	–	0.49 / 0.80	0.76 / 0.94
60 Hz line left in the noise (norm 3 / 6)	0.0085	0.17 / 0.29	0.66 / 0.65	0.87 / 0.88
signals at the window edges (t_0 outside 0.25–0.75)	–	–	0.02	0.74
PSD from the mean / 1 s / 16 s segments	0.0064 / 0.0075 / 0.0097	0.0035 / 0.0057 / 0.0048	0.64–0.66	0.86–0.87

Table 8. Robustness of the bank and of the CNN to one perturbation at a time (nominal thresholds).

- A ± 10 % error on the calibration of the signal amplitude moves the efficiency at $\rho = 6$ by about ± 0.1–0.15 for both detectors, as any ρ rescaling does; the alternative PSD estimates change the FAP by a factor < 1.5 and the efficiencies by < 0.02 for both.
- **A spectral line left in the data breaks the CNN, not the bank**: with a 60 Hz line of norm 3 the CNN FAP rises from 0.4 % to 17 % (29 % at norm 6), the bank is unchanged. The network has never seen a persistent line and responds to a feature that the template statistic, with its narrow-band but short-lived templates, ignores. Retraining with augmentation and the robust tests of Section 6.6 are the remedies; the line itself is removed in practice by the PSD whitening.
- The bank cannot see signals whose lag lies outside the range it scans (0.15–0.85 s), an explicit design choice, while the CNN keeps 0.74 out of its training prior of the lag.

Conformal prediction. A split-conformal threshold (the $\lceil (n+1)(1-\alpha) \rceil$-th calibration score) is guaranteed to give FAP $\leq \alpha$ for exchangeable data; across epochs it is not: calibrating on one validation chunk and testing on the other gives observed/nominal ratios of 0.5–1.9 for the CNN and 0.5–3.5 for the bank at $\alpha = 10^{-2}, 10^{-3}$, i.e. the guarantee holds only within an epoch (this is the epoch dependence of Section 4.9 seen from another side). Conformal sets of the 4-class morphology network at 90 % nominal coverage have an average size 1.00–1.04 and marginal coverage 0.98 on noise, but the *conditional* coverage depends on the SNR (sine-Gaussian: 0.07 at $\rho = 3$, 0.69 at 6, 0.95 at 8): a faint signal receives a confident wrong label rather than a larger set, and the 90 % holds only on average over the training SNR prior.

Interpretability. The input gradient times the input of the detection logit puts 30 % of its mass within $\pm 4\sigma_t$ of an injected sine-Gaussian, against 10 % for a uniform saliency; the 16 first-layer filters peak between $\approx$ 40 and 500 Hz (Fig. 22), covering the injection prior as a filter bank would.

Latency. Per one-second window, the CNN takes 0.17 ms in batches of 1024 (and 88 ms for a single call, Keras overhead), the hybrid 1.9 ms, the Transformer 1.0 ms and the 160-template bank 3.5 ms (numpy FFT): the network is 20 times faster than the bank on a batch.

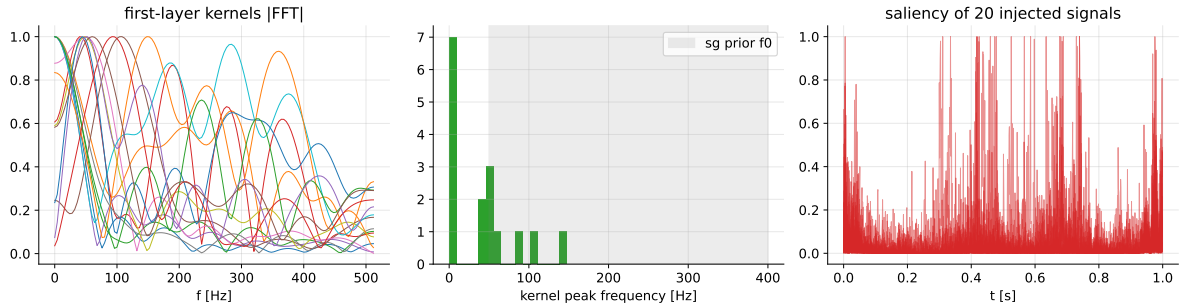


Figure 22. Left: spectra of the 16 first-layer kernels. Centre: their peak frequencies against the sine-Gaussian prior. Right: saliency of twenty injected signals.

Adversarial perturbations. A one-step FGSM attack with $\epsilon = 0.1\sigma$ per sample (perturbation norm 3.2, a third of the signal norm 10) lowers the CNN efficiency at $\rho = 10$ from 1.0 to 0.33 (0.02 at 0.2σ), while the bank, given the same perturbation, stays at 0.998; in the opposite direction $\epsilon = 0.05\sigma$ takes the CNN FAP from 0.5 % to 7 %. The CNN is not adversarially robust, which matters only if the noise could be chosen by an adversary; it matters in practice as a statement about sensitivity to small coherent structure.

6.8 Scaling with the data, augmentation, label noise and transfer to L1

Notebook 26 retrains the CNN of notebook 07 with fewer or differently prepared data (one seed per variant; mean ρ_{50} over the three morphologies, FAP $10^{-2}/10^{-3}$, validation thresholds).

variant	FAP 10^{-2}	FAP 10^{-3}
static injections, 1000 / 4000 / 16 224 noise windows	6.70 / 6.16 / 5.34	9.61 / 7.39 / 6.48
fresh injections, 1000 / 4000 / 16 224 noise windows	6.53 / 5.83 / 4.49	9.33 / 7.11 / 5.67
static, 16 224 windows, sign flip + circular shift	5.32	6.15
static, 4000 windows, sign flip + circular shift	5.94	7.23
fresh, 4000 windows, 5 % / 20 % wrong labels	5.77 / 6.21	6.72 / 8.47
H1-trained CNN on H1 / zero-shot on L1	4.57 / 5.55	5.66 / 6.97
fine-tuned on L1 (5 epochs)	5.49	6.94
trained from scratch on L1 (two chunks)	8.45	9.95

Table 9. Mean ρ_{50} (bootstrap error 0.04–0.07 at 10^{-2}, up to 0.5 at 10^{-3} for the smallest sets).

- **The network is data-limited**: going from 10^3 to $1.6 \cdot 10^4$ noise windows lowers ρ_{50} by 1.4 (static) to 2.0 (fresh injections) at FAP 10^{-2} and by 3.1 to 3.7 at 10^{-3}, with no sign of saturation: more background is the most direct way to improve the detector, and the fresh-injection advantage grows with the amount of noise (0.2 at 10^3, 0.9 at $1.6 \cdot 10^4$).
- **Flip-and-shift augmentation** does not replace fresh injections: it changes nothing at full size (5.32 against 5.34) and gains 0.2 at 4000 windows; the over-fitting of the static set is a lack of signal diversity, not of translation or sign invariance.
- **Label noise is tolerated**: 5 % wrong labels cost nothing measurable, 20 % cost 0.4 (FAP 10^{-2}) and 1.4 (10^{-3}).
- **Transfer works**: the H1-trained network applied to L1 gives 5.55 (the noise and the thresholds are those of L1, from two chunks), fine-tuning on L1 does not improve it (5.49), while

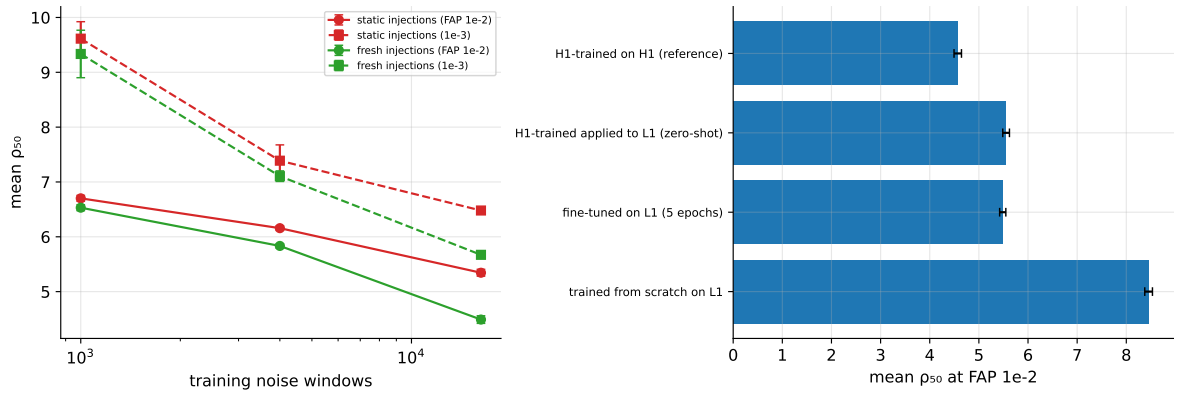


Figure 23. Left: ρ_{50} against the number of training noise windows. Right: transfer from H1 to L1.

a network trained from scratch on the two L1 training chunks is far worse (8.45): the value of the H1 training is in the number of noise windows, not in the specifics of H1.

6.9 Unsupervised noise models: a normalising flow and a variational autoencoder

Notebook 33 uses models of the noise only (no signal is ever seen) as model-independent detectors. A RealNVP flow (6 affine coupling layers) learns the density of a 16-dimensional description of each window (8 log band energies, the maxima of the sliding energy at 5 durations, max $|v|$, skewness, kurtosis); its score is $-\ln p$. A 1-D convolutional VAE (latent dimension 16) reconstructs the whitened samples; scores are the largest local (50 ms) reconstruction error and the window ELBO. Thresholds are those of the validation noise. The local VAE score reaches

detector	sine-Gaussian	pulse	chirp	IMR	AUC($\rho = 8$, sg)
normalising flow ($-\ln p$)	7.45	7.24	9.25	9.45	0.97
VAE, local reconstruction error	6.74	6.52	8.50	7.69	0.98
VAE, window ELBO	12.2	12.4	12.3	12.4	0.79
excess power (Section 6)	6.54	5.77	10.10	8.28	—
CNN, fresh injections (Section 7)	4.7	4.2	4.8	4.45*	—

Table 10. ρ_{50} at FAP 10^{-2} of the unsupervised detectors (IMR: held-out IMRPhenomD library; * network trained with IMR signals).

the level of excess power on bursts (6.5–6.7 for sine-Gaussians and pulses) and is better on chirps and IMR signals (8.5 and 7.7 against 10.1 and 8.3); it improves on the deterministic autoencoder of Section 6. The flow, on hand-built features, is 0.7–1 unit worse than the VAE. The window ELBO is useless as a detector: the KL term and the global average dilute a 50 ms burst. At FAP 10^{-3} the gap to the supervised detectors widens (9.2–12.5 against 5.7 for the CNN): the price of seeing no signal during training is ≈ 2 units at 10^{-2} and ≈ 4 at 10^{-3}, in return for a detector that has no prior on the morphology (the IMR signals are detected as well as the three burst families). The alarms of the unsupervised models are not simply the glitches that fool the bank: only 20–26 % of the bank alarms are also flagged (Spearman correlation of the scores with the bank statistic 0.17–0.33), so they are complementary background rejectors rather than copies of the matched filter.

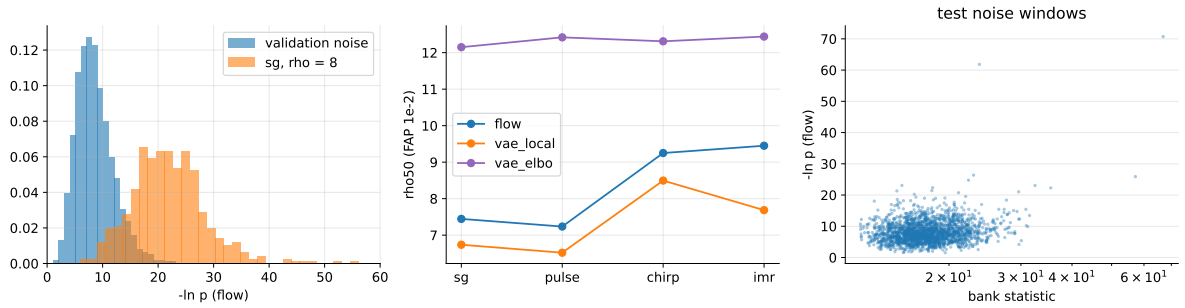


Figure 24. Left: flow score of validation noise and of $\rho = 8$ sine-Gaussians. Centre: ρ_{50} per family. Right: flow score against the bank statistic on test noise.

7 Sequence models and signal hypothesis testing

Section 6 used a small convolutional network. Two natural questions follow, both asked on the same data, the same split by chunk, the same test injections and the same thresholds on the validation noise: (i) does a structure designed for sequences — a recurrent network or a Transformer [18] — detect bursts better than the CNN, and why or why not; (ii) for a signal that is above the detection threshold and incompatible with the noise, which method can tell *which morphology* produced it? Because the AUC of the CNN ensemble is already ≈ 0.99, the detectors are judged by the efficiency at a fixed low false-alarm probability (ρ_{50}, ρ_{90} at FAP 10^{-2} and 10^{-3}) and by their behaviour on glitches, not by the AUC.

7.1 Transformer and GRU against the CNN

Architectures. The 1024 samples are cut into 64 non-overlapping patches of 16 samples (15.6 ms) [19] and embedded linearly ($d = 64$). The *Transformer* has a learned positional embedding, three pre-norm blocks (4 heads, MLP 128, dropout 0.1) and a head that concatenates max- and average-pooling over the tokens (1.1×10^5 parameters). The *GRU* [20] has the same patch embedding and head, and two stacked bidirectional GRU layers of 48 units (8.8×10^4 parameters): it isolates the sequence mixer (recurrence against attention). The CNN of Section 6 has 1.8×10^4 parameters. All are trained with AdamW [21], two epochs of warm-up and cosine decay, batch 128, early stopping on the validation loss, at most 12 epochs (CPU budget); the first seed of the Transformer was also trained for 25 epochs and ends with the same validation loss (0.312 against 0.310).

Pre-training. A masked-patch self-supervised pre-training [22] (40 % of the patch embeddings replaced by a mask token, a linear decoder reconstructs the raw samples) on the *noise-only* training windows is tested as a way to save labelled injections.

Results. Table 11 and Fig. 25 give the efficiency. The CNN ensemble and the Bayes factor of the three families are the best at FAP 10^{-2}; the GRU and the Transformer are behind the CNN by 0.7 to 2.1 SNR units (the smallest gap is on pulses, the largest on chirps for the Transformer); at FAP 10^{-3} the ranking is unchanged and the gap grows (on chirps the Transformer needs $\rho_{50} = 8.8$ against 6.2). Four further observations:

- **The number of parameters is not the explanation.** The Transformer has six times the parameters of the CNN. The CNN's local filters followed by a global max-pool encode the two facts the task needs — a burst is localised and can be anywhere in the window — and at this training-set size ($\sim 1.6 \times 10^4$ windows per class) that prior is worth more than the flexibility of attention. The label-efficiency curve (Fig. 26) agrees: with 1000 labelled windows per class the CNN is at AUC 0.91 and the Transformer at 0.54.
- **Pre-training learns almost nothing.** The masked-patch error is 0.995 of the error of the trivial predictor "zero": after whitening the noise is nearly white, so there is little to predict from the neighbours (the same reason why the autoencoder of Section 6 does not beat excess power). The pre-trained encoder is a better starting point only with 1000 labels per class (AUC 0.76 against 0.54) and equal to the random start with 4000 labels or the full set; at the level of ρ_{50} it changes nothing (6.29/4.87/7.29 against 6.38/4.88/7.35).
- **The attention is informative** (Fig. 27). The most attended patch is within ± 1 patch of the burst time in 48 % (sine-Gaussian), 61 % (pulse) and 44 % (chirp) of the windows, against 4.7 % by chance; chirps, which last 0.1–0.2 s, are the least localised and are where the Transformer is weakest.
- **Glitch rejection is not better.** Taking the 5 % of test noise windows with the largest $\max|v|$, the alarm rate inside this subset at the threshold of FAP 10^{-2} is 2.7 % (CNN ensemble), 4.9 % (Transformer), 5.2 % (GRU), 5.7 % (Bayes) and 6.9 % (bank); the efficiency for $\rho = 8$ injections is 0.96–0.98 (CNN), 0.83 (Transformer), 0.995 (Bayes). A glitch does not hide a $\rho = 8$ signal for any method (the efficiency inside the glitchy windows equals the overall one).

	FAP 10^{-2}: ρ_{50}			FAP 10^{-3}: ρ_{50}		
statistic	SG	pulse	chirp	SG	pulse	chirp
matched-filter bank (SG templates)	4.87	5.34	6.28	6.80	7.37	9.01
Bayes factor (3 families)	5.21	4.78	5.34	7.56	7.24	7.72
CNN ensemble (5)	**5.05**	**4.21**	**5.28**	**6.09**	**5.04**	**6.24**
Transformer (2 seeds)	6.38	4.88	7.35	7.46	5.73	8.78
Transformer, pre-trained encoder	6.29	4.87	7.29	7.26	5.73	8.77
GRU	6.27	5.16	6.36	7.44	6.05	7.70

Table 11. Optimal SNR at 50 % efficiency of the sequence models, the CNN and the model-based statistics (thresholds on the validation noise; test injections). Bold: best neural network in each column.

7.2 Signal hypothesis testing: which morphology?

Given a signal above threshold, the decision among the three families is compared across methods (Fig. 28). The frequentist rules use the maximum matched-filter statistic of each family (a bank of 160 sine-Gaussian, 14 pulse and 120 chirp templates), either raw or converted into a background p-value to take the different trial factors into account; the Bayesian rule uses the marginal likelihood (Bayes factor) of each family with $\sigma_\rho = 10$ and a uniform prior over the templates, so that the Occam factor penalises the larger families; the networks are trained as four-class classifiers (noise, sine-Gaussian, pulse, chirp) on the same data as Section 6. All methods use the same rule for the four-class decision: a window is called noise if its own detection statistic is below the FAP 10^{-2} threshold of the validation noise, otherwise the family with the largest score is chosen.

- **Bayesian model comparison is the best classifier from $\rho \approx 6$ among the networks**

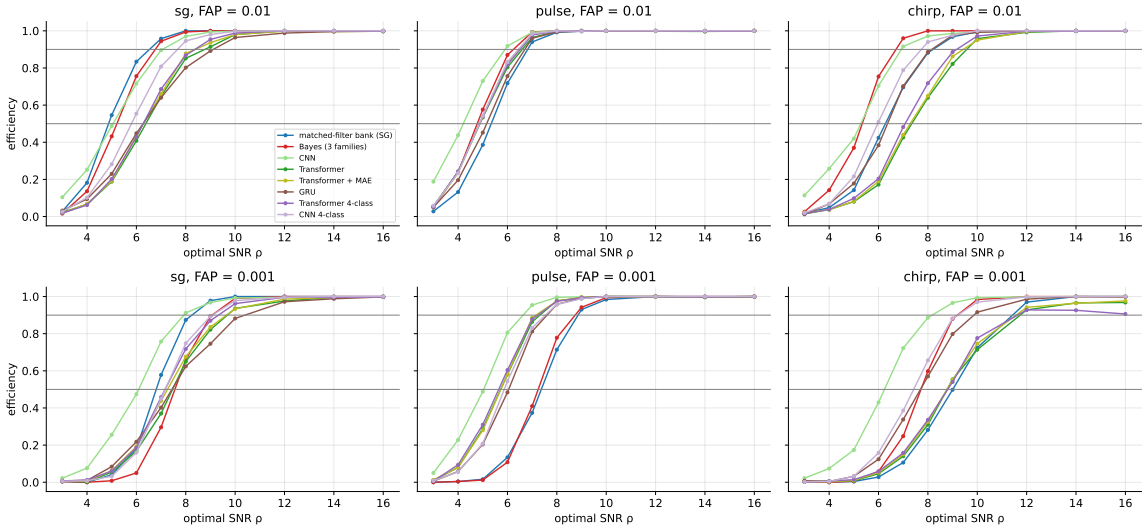


Figure 25. Efficiency against optimal SNR at FAP 10^{-2} (top) and 10^{-3} (bottom) for the model-based statistics, the CNN ensemble, the Transformer (with and without the pre-trained encoder) and the GRU.

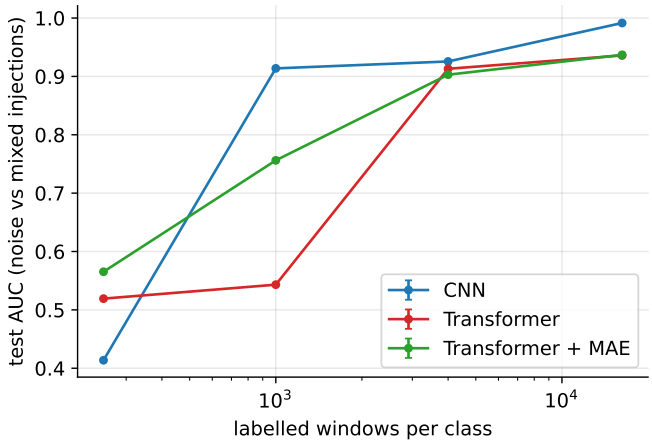


Figure 26. Test AUC against the number of labelled windows per class for the CNN, the Transformer and the Transformer with the pre-trained encoder.

trained on the static set (tied with the CNN above $\rho \approx 10$) (0.969 at $\rho = 8$; this ranking changes when the networks are trained with fresh injections, Section 7.3) and has the best four-class balanced accuracy in both SNR ranges (0.778 for $\rho = 4$–7, 0.988 for $\rho = 8$–16). The CNN is close at high SNR (0.960 at $\rho = 8$) and 0.04–0.05 behind at $\rho = 5$–6; then come the GRU, and last the Transformer (0.867–0.873), pre-trained or not. The answer to the question is therefore: among the networks the convolutional structure (local filters plus global pooling) does best, a bidirectional recurrent encoder is second and attention on 16-sample patches third; none beats the Bayesian comparison, which uses the exact likelihood of each family.

- **The frequentist rule depends on how the trial factors are treated.** The largest raw ρ_f^2 is 0.05 worse than the Bayes factor at $\rho = 8$; converting each into a background p-value is the best rule at $\rho \leq 5$ but its accuracy *falls* with SNR beyond $\rho \approx 9$ (0.82 at 14) because the empirical p-values of loud signals saturate at $1/(N+1)$ for every family and the tie is broken by the raw ρ^2, which favours the families with more templates. This is a limit of the 8112-window background, not of the method.

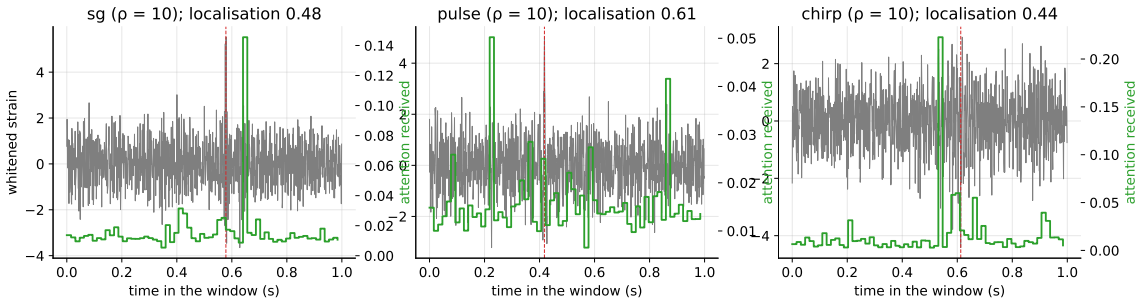


Figure 27. Attention received by each patch (green) in the last block of the Transformer for one $\rho = 10$ injection of each morphology; dashed: burst time.

3-way accuracy at $\rho =$	5	6	8	10	14
Bayes (3 families)	0.771	**0.906**	**0.969**	0.977	0.989
frequentist, max ρ^2	0.724	0.837	0.921	0.948	0.981
frequentist, background p-value	**0.825**	0.900	0.913	0.889	0.816
CNN 4-class (2 seeds)	0.725	0.863	0.960	**0.980**	**0.991**
GRU 4-class	0.665	0.813	0.925	0.969	0.989
Transformer 4-class	0.643	0.759	0.867	0.925	0.955
Transformer 4-class, pre-trained	0.665	0.765	0.873	0.924	0.955

Table 12. Accuracy of the choice among the three families for injections of each SNR (chance: 0.333). The p-value row is plotted in Fig. 28; bold: best in each column (at $\rho = 10$ and 14 the CNN and the Bayes factor differ by less than the statistical error, ≈ 0.01).

- **Detection comes first.** Only $\approx 46\,\%$ of the $\rho = 5$ injections and 97 % of those at $\rho = 7$ pass the common Bayesian gate, so the morphology can be discussed only for signals loud enough to be detected. In the range $\rho = 4$–7 the fraction of signals called noise is 0.40 (Bayes, frequentist), 0.51 (CNN), 0.54 (GRU), 0.60 (Transformer).
- **A signal incompatible with every hypothesis.** A Gaussian-enveloped burst of band-limited stochastic noise ($\sigma_t = 15\,\text{ms}$) belongs to none of the families. It is detected when loud (94–98 % at $\rho = 12$) and every method gives it a family, mostly sine-Gaussian (0.93 max-ρ^2, 0.95 CNN, 0.80 Bayes, 0.82–0.85 Transformers, 0.90 GRU) and then pulse. No method can say "none of the above": a softmax is normalised over the classes it knows and the Bayesian posterior over the families it contains. Flagging such signals would need an explicit goodness-of-fit (the residual after subtracting the best template) or an open-set head; neither was tried.

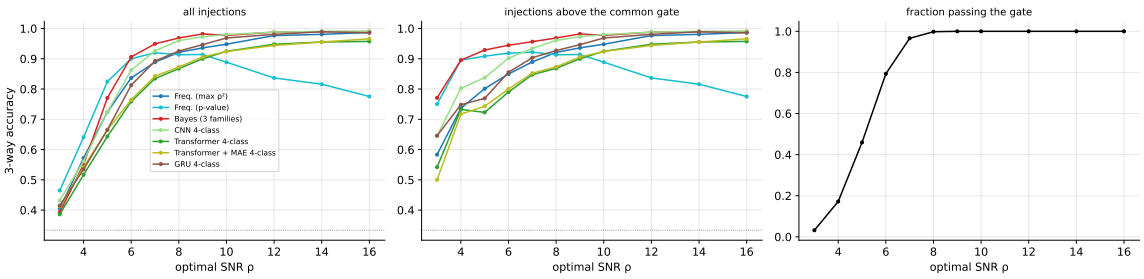


Figure 28. Accuracy of the morphology choice against SNR: all injections (left), injections above the common Bayesian detection threshold (centre) and the fraction passing it (right).

7.3 Is the Transformer's loss to the CNN real? A tuning study

The comparison above used a small budget (at most 12 epochs, two seeds, no tuning of depth, width or patch size), so "the Transformer loses" is a statement about that budget. It is stress-tested by changing *one lever at a time*, with the same data, thresholds and test injections (verified identical to those of the previous sections), and by giving the CNN the same levers so that the comparison stays fair.

Diagnosis. At the best-validation epoch every model has a small train–validation gap (0.01–0.03), but the *level* of the loss differs: 0.15–0.17 for the CNN of Section 6, 0.29–0.31 for the Transformers and 0.29 for the GRU (Fig. 29, left). The sequence models are limited by what they can fit, not by over-fitting. The CNN trained with the same loop on the static data does the opposite: it over-fits (training loss 0.07–0.09, validation 0.17–0.19, and a validation loss that rises again after epoch 15). So the CNN is *data-limited*, the patch Transformer is *capacity/bias-limited*.

Levers. (i) *Data*: fresh injections — new morphology, SNR, parameters and phase — at every epoch, for the CNN, the Transformer and the hybrid; (ii) *optimisation*: peak learning rate 3×10^{-3} instead of 10^{-3}, and the 25-epoch run; (iii) *resolution*: patches of 8 samples (128 tokens); (iv) *inductive bias*: a *convolutional stem* (the backbone of the CNN, down-sampling by 8) followed by two Transformer blocks on 128 tokens (the "hybrid"). Budget: 12 epochs for the sequence models, 20 for the CNN; the Transformer variants take 23–81 min each on a laptop.

model	train loss	val. loss	$\overline{\rho_{50}}$ @10^{-2}	$\overline{\rho_{50}}$ @10^{-3}
Transformer (Sec. 7, 2 seeds)	0.30	0.31	6.20	7.32
+ fresh injections (2 seeds)	0.30	0.30	6.27	7.38
+ learning rate 3×10^{-3}	0.30	0.32	6.36	7.68
+ patches of 8 samples	0.31	0.31	6.59	7.58
hybrid (conv. stem + 2 blocks), static (2 seeds)	0.22	0.27	5.64	6.79
hybrid, fresh injections (2 seeds)	0.23	0.25	5.33	6.21
CNN, static data (2 seeds)	0.08	0.18	5.22	6.26
CNN, fresh injections (2 seeds)	0.09	**0.117**	**4.60**	**5.65**
CNN ensemble of Sec. 6 (5, static)	–	–	4.85	5.79

Table 13. Tuning study: training and validation loss at the best epoch and mean over the three morphologies of ρ_{50} at FAP 10^{-2} and 10^{-3} (thresholds on the validation noise). Differences below ≈ 0.3 are not significant: the half-range between seeds of the same configuration is 0.11/0.07 (static CNN), 0.03/0.01 (fresh CNN), 0.03/0.13 (Transformer, fresh), 0.02/0.18 (hybrid, static) and 0.03/0.07 (hybrid, fresh) at FAP $10^{-2}/10^{-3}$; the lr-3×10^{-3} and patch-8 variants have a single seed.

Findings.

1. **The patch Transformer does not recover with any single lever.** Fresh injections (6.27, two seeds), a three times higher learning rate, smaller patches (four times the cost) and the 25-epoch run leave $\overline{\rho_{50}}$ at 6.3–6.6 and the validation loss at 0.30–0.32: the limitation is the embedding, not the data or the optimisation time. Chirps are the worst case ($\rho_{50} \approx 7.4$–8.0 against 5.5–5.8 for the CNN): a 30–80 ms oscillation needs frequency-selective filters, which a linear map of 16 raw samples does not provide.

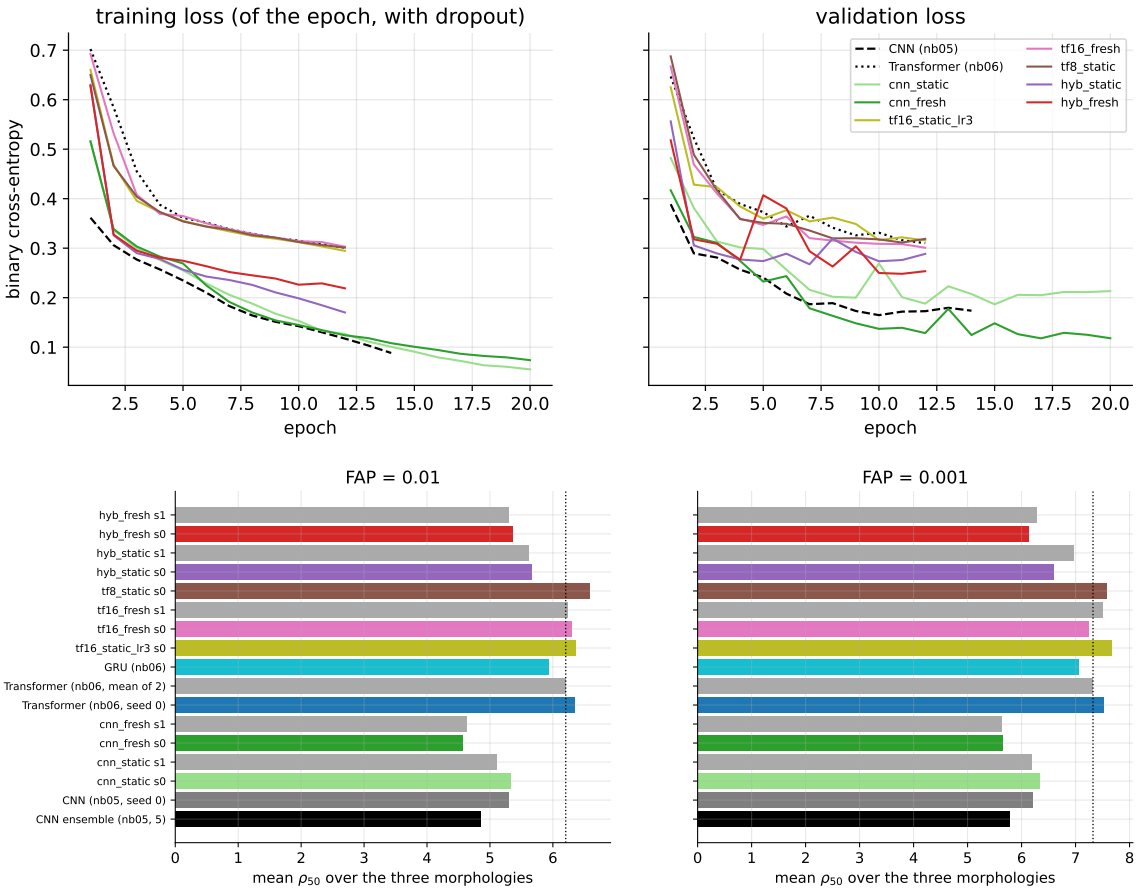


Figure 29. Top: training and validation loss of the variants. Bottom: mean ρ_{50} over the three morphologies for every model, at FAP 10^{-2} and 10^{-3} (dotted: the Transformer of Section 7).

2. **A convolutional stem is the lever that works.** The hybrid reaches 5.64 (static) and 5.33 (fresh), 0.6–0.9 better than the patch Transformer, well above the seed scatter, and a validation loss of 0.27/0.25; it is the only sequence model competitive at FAP 10^{-3} (6.21 against 7.3–7.7).
3. **But the CNN stays ahead with the same data.** Static data: hybrid 5.64 against CNN 5.22; fresh data: 5.33 against 4.60. Attention on top of a good convolutional front-end does not add detection power on one-second windows with a localised burst.
4. **The CNN of Section 6 was data-limited, and fresh injections are the best lever of the whole study.** Fresh injections cut $\overline{\rho_{50}}$ of the CNN from 5.22 to 4.60 at FAP 10^{-2} and from 6.26 to 5.65 at 10^{-3}, of the hybrid by 0.31 and 0.58, and of the patch Transformer by nothing. A *single* fresh CNN is better than the five-member static ensemble (4.85, 5.79), with the same ρ_{90} (6.6); its validation loss falls from 0.17–0.19 to 0.117. This is the reference CNN for any future comparison; the numbers of Section 6 are unchanged but should be read as those of a data-limited network.
5. **The comparison does not favour the CNN.** At a nominal FAP of 1 % the observed false-alarm probability on the test chunks is 0.4–0.5 % for the fresh CNNs, 0.7–1.0 % for the patch Transformers and 1.0–1.1 % for the hybrids: the CNN threshold is the more conservative one, so at equal observed FAP the gap would be larger.

Intuition (interpretation, not tested separately). The burst is a short, localised oscillation in white noise; seeing it requires a filter tuned to its frequency, which is what a convolution does (a learned matched filter). A patch Transformer first mixes 16 raw samples with a single linear map, which cannot build such filters, and nothing downstream recovers what is lost: this is consistent with the high training loss. Once the filters are provided by a convolutional stem the Transformer improves a lot, but a burst that sits in one place of a one-second window leaves little long-range structure for attention to exploit, and recurrence, whose strength is remembering distant context, is similarly wasted.

Multiclass networks with fresh injections change a conclusion of Section 7. The four-class networks of the previous subsection were trained on the static set. Trained with fresh injections (Table 14) the CNN rises from 0.960 to 0.977 three-way accuracy at $\rho = 8$ and from 0.983 to 0.991 balanced four-class accuracy for $\rho = 8$–16 (0.796 against 0.722 for $\rho = 4$–7): it ties with the Bayesian model comparison at low SNR (0.768 against 0.771 at $\rho = 5$) and is slightly *ahead* of it at high SNR (0.977 against 0.969 at $\rho = 8$, 0.988 against 0.977 at $\rho = 10$; differences of ≈ 0.01, about two standard errors with 1500 injections per point). The hybrid network is level with the Bayes factors; the patch Transformer is not helped (0.844 at $\rho = 8$ against 0.867 static). The statement of Section 7 that Bayesian model comparison is the best morphology classifier was therefore a statement about a data-limited CNN: with the same budget and fresh injections the CNN is as good.

3-way accuracy at $\rho =$	5	6	8	10	14	bal. 4–7	bal. 8–16
Bayes (3 families)	0.771	0.906	0.969	0.977	0.989	0.778	0.988
CNN 4-class, static	0.725	0.863	0.960	0.980	0.991	0.722	0.983
CNN 4-class, fresh	0.768	0.900	**0.977**	**0.988**	**0.997**	**0.796**	**0.991**
hybrid 4-class, fresh	0.769	0.888	0.965	0.986	0.997	0.760	0.987
Transformer 4-class, static	0.643	0.759	0.867	0.925	0.955	0.661	0.944
Transformer 4-class, fresh	0.635	0.736	0.844	0.899	0.950	0.656	0.933
GRU 4-class, static	0.665	0.813	0.925	0.969	0.989	0.700	0.966

Table 14. Three-way morphology accuracy at several SNRs and balanced four-class accuracy for $\rho = 4$–7 and 8–16, with static and fresh-injection training (the Bayes row is from Table 12).

Caveats. A single seed for the lr-3 $\times 10^{-3}$ and patch-8 Transformers and for the four-class Transformer and hybrid; a budget of 12 epochs (the hybrid reaches its best validation epoch at 7–12, so the fresh-data runs might gain a little more); depth, width, number of heads and the pre-training were not swept; the unit tested is one second of whitened data with one localised burst — longer windows, or several overlapping signals, are the setting where attention could be expected to matter.

7.4 Seed scatter and paired tests

The comparisons above used one or two seeds per architecture. Notebook 30 trains more seeds with the same recipe (CNN with fresh injections: 5; hybrid: 3; Transformer: 3) and evaluates all on the same test injections with the validation thresholds (mean ρ_{50} over the three morphologies). The scatter of the CNN at FAP 10^{-2} (0.40; two seeds at 3.72 and 4.08 against 4.56–4.63 for the other three) is much larger than the 0.03 half-range of the two seeds of notebook 07 and

architecture (seeds)	FAP 10^{-2}: mean $\pm$ std (range)	FAP 10^{-3}: mean $\pm$ std
CNN, fresh injections (5)	4.31 ± 0.40 (3.72–4.63)	5.67 ± 0.03
hybrid, fresh (3)	5.35 ± 0.05	6.29 ± 0.16
Transformer, fresh (3)	6.29 ± 0.05	7.47 ± 0.20
average of the 5 CNN logits	3.79	5.32

Table 15. Seed-to-seed scatter of the detectors (the test-set bootstrap error of a single network is 0.08).

larger than the test-set error, while at 10^{-3} all seeds agree to 0.03: a single-seed number at 10^{-2} is uncertain by about ± 0.4. The architecture differences are nevertheless larger than the scatter and significant at the level of seeds (permutation and Welch tests): hybrid minus CNN $+1.04$ ($p = 0.018$ permutation, 0.004 Welch) and Transformer minus CNN $+1.98$ ($p = 0.018$, 0.0003) at 10^{-2}; $+0.62$ and $+1.79$ at 10^{-3}. Averaging five CNNs lowers ρ_{50} by a further 0.5 (3.79, 5.32), the largest single gain obtained in this study at fixed training data. The earlier statement that the ranking CNN $<$ hybrid $<$ Transformer is robust is confirmed; its margin is 1 to 2 units, not tenths.

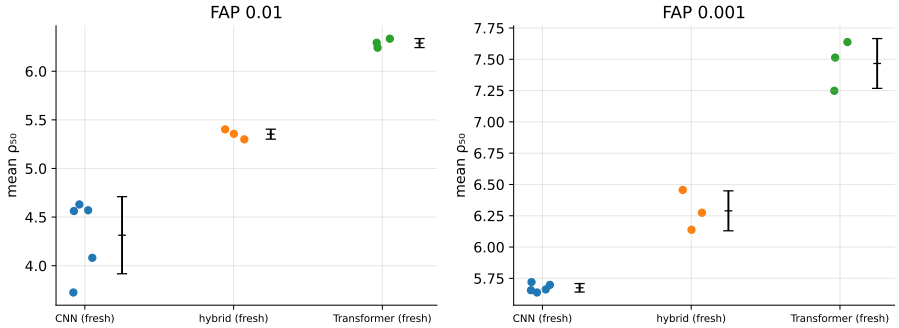


Figure 30. Mean ρ_{50} of every seed (dots) with mean and standard deviation (bar).

7.5 Other architectures: ResNet-1D, TCN, spectrogram CNN, graph network

Notebook 34 trains four more detectors with the recipe of notebook 07 (fresh injections, 12 epochs, one seed): a ResNet-1D (stem and four residual stages), a temporal convolutional network (residual dilated convolutions, dilation 1–32), a 2-D CNN on the log-spectrogram (33×121) and a graph network (EdgeConv with 8 nearest neighbours) on the 64 time-frequency pixels of largest power, the construction behind the clustering of coherent WaveBurst. Mean ρ_{50} over the three morphologies at FAP 10^{-2} (10^{-3}): ResNet-1D 5.01 (5.77), TCN 5.09 (5.87), 2-D CNN 5.20 (5.97), graph network 5.48 (6.46), against 4.31 ± 0.40 (5.67 ± 0.03) for the project CNN over five seeds, 5.35 (6.29) for the hybrid and 6.29 (7.47) for the Transformer; the bootstrap error of each is 0.04–0.05 at 10^{-2} and the seed scatter of the CNN is 0.4. No architecture beats the plain 1-D CNN, the four are within ≈ 0.2–0.5 of each other and of the matched-filter bank (5.5), and all are better than the patch Transformer: the global max-pooling over localised convolutional features is what the detection of a localised burst needs, and adding depth, dilation or a second dimension does not change that. The graph network, which sees only 64 pixels, loses ≈ 0.5 against the best and is the cheapest to evaluate; per morphology the four differ by less than 1 (4.5–5.8 over the three morphologies), without a consistent winner.

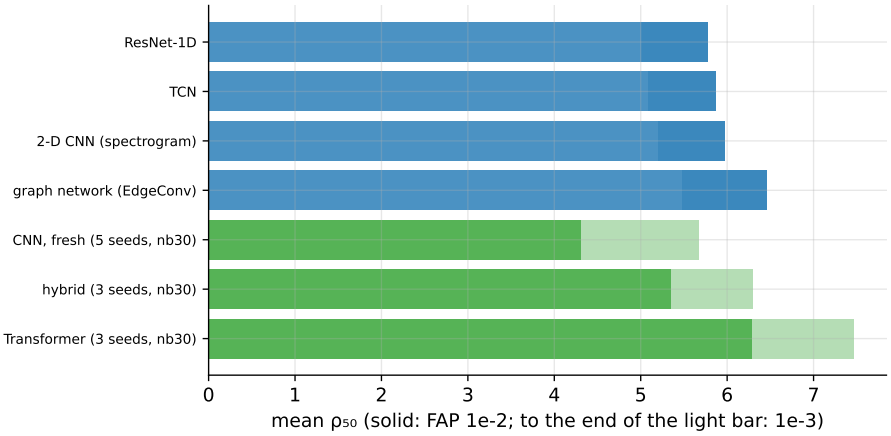


Figure 31. Mean ρ_{50} of the four additional architectures (blue) and of the references of Section 7.4 (green); light part: increase at FAP 10^{-3}.

7.6 Transformer tuning, second round: depth, width, length

The levers left open by notebook 07 were run in notebook 39 (patch 16, fresh injections, one seed each): six blocks instead of three (214 k parameters), width 128 with MLP 256 (425 k) and 30 epochs instead of 12. Mean ρ_{50} at FAP 10^{-2} (10^{-3}): baseline 6.29 (7.47; three seeds, seed scatter 0.05 (0.2)), depth 6.23 (7.21), width 6.11 (6.97), 30 epochs 5.81 (7.15), with validation losses 0.301, 0.299, 0.301 and 0.255. Deeper and wider networks gain 0.06–0.2 at 10^{-2} (within the seed scatter for the depth) and 0.3–0.5 at 10^{-3}; training longer is the only lever whose gain at 10^{-2} (0.48) is far beyond the scatter, and it still leaves the Transformer 1.5 units behind the CNN (4.31) and 0.5 behind the hybrid (5.35). Combined with the earlier levers (learning rate, patch size, fresh data, convolutional stem), the Transformer on one-second windows with a localised burst does not reach the CNN with any budget tried here; contrastive pre-training and overlapping windows, also on the original list, were not run.

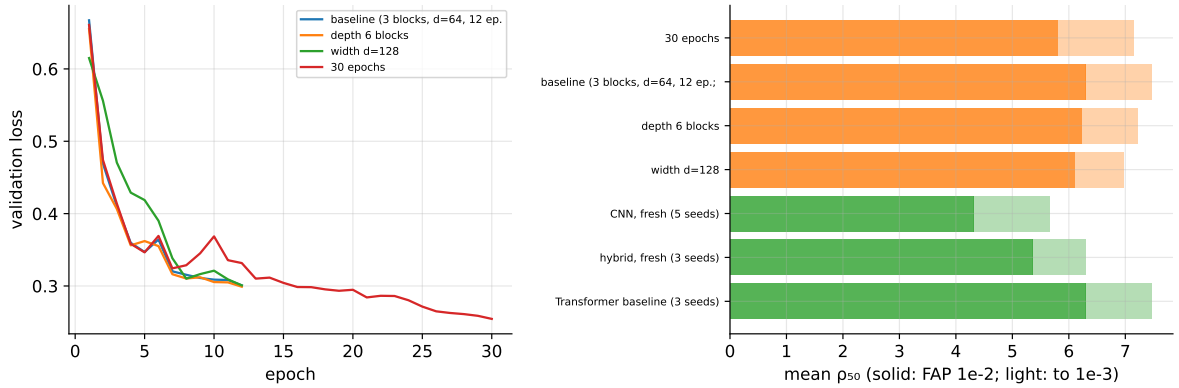


Figure 32. Left: validation loss of the variants. Right: mean ρ_{50} of the variants and of the references.

8 REALISTIC SIGNALS: BINARY BLACK HOLES

The three morphologies of Sections 3–7 are phenomenological: standard benchmarks for burst searches, but not what LIGO observes. The signals detected so far are compact-binary coalescences [1, 23]: an inspiral whose frequency sweep accelerates up to a merger and a ringdown. This section asks (i) how close the project's chirp is to a real inspiral, and (ii) whether the detectors built on the phenomenological bursts still work on realistic signals.

8.1 The linear chirp as a binary

At leading order a binary of chirp mass $\mathcal{M}_c$ sweeps in frequency at $\dot{f} = \frac{96}{5}\pi^{8/3}(G\mathcal{M}_c/c^3)^{5/3}f^{11/3}$ and merges a time $\tau = \frac{5}{256}(G\mathcal{M}_c/c^3)^{-5/3}(\pi f)^{-8/3}$ after passing f. Inverting the first relation for the $(f_0, \dot{f})$ prior of the linear chirp gives the chirp mass whose local sweep rate it mimics (Table 16, left): $\mathcal{M}_c \approx$ 4–10 $M_\odot$ at $f = 100$ Hz, 2–4 $M_\odot$ at 150 Hz and 20–47 $M_\odot$ at 50 Hz, i.e. binaries within the reach of LIGO. But a real signal of that chirp mass is much longer than the 30–80 ms of the chirp's envelope and its sweep *accelerates*, whereas the linear chirp has a constant $\dot{f}$, a symmetric envelope and no merger. A one-second window contains the whole in-band inspiral (from 30 Hz) only for $\mathcal{M}_c \gtrsim 13\,M_\odot$ (total mass $\gtrsim 30\,M_\odot$); neutron-star binaries last about a minute in band and merge above the 500 Hz cut, so they are out of reach of this setup (Table 16, right).

$\mathcal{M}_c$ [$M_\odot$]	$\dot{f} = 150$	300	600 Hz/s
$f = 50$ Hz	20.4	31.0	46.9
$f = 100$ Hz	4.4	6.7	10.2
$f = 150$ Hz	1.8	2.8	4.2

system	$\mathcal{M}_c$	$\dot{f}_{100}$ [Hz/s]	$\tau_{30\text{Hz}}$ [s]
BNS 1.4+1.4	1.2	17	54
BBH 8+8	7.0	317	2.9
BBH 15+15	13.1	904	1.03
GW150914 (36+29)	28.1	3242	0.29

Table 16. Left: chirp mass of the leading-order inspiral whose sweep rate equals the $(f, \dot{f})$ of the linear-chirp prior. Right: sweep rate at 100 Hz and time from 30 Hz to merger of real systems.

8.2 IMRPhenomD library and injection

A library of black-hole binaries is generated with LALSuite [24] (IMRPhenomD [25], aligned spins) as the strain of one interferometer (random sky position, polarisation and inclination) in the frequency bins of the one-second window: total mass $U(30, 100)\,M_\odot$ ($\mathcal{M}_c =$ 10–43 $M_\odot$), mass ratio $q \sim U(0.25, 1)$, spins $\chi_{1,2} \sim U(-0.5, 0.5)$, merger at a random time of the window. Two **disjoint** libraries with the same prior are built: 4000 waveforms for training and 2000 *held-out* waveforms used for every evaluation (so that a network trained with IMR signals cannot have memorised the test waveforms). The injection is exactly as for the other morphologies, at a given optimal SNR (Fig. 33). As validation, a matched-filter bank of 24 IMR templates (total mass 30–100 $M_\odot$, $q = 1, 0.6, 0.3$, no spin, face-on) recovers on average 93 % of the injected amplitude (fitting factor 0.93, 0.76 at the 10th percentile) and the merger time to 1.6 ms; the sine-Gaussian bank recovers only 65 %: on a binary-black-hole signal a sine-Gaussian bank is a poor matched filter.

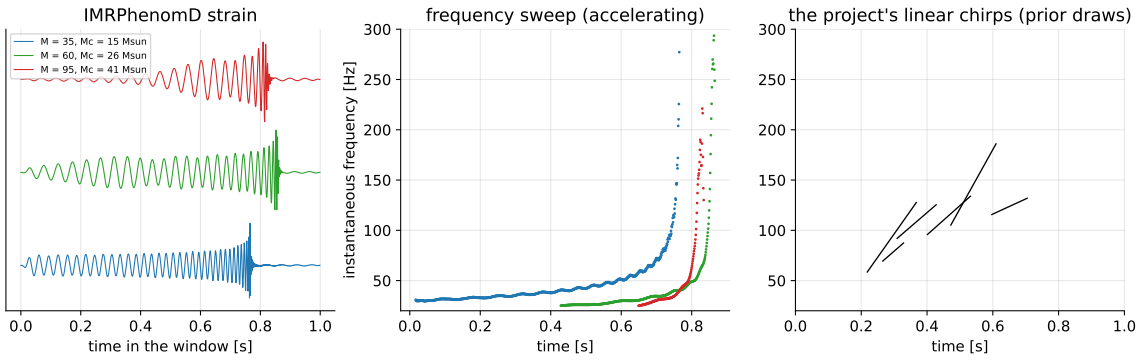


Figure 33. Left: IMRPhenomD strain of three binaries of increasing mass in the window. Centre: instantaneous frequency against time (accelerating sweep and merger). Right: draws of the linear-chirp prior (constant sweep rate).

8.3 Detection of binary-black-hole signals

Test set: 500 windows of the test chunks per SNR with a held-out library signal injected, thresholds on the validation noise as in the previous sections (Table 17, Fig. 34). The *oracle* knows the exact waveform and its time: the Neyman–Pearson statistic of a simple hypothesis and the floor that no detector can beat.

- **The detectors designed on the burst morphologies do not transfer for free.** The sine-Gaussian bank needs $\rho_{50} = 7.4$ / 11.1 at FAP 10^{-2} / 10^{-3}, i.e. 2.0 / 4.6 more than the IMR bank (5.4 / 6.4); excess power 8.3 / 10.2; the hybrid Transformer trained without IMR signals 7.6 / 9.4. The CNNs trained on the three morphologies without IMR signals generalise better (5.5 / 7.2 with fresh injections, 6.0 / 7.3 for the static ensemble): on par with the IMR bank at FAP 10^{-2}, worse at 10^{-3}.
- **Training with IMR signals closes the gap and goes beyond the matched filter.** The CNN trained with fresh injections of the three morphologies *and* of the training library reaches 4.45 / 5.91 against 5.40 / 6.42 for the IMR bank and 3.00 / 3.50 for the oracle; the efficiency at $\rho = 8$ is 0.98 against 0.91. It does so on waveforms never seen in training (4.45 on the held-out library against 4.40 on the training one: no memorisation) and at no cost on the three phenomenological morphologies ($\rho_{50} = 4.32/3.65/4.36$ against $4.63/3.87/4.47$ at FAP 10^{-2}). Two contributions to the advantage over the bank are consistent with the data but were not separated: the bank's fitting factor of 0.93 (a coarse bank of non-spinning face-on templates), and the glitch tail of its statistic, which raises its threshold, while the network has learned from real noise which transients are not signals. The oracle–bank gap (2.4 at FAP 10^{-2}) contains the look-elsewhere effect, the template mismatch and the glitches.
- **Mass dependence** (Fig. 35). The IMR bank is flat (0.87–0.93 efficiency at $\rho = 8$). The methods built on bursts are worst for the lightest systems (30–45 $M_\odot$: sine-Gaussian bank 0.45, excess power 0.25), whose signal is a long slow chirp, and better for the heaviest (80–100 $M_\odot$: 0.75 and 0.53), dominated by a short merger that looks like a burst. The CNN trained with IMR signals is at 0.93 or above for every mass.

Caveats. Single detector; IMRPhenomD only (aligned spins, no precession, higher modes or eccentricity); total masses $\geq 30\,M_\odot$; the network was trained on the same *prior* as the test waveforms (the waveforms themselves are disjoint); a real search works at false-alarm rates

statistic	FAP 10^{-2} ρ_{50}	FAP 10^{-2} ρ_{90}	FAP 10^{-3} ρ_{50}	FAP 10^{-3} ρ_{90}	$\varepsilon(\rho = 8)$
oracle (exact waveform known)	3.00	4.35	3.50	4.92	1.00
CNN, trained with IMR signals (2 seeds)	**4.45**	**6.72**	**5.91**	**8.00**	0.98
IMR matched-filter bank (24 templates)	5.40	7.73	6.42	8.71	0.91
CNN, fresh injections, no IMR (2 seeds)	5.52	8.11	7.16	9.69	0.89
CNN ensemble of Sec. 6 (static, no IMR)	5.97	8.30	7.31	9.72	0.88
CNN, static, no IMR (2 seeds)	6.18	8.75	7.67	10.00	0.81
sine-Gaussian bank	7.38	9.98	11.05	13.98	0.64
hybrid Transformer, fresh, no IMR	7.55	10.52	9.41	13.51	0.57
excess power	8.28	11.06	10.16	13.19	0.45

Table 17. Optimal SNR at 50 % and 90 % efficiency on binary-black-hole injections of the held-out library (thresholds on the validation noise) and efficiency at $\rho = 8$ and FAP 10^{-2}.

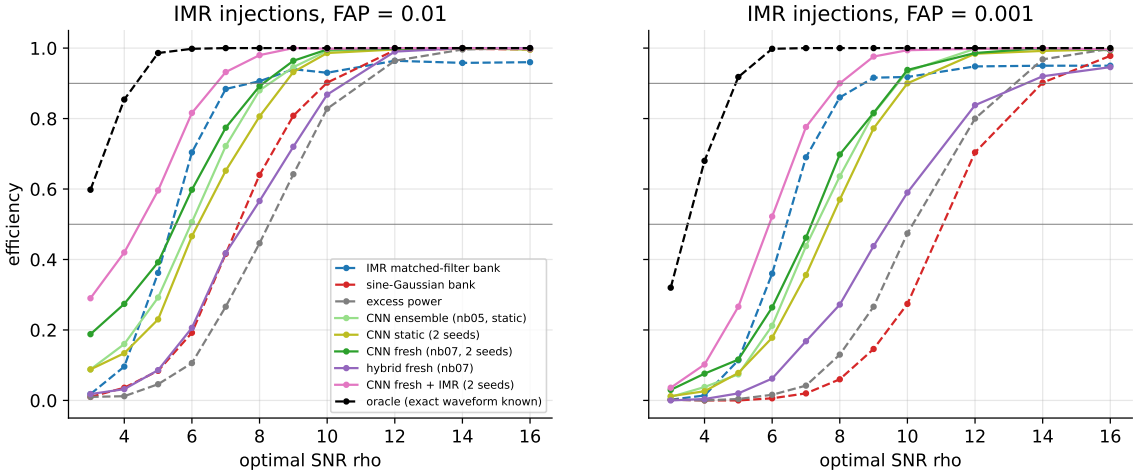


Figure 34. Efficiency against optimal SNR on binary-black-hole injections, at FAP 10^{-2} (left) and 10^{-3} (right); dashed: model-based statistics.

several orders of magnitude lower (next section).

8.4 Three real events as a blind-ish check

The unchanged pipeline (whitening with the median-Welch PSD, sine-Gaussian and IMR banks, per-detector p-values on the background of Section 10, Fisher combination with the 15 ms gate) is applied to the one-second window around GW150914, GW151012 and GW151226 in H1 and L1 (notebook 13, Fig. 36). Where the GWOSC file contains out-of-science gaps (NaN) only the common clean stretch around the event is used. GW150914 is the loudest window of H1 and the second of L1 in its chunk, the best-fit total mass ($72\,M_\odot$, detector frame) and the H–L delay (5.9 ms) agree with the catalogue and the matched-filter network SNR (23.4) with the published one (≈ 24); the sine-Gaussian bank sees the same event with $\rho_{\rm net} = 14.9$, the price of the wrong family. GW151012 (network $\rho = 9.1$, catalogue ≈ 9.5) is a weak but coincident detection with the IMR bank only. GW151226 is *not* recovered: it lies partly outside the bank (source-frame total mass $\approx 22\,M_\odot$, long inspiral inside the cut band and window), and in this chunk only 540 s of clean two-detector data are available. The test does not tune anything and is a plausibility check of the pipeline, not a detection claim.

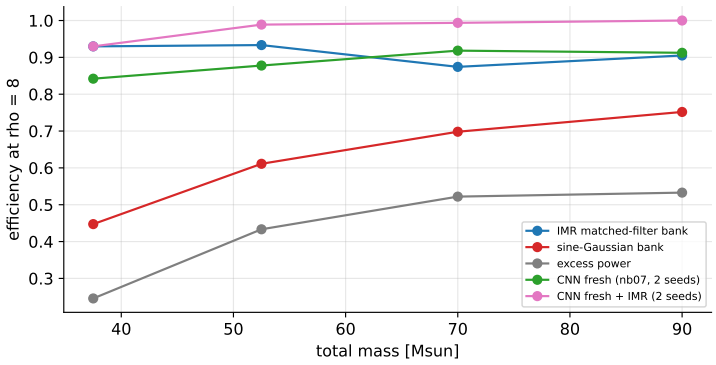


Figure 35. Efficiency at $\rho = 8$ (FAP 10^{-2}) against the total mass of the binary.

event	ρ_H	ρ_L	Δt_{HL} [ms]	$M_{\rm tot}^{\rm best}$	p_H, p_L	$\mathrm{FAP}_{\rm coinc}$
GW150914 (IMR bank)	19.2	13.3	5.9	72	$1.4 \cdot 10^{-3}$, $3.7 \cdot 10^{-4}$	$< 10^{-5}$
GW150914 (sine-Gaussian)	12.1	8.8	6.8	–	$2.4 \cdot 10^{-3}$, $1.0 \cdot 10^{-3}$	$< 10^{-5}$
GW151012 (IMR bank)	6.5	6.4	−1.0	42	$4.9 \cdot 10^{-3}$, $3.9 \cdot 10^{-3}$	$2 \cdot 10^{-4}$
GW151012 (sine-Gaussian)	4.5	4.1	5.9	–	0.19, 0.67	0.39
GW151226 (IMR bank)	4.7	4.8	−153	35/50	0.09, 0.11	gate fails

Table 18. Matched-filter SNR $\rho = \sqrt{\max \rho^2}$ of the banks on three real events. p: empirical p-values on the 16 224-window background of notebook 09 (floor $6 \cdot 10^{-5}$, the H1 tail contains glitches). $\mathrm{FAP}_{\rm coinc}$ uses $(1 + S)e^{-S}$ and is conservative; it is not an astrophysical rate.

8.5 Precessing spins and higher modes

Notebook 29 repeats the detection with 600 signals per family and the same mass, inclination and sky prior: IMRPhenomD with aligned spins ($|\chi| \leq 0.5$, a control regenerated here), IMRPhenomPv2 (precessing, $|\chi| \leq 0.8$ in random directions) and IMRPhenomXPHM (precession and higher modes). The networks are not retrained. At FAP 10^{-2} (10^{-3}) the ρ_{50} is 5.30 (6.42), 5.57 (6.56) and 5.59 (6.57) for the IMR bank, 4.44 (5.74), 4.44 (6.00) and 4.52 (5.91) for the CNN trained with IMRPhenomD, and 5.32, 5.62, 5.25 for the CNN trained without IMR signals: the loss is at most 0.3 and often inside the precision of 0.1–0.2. The fitting factor of the non-spinning, face-on bank falls from a median of 0.971 (10th percentile 0.80) to 0.937 (0.66) with precession and 0.931 (0.66) with higher modes, but this reflects mainly the large spins and the orientation, and it costs the detector only a few per cent of SNR for the *short* signals (30–100 $M_\odot$, one second) considered here; the ranking of the methods (CNN with IMR $<$ IMR bank $\approx$ CNN without IMR $<$ sine-Gaussian bank $<$ excess power) does not change. The situation would differ for lower masses and longer signals, where the precession cycles and the higher-mode content are within the window, which this analysis does not cover.

8.6 Where the pipeline stops: low masses, neutron stars, NSBH and eccentricity

The analysis unit (one second, 30–500 Hz) is made for black-hole binaries of 30–100 $M_\odot$. Notebook 37 quantifies what leaves it. For equal-mass IMRPhenomD signals the fraction of the optimal SNR retained by the band *and* by the last second of the signal, with the PSD of the test chunks, is 0.49 for $3 + 3\ M_\odot$, 0.73 for $5 + 5$, 0.87 for $7.5 + 7.5$, 0.93 for $10 + 10$ and 0.96–0.97 for 15–25 $M_\odot$ per component (0.93 at $50 + 50$, where the signal is too short and the 20 Hz cut

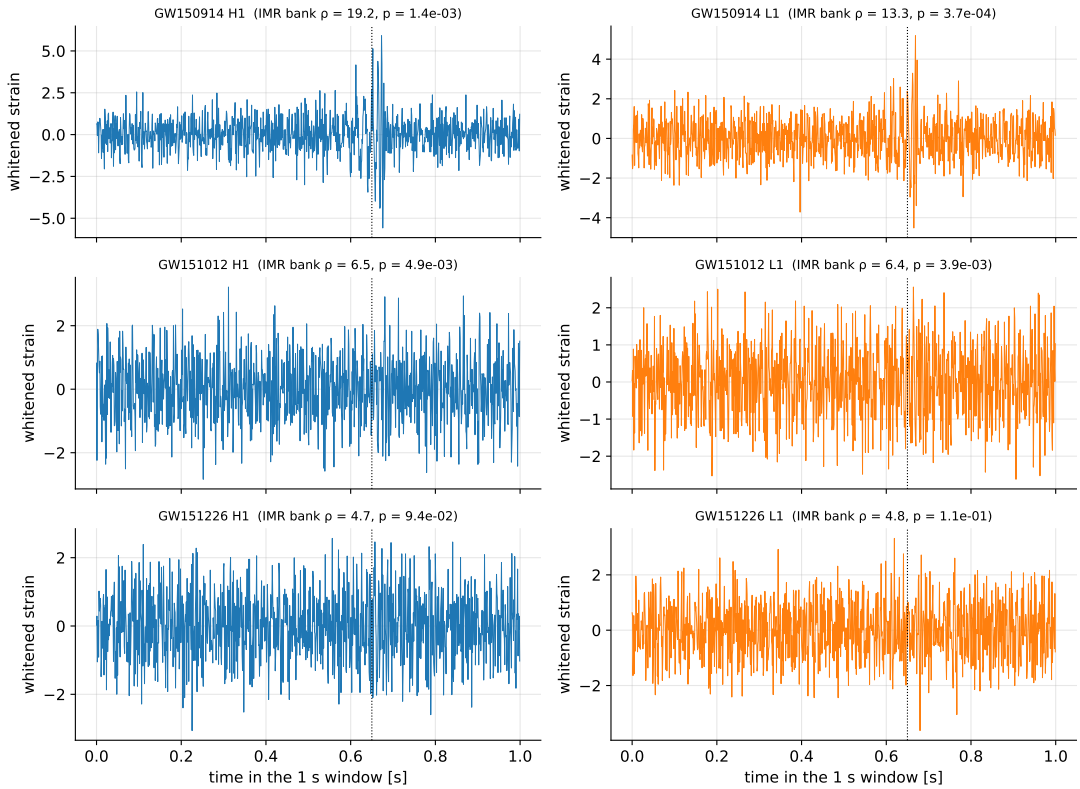


Figure 36. Whitened strain of the one-second window containing each event (dotted: catalogue time).

matters); in the band alone 0.97–0.98 of the SNR^2 is kept above 15 $M_\odot$ total (0.84 at 10, 0.46 at 8). A binary neutron star $1.4 + 1.4\,M_\odot$ keeps only 3.6 % of its SNR^2 in the band and 0.8 % in the last second: the pipeline cannot see it (8.7 % of the SNR), which requires the full 20–2000 Hz band, windows of minutes and a bank of inspiral templates. Four- to eight-second windows would recover the systems down to $\approx 10\,M_\odot$ total at a cost in background and bank size that was not explored here. NSBH signals cost the IMR bank, whose templates are non-spinning

detector	NSBH	IMRPhenomD 30–60	ecc. $e_0 = 0.2$	ecc. $e_0 = 0.4$	IMRPhenomD 15–40
IMR bank (30–100 $M_\odot$, non-spinning)	10.8	5.29	7.38	7.39	6.53
sine-Gaussian bank	8.99	8.08	8.57	8.51	8.93
CNN without IMR training	6.23	5.48	5.86	5.68	6.15
CNN trained with IMRPhenomD	5.44	4.56	5.44	5.19	5.24
excess power	11.0	9.03	9.97	9.66	10.8

Table 19. ρ_{50} at FAP 10^{-2} (single H1) for NSBH (BH 20–60 $M_\odot$, NS 1.2–2 $M_\odot$, $\Lambda = 0$–800, aligned spin, IMRPhenomNSBH) and eccentric inspirals (EccentricFD, 15–40 $M_\odot$ total, inspiral only) against IMRPhenomD controls of the same mass ranges.

IMRPhenomD with total masses $\geq 30\,M_\odot$ and equal to or comparable component masses, 5.5 units (10.8 against 5.3 for the control), while the CNN trained with IMRPhenomD loses 0.9 (5.44 against 4.56) and the CNN without IMR training 0.8. Eccentricity up to 0.4 at 20 Hz costs the IMR bank $\approx$ 0.4–0.9 with respect to the control (the signal is inspiral-only) and the CNN trained with IMR signals nothing. The sine-Gaussian bank and excess power are unaffected, because they are equally poor for all of them. The conclusion for the detectors designed on the project's families is the same as in Section 8: the network trained on a family generalises to neighbouring ones, the template statistic does not.

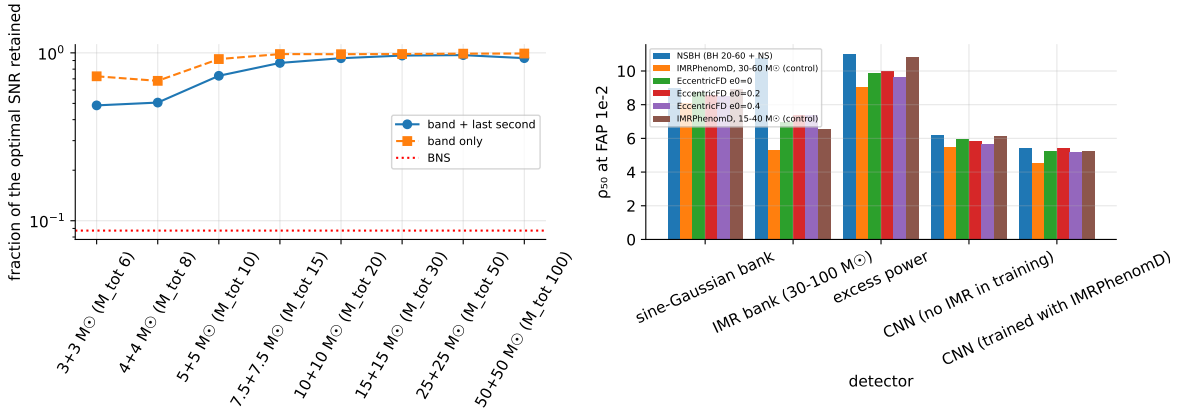


Figure 37. Left: fraction of the optimal SNR retained by the 1 s, 30–500 Hz window against the total mass. Right: ρ_{50} for the NSBH and eccentric libraries.

9 Physical units: $h_{\rm rss}$, other sources and sensitive range

Notebook 28 converts ρ_{50} into physical quantities. Thresholds are those of the validation noise at FAP 10^{-2}, the PSD is the median of the test chunks, the detector is H1 alone, optimally oriented and linearly polarised for the bursts.

$h_{\rm rss}$ benchmarks. The root-sum-square strain at 50 % efficiency is $\rho_{50}/\rho(h_{\rm rss}{=}1)$ with the SNR per unit $h_{\rm rss}$ computed from the PSD (the SNR of $h_{\rm rss} = 10^{-22}\,\mathrm{Hz}^{-1/2}$ is 13–18 for 70–153 Hz sine-Gaussians). Table 20 gives $h_{\rm rss,50}$ in units of $10^{-22}\,\mathrm{Hz}^{-1/2}$. The best value is $\approx 2.5\cdot$

waveform	bank	CNN	excess power
sine-Gaussian $Q=9$, f_0 =70 Hz	0.38	0.32	0.58
sine-Gaussian $Q=9$, f_0 =100 Hz	0.29	0.25	0.41
sine-Gaussian $Q=9$, f_0 =153 Hz	0.27	0.25	0.34
sine-Gaussian $Q=9$, f_0 =235 Hz	0.29	0.29	0.34
sine-Gaussian $Q=9$, f_0 =361 Hz	0.34	0.38	0.37
Gaussian pulse, σ =0.1 ms	0.38	0.35	0.35
Gaussian pulse, σ =0.5 ms	0.38	0.31	0.35
Gaussian pulse, σ =2.5 ms	0.67	0.47	0.71
ringdown f =100 Hz, τ =10 ms	0.33	0.27	0.37
ringdown f =200 Hz, τ =5 ms	0.31	0.28	0.31
ringdown f =300 Hz, τ =3 ms	0.33	0.31	0.33
cosmic-string cusp, $f_{\rm high}$ =150 Hz	0.57	0.38	0.58
cosmic-string cusp, $f_{\rm high}$ =300 Hz	0.56	0.37	0.52
cosmic-string cusp, $f_{\rm high}$ =500 Hz	0.56	0.37	0.53
white-noise burst 100-200 Hz	0.83	0.57	1.04
white-noise burst 100-500 Hz	1.39	—	1.22

Table 20. $h_{\rm rss}$ at 50 % efficiency, in units of $10^{-22}\,\mathrm{Hz}^{-1/2}$, FAP 10^{-2} (single H1, optimally oriented); "—": does not reach 50 % up to $\rho = 24$.

$10^{-23}\,\mathrm{Hz}^{-1/2}$ at 100–153 Hz, a factor ~ 10 below the network values of LVK burst searches, because of the optimal orientation, the single detector, the FAP per window (10^{-2}) and the

absence of an astrophysical false-alarm requirement: the numbers rank the methods, they are not search sensitivities. The CNN matches or beats the bank on every shape but the highest frequency (361 Hz), including shapes it never saw (ringdowns, cusps: $\rho_{50} = 3.8$–5.0 against 5.1–5.8 for the bank); it fails on the broadband white-noise bursts of 400 Hz bandwidth (no 50 % up to $\rho = 24$), the one family far outside its prior, where excess power is as good as the bank.

Sensitive range for black-hole binaries. 600 IMRPhenomD sources of the library prior (spins, inclination, sky position, H1 antenna pattern) are generated at 1 Mpc with their physical amplitude, giving a median optimal SNR of $4.1 \cdot 10^3$ at 1 Mpc (an optimally oriented $36 + 29\,M_\odot$ system has $1.5 \cdot 10^4$, i.e. SNR 36 at 410 Mpc against 19–20 measured in H1 for GW150914, reduced by the inclination and the antenna pattern). Folding $\rho(D)$ with the efficiency curves of Section 8 gives, in Euclidean space, a volume-averaged range of 1.31 Gpc for the IMR bank and 1.57 Gpc for the CNN trained with IMR signals at FAP 10^{-2} (1.10 and 1.26 Gpc at 10^{-3}): 0.80/1.27/1.56 Gpc for total masses 30–50/50–75/75–100 $M_\odot$ with the bank, and a volume per year of observing time of 9.3 and 16.3 Gpc^3 for single H1. These are detector-frame masses with no cosmology (the largest distances have $z \approx 0.2$ or more), no duty cycle and no network, and a FAP per window that no real search would accept: they only show that the SNR scale of the project corresponds to a realistic reach.

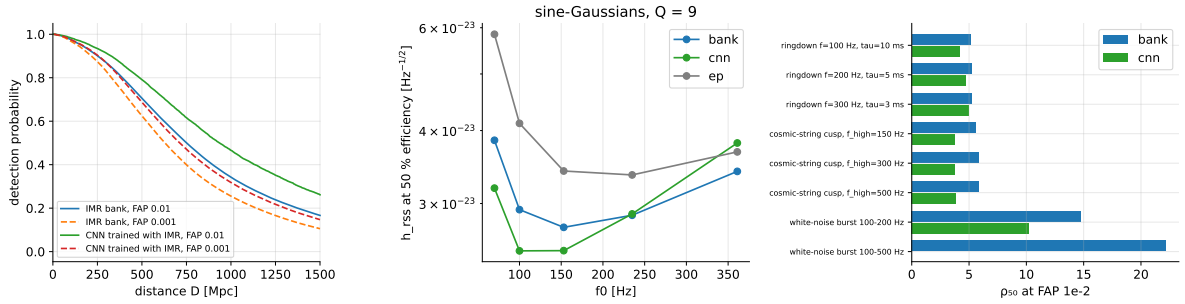


Figure 38. Left: detection probability against distance for black-hole binaries. Centre: $h_{\mathrm{rss},50}$ of Q= 9 sine-Gaussians. Right: ρ_{50} of the other burst shapes.

10 Two detectors: coincidence and time slides

Every false-alarm probability so far is per one-second window of H1 alone: the glitch floor of a single interferometer (a few 10^{-3} per window, Section 4) prevents going below FAP $\approx 10^{-3}$. A real search requires a candidate in *both* LIGO detectors with a compatible arrival time and measures its background with *time slides*: data of one detector are shifted by more than the light-travel time so that no real signal is coincident, and the pairs sample the accidental coincidences of independent noise.

10.1 Data, statistic and background

Six chunks of 4096 s with H1 *and* L1 in science mode (two of them are chunks 0 and 6 of the single-detector study; four new ones evenly spread over the run, $\geq$ 2000 s from every event) are whitened exactly as before; the windows with the same index of the two detectors are simultaneous. Four chunks give the background (4×4056 windows), two (2×4056) the zero-lag check and the injections. Each single-detector statistic (sine-Gaussian bank, IMR bank, the CNN trained with IMR signals, excess power) is converted into the empirical p-value p_d of that detector (rank in its own background, $p \geq 6 \times 10^{-5}$) and the coincident statistic is the Fisher combination $S = -\ln p_H - \ln p_L$ (for the banks a pair is rejected unless $|\tau_H - \tau_L| \leq 15$ ms). A plain sum of the statistics was tried first and discarded: it is dominated by a loud glitch of one detector (with the plain sum the FAP 10^{-5} threshold of the sine-Gaussian bank was a statistic of $\approx 3 \times 10^5$, and still $\approx 10^3$ after capping each detector's contribution at its FAP 10^{-3} value): the tail of the sum was the glitch tail of H1, not that of accidental coincidences. Real pipelines use signal-consistency tests (χ^2 vetoes) for the same purpose; they are not implemented here. The background is made by pairing every H1 window of a chunk with every L1 window of the same chunk at a different time (offsets of 2 to $N - 2$ windows): 6.6×10^7 pairs, enough for FAP down to 10^{-6}.

Calibration: are the noises independent? Zero-lag (simultaneous) windows of the two test chunks against time slides *of the same chunks*: 71–90 coincidences above the FAP 10^{-2} threshold against 81 expected and 5–9 above the 10^{-3} threshold against 8.1 expected (ratio 0.6–1.1 for the four statistics): the two noises are independent at this level. Against the slides of *other* epochs the ratio is 1.3–1.7 for the CNN and the banks and 5 for excess power: the epoch dependence of the background survives coincidence. Figure 39 shows the background: it follows $(1 + S)e^{-S}$, the law of two independent detectors, for the CNN, and lies below it for the banks because of the arrival-time requirement (a random pair is coincident within 15 ms about 4 % of the time).

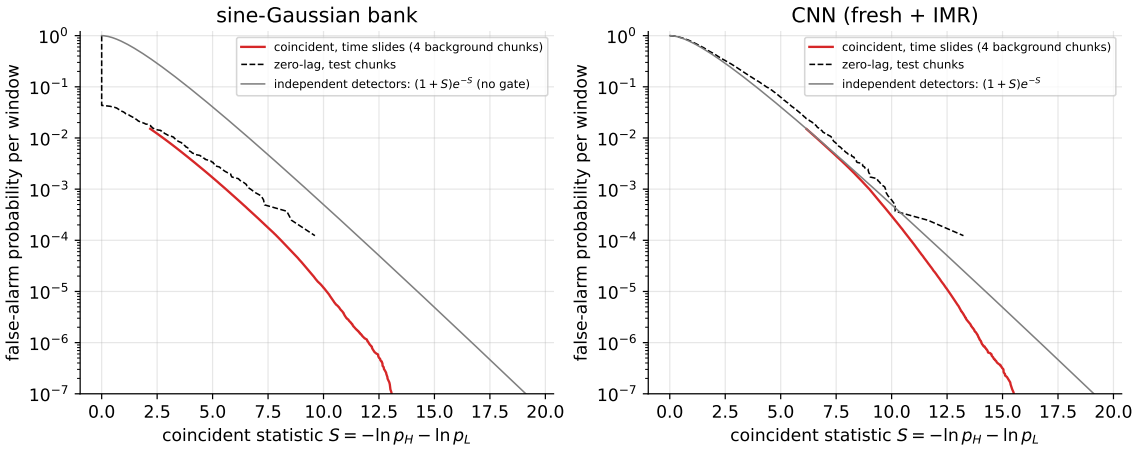


Figure 39. False-alarm probability of the coincident statistic S: time slides of the background chunks (red), zero-lag of the test chunks (dashed) and the expectation for independent detectors (grey).

10.2 Detection at low false-alarm probability

Injections in the test chunks, in the same window of H1 and L1: a held-out IMR signal (or a sine-Gaussian) with network optimal SNR $\rho_{\mathrm{net}} = \sqrt{\rho_H^2 + \rho_L^2}$ split as $\rho_H = \rho_{\mathrm{net}} \cos\alpha$, $\rho_L = \rho_{\mathrm{net}} \sin\alpha$ with $\alpha \sim U(0.45, 1.12)$ (amplitude ratio between 0.5 and 2), and a random relative delay of ± 10 ms. A single detector sees only its own share of the network SNR (on average about 0.7 of it), which is part of what a network gains. Table 21 and Fig. 40 give ρ_{50} in terms of the network SNR.

- **Coincidence moves the false-alarm floor from 10^{-3} to 10^{-6} and below, at a small cost in SNR.** Lowering the FAP by four orders of magnitude costs $+3.7$ (IMR bank) to $+6.9$ (CNN) in network SNR. A single detector cannot follow: H1 alone, the CNN needs 6.9 at 10^{-2} and 9.8 at 10^{-3}; the IMR bank of H1 alone does not reach 50 % efficiency at 10^{-3} up to a network SNR of 30, because of the glitch tail of the H1 data.
- **At FAP 10^{-6} per window the network SNR needed is ≈ 11** for both the CNN and the IMR bank, the same order as the thresholds of real searches (which work at much lower rates).
- **The ranking of Section 8 survives coincidence.** On binary-black-hole signals the burst statistics are 2–3 times worse (sine-Gaussian bank 12.4 at FAP 10^{-2} and 20.4 at 10^{-6}; excess power 10.5 and no 50 % efficiency at 10^{-6} up to 30); on sine-Gaussians the sine-Gaussian bank and the CNN are the best, and the IMR bank, of the wrong family, is useless (13.2 at 10^{-2}).

		coincident, FAP					H1 alone	
injection	statistic	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-2}	10^{-3}
IMR	CNN, trained with IMR	4.05	6.78	8.22	9.74	10.95	6.89	9.75
	IMR bank	7.38	7.63	8.48	9.72	11.04	7.53	–
	sine-Gaussian bank	12.38	12.86	14.28	16.62	20.40	11.08	–
	excess power	10.45	12.88	15.32	18.86	–	11.74	–
sine-Gaussian	CNN, trained with IMR	4.14	6.80	8.05	9.56	11.00	7.11	9.65
	sine-Gaussian bank	7.08	7.57	8.47	10.34	12.19	7.29	–
	IMR bank	13.16	13.84	15.73	18.38	21.51	11.77	–
	excess power	8.52	10.88	13.61	15.91	–	9.40	–

Table 21. ρ_{50} in terms of the network optimal SNR, coincident (H1+L1) at FAP 10^{-2}–10^{-6} and H1 alone at FAP 10^{-2}, 10^{-3} (a dash: 50 % efficiency not reached up to a network SNR of 30).

Caveats. No coherent (phase-consistent) network statistic and no antenna-pattern consistency between the detectors (the same waveform shape is injected in both, with different amplitudes); the CNN was trained on H1 and applied to L1 as it is; the p-values are resolved down to 6×10^{-5} per detector (16 224 background windows), which limits the Fisher statistic to $S \leq 19.4$; six chunks of 4096 s; and the false-alarm probability is per one-second window pair ($10^{-6} \approx$ one accidental coincidence every 11.6 days of coincident data), not yet a rate comparable to the one-per-century level of real detections.

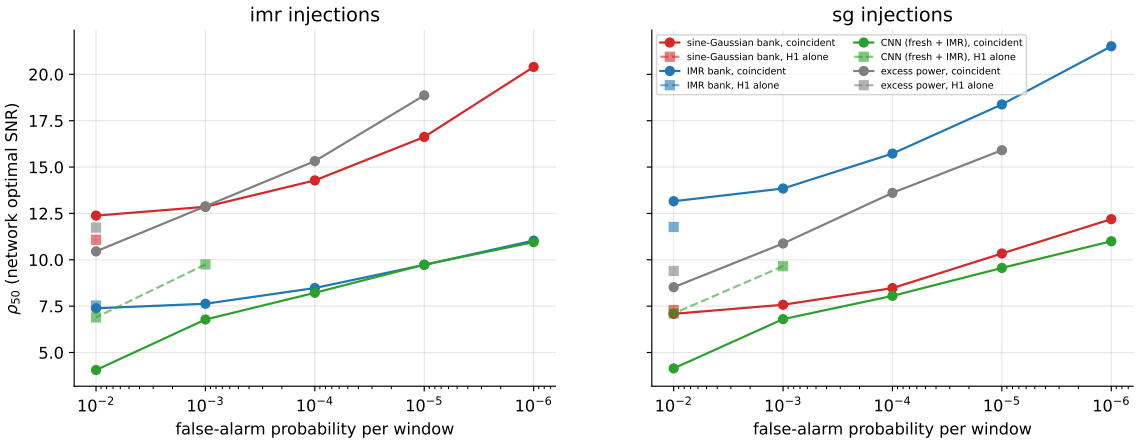


Figure 40. ρ_{50} (network SNR) against the false-alarm probability: coincident (circles) and H1 alone (squares) for the four statistics, on binary-black-hole (left) and sine-Gaussian (right) injections.

10.3 A mini continuous search

The per-window analysis is turned into a search (notebook 14) on the two test chunks (2.25 h of H1+L1): windows of 1 s every 0.5 s, IMR bank in both detectors, coincident statistic of this section with the 15 ms gate, one trigger per ±1 s (clustering), and a background built from every shift of L1 by $k \times 0.5$ s of the same data (2.08 years of equivalent slid time). A first version of the slides lost the pairs that wrap around the end of the chunk and under-estimated the background by a factor ≈ 4; with the alignment fixed the number of zero-lag triggers above $S = 6$ is 62 against 63.1 expected from the slides (chunk-by-chunk ratios 0.8–1.0 at $S \geq 6$–8), i.e. the data contain no foreground at these significances, as expected from chunks selected ≥ 2000 s from catalogued events. The loudest candidate has $S = 9.8$, a false-alarm rate of $\approx 7 \cdot 10^3\,\mathrm{yr}^{-1}$ (1.7 such triggers expected in the data), and the top of the list is made of loud H1 glitches (ρ_H up to 296) paired with a noise-level L1 window. Injections of 40 IMR signals into the same data are recovered with S above the $10^3\,\mathrm{yr}^{-1}$ threshold in 10, 40, 70 and 100 % of the cases at network $\rho = 8, 10, 12, 15$ but never below $1\,\mathrm{yr}^{-1}$: two years of slid background cannot resolve it, and reaching it requires far more data and chi^2 vetoes.

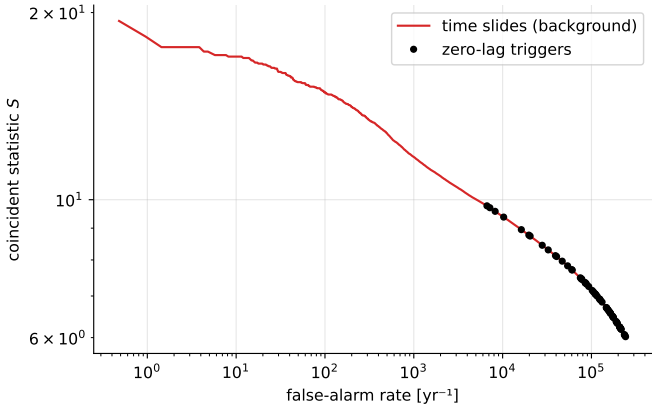


Figure 41. False-alarm rate of the coincident statistic: time-slide background and zero-lag triggers.

10.4 A coherent statistic and physical network injections

Notebook 31 asks whether combining the two detectors coherently, rather than through p-values, helps, and replaces the balanced amplitude split of the injections of this section by the real antenna patterns. Six more H1+L1 chunks (download_coinc_extra.py) extend the background to ten chunks and $1.64 \cdot 10^8$ time-slide pairs (FAP floor $6 \cdot 10^{-9}$). For every IMR template the complex peak outputs z_H, z_L are combined as λ_1, the largest eigenvalue of $\mathrm{Re}(zz^\dagger)$ (the profile likelihood of a linearly polarised signal: same waveform, real amplitudes of either sign), with the 15 ms gate and the same template in both detectors; the incoherent counterpart is $|z_H|^2 + |z_L|^2$. The injections place the same IMRPhenomD source (aligned spins, random inclination) in H1 and L1 with the real F_+, $F_\times$ of each site for a random sky position and polarisation and the real delay.

- **The raw statistics fail in non-Gaussian noise**, as for the plain sum of Section 10: one enormous glitch ($\rho \sim 10^4$) paired with each window of the other detector gives a threshold of $3 \cdot 10^8$ at FAP 10^{-6}. Rescaling each output (keeping its phase) so that $|z|^2 \to -2 \ln p$ with p the single-detector empirical p-value cures this (threshold 28 for λ_1, 30 for the sum, 13 for the Fisher statistic at 10^{-6}).
- **Coherence does not help in this implementation**: at FAP 10^{-4} the network ρ_{50} is 9.36 (Fisher of the best templates), 9.77 (incoherent sum, same template) and 10.20 (λ_1); at 10^{-6} only the Fisher statistic reaches 50 % within network $\rho = 20$ ($\rho_{50} = 15.0$). The requirement of the *same* template and of both peak lags within the gate costs more than the coherence gains: in the weaker detector the best template is often a noise one, and elliptically polarised signals do not satisfy the real-amplitude hypothesis.
- **The injections of Section 10 were optimistic**: with the real antenna patterns (including orientations with a very unequal response in the two sites) the 50 % point at FAP 10^{-6} moves from $\rho \approx 11$ to ≈ 15 network SNR for the same IMR bank and Fisher statistic.
- Zero-lag against slides of the same chunks: 9 against 8.1 expected at FAP 10^{-3} and 2 against 0.8 at 10^{-4} for all three statistics: no excess.

A proper coherent search (sky grid, both polarisations, null stream, χ^2 vetoes) is beyond this analysis; the result shows that the naive step is not enough.

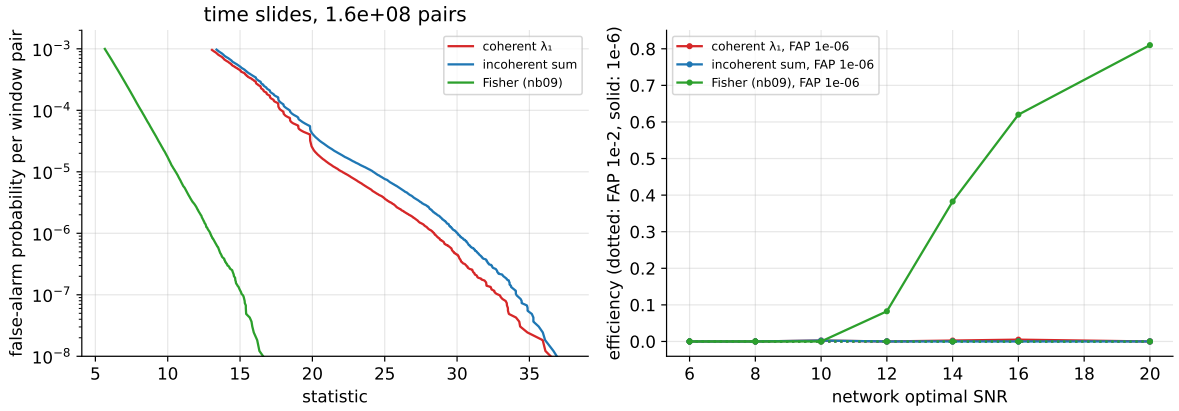


Figure 42. Left: false-alarm probability of the three statistics from $1.6 \cdot 10^8$ slide pairs. Right: efficiency against network SNR for physical injections (dotted: FAP 10^{-2}, solid: 10^{-6}).

10.5 Adding Virgo

A third detector can only be assessed here in the *ideal* way, because no three-detector background was built (notebook 38, one chunk of August 2017 with H1, L1 and V1 in science mode, away from events). With the real PSDs, antenna patterns and times, 20 000 random sources of the notebook 31 library give the optimal SNR in each detector. The median amplitude spectral density of V1 in the band is 3.7 times that of H1 (0.93 for L1). Virgo contributes on average 9.6 % of the network SNR^2 (median 4.9 %): at an equal network threshold the sensitive volume grows by 6.3 % (5.8 for HL against H1 alone, 6.2 for HLV), and of the 23 % of orientations in which the H1–L1 network SNR is below 0.6 of its median only 14 % are rescued above that level. The benefit of Virgo in O2 for this detection-only analysis is therefore modest; its main value would be sky localisation, which this project does not attempt, and a real gain would have to exceed the cost of a higher threshold from the additional background trials, not estimated here.

11 Open set: signals outside the hypotheses

In Section 7 every method detected a burst of band-limited stochastic noise — which belongs to none of the three families — and assigned it a family, mostly sine-Gaussian: a softmax is normalised over the classes it knows and the Bayesian posterior over the families it contains, so none of them could say "none of the above". This section tests that limitation directly with two kinds of out-of-distribution signals injected at the same SNR as the in-family ones: the *stochastic burst* of Section 7 (Gaussian envelope, $\sigma_t = 15\,\mathrm{ms}$) and a *binary-black-hole signal* of Section 8 (a real chirp that is not one of the three phenomenological families). The networks were trained *without* IMR signals.

Scores. (i) *Goodness-of-fit of the matched filter*: the three families are fitted (best template, lag and complex amplitude), the best fit of the family with the largest ρ^2 is subtracted, and the energy of the residual of the whitened series in $\pm 60\,\mathrm{ms}$ around the fitted time is the score: a signal that the family describes leaves only noise, one that it does not leaves excess power. (ii) For the four-class CNNs (static and fresh-injection training): the maximum softmax probability among the signal classes ($1 - \mathrm{MSP}$), the energy score $-\log \sum_c e^{z_c}$, and the Mahalanobis distance of the penultimate-layer features from the three in-family class means (shared covariance, training signals of $\rho \geq 6$). Only detected signals are compared (any-signal statistic of the network above the FAP 10^{-2} threshold; 96 % of the in-family signals at $\rho = 8$ and 100 % at $\rho \geq 12$ pass it, against 46 % of the stochastic bursts and 74 % of the IMR signals at $\rho = 8$). Performance is the AUROC for separating in-family from out-of-family signals and the fraction of out-of-family signals flagged at 5 % of in-family signals wrongly flagged (Table 22, Fig. 43).

- **The goodness-of-fit of the matched filter solves the problem of Section 7 for stochastic bursts.** AUROC 0.94 / 0.998 / 1.000 at $\rho = 8$ / 12 / 16 and 77 % of the stochastic bursts flagged at $\rho = 8$ and 100 % at $\rho \geq 12$, at 5 % false flagging. The median residual energy of in-family signals is 108–114 (noise 104); that of a stochastic burst 144 / 202 / 281.
- **For a binary-black-hole signal the best open-set score is the energy score of the network** (0.93 / 0.98 / 0.99; 64 / 92 / 94 % flagged), followed by the maximum softmax probability of the fresh-injection network; the residual of the matched filter is weaker (0.79 / 0.90 / 0.93), because an IMR chirp is *partly* described by the chirp and sine-Gaussian families

AUROC at $\rho =$	stochastic burst 8	12	16	IMR (BBH) signal 8	12	16
matched-filter residual (±60 ms)	**0.935**	**0.998**	**1.000**	0.793	0.903	0.925
energy score, CNN fresh	0.838	0.966	0.986	**0.932**	**0.982**	**0.989**
energy score, CNN static	0.796	0.944	0.975	0.892	0.974	0.990
1−MSP, CNN fresh	0.733	0.891	0.934	0.910	0.967	0.981
1−MSP, CNN static	0.715	0.833	0.863	0.851	0.921	0.938
Mahalanobis, CNN fresh	0.882	0.805	0.612	0.692	0.519	0.377
Mahalanobis, CNN static	0.747	0.477	0.300	0.672	0.517	0.408
flagged at 5 %: residual	0.77	1.00	1.00	0.37	0.75	0.81
flagged at 5 %: energy, CNN fresh	0.38	0.83	0.92	0.64	0.92	0.94

Table 22. Open-set scores: AUROC of separating in-family from out-of-family detected signals at the same SNR, and fraction of out-of-family signals flagged at 5 % of in-family signals flagged.

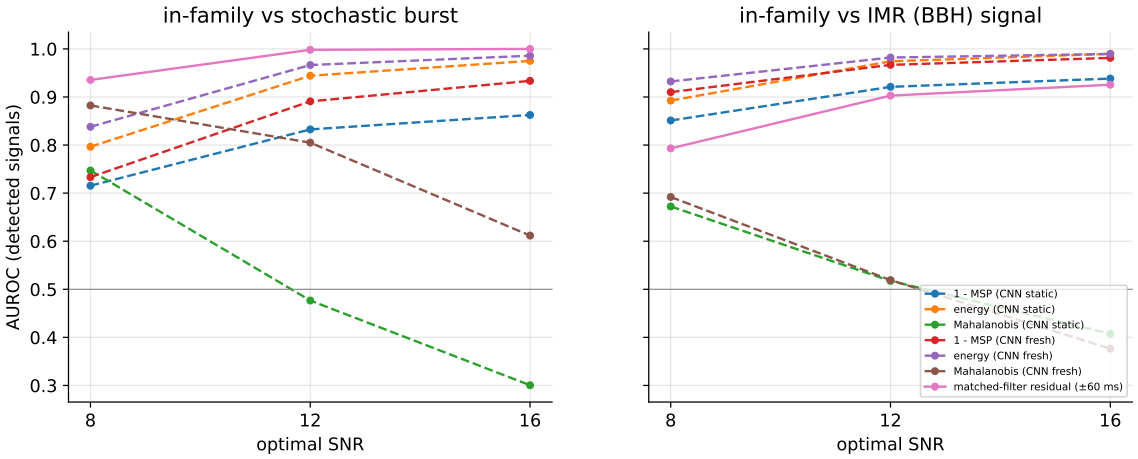


Figure 43. AUROC of every open-set score against the SNR, for stochastic bursts (left) and binary-black-hole signals (right).

(the best fit is called a sine-Gaussian or a chirp in 99 % of the cases): its residual is smaller than that of a noise-like signal.

- **Fresh-injection training also improves the open-set behaviour of the networks** (stochastic bursts, energy score 0.84 / 0.97 / 0.99 against 0.80 / 0.94 / 0.98 for the static network).
- **The Mahalanobis distance does not work here:** its AUROC *falls* with SNR, to 0.61 (fresh) and 0.30 (static) at $\rho = 16$, below 0.5. Louder in-family signals move away from the class means along the amplitude direction, so the distance measures loudness rather than membership; it would need a normalisation of the features.
- **The two kinds of score are complementary**: the residual of the matched filter flags what no family can describe (noise-like, unmodelled power), the network's energy score also flags a coherent chirp of the wrong family. A search should report both together with the family decision.

Caveats. The networks were not trained with an explicit unknown class; the flagging threshold is the 95 % quantile of the in-family scores *at the same SNR*, an oracle on the SNR (the SNR of a real candidate is not known); the ±60 ms residual window assumes a burst of comparable duration (a long signal spills outside it); only two kinds of out-of-family signal were tried; no

retraining.

12 Parameter estimation of binary black holes

Detecting a merger is half of the problem; the other half is to say what it is. Notebook 22 estimates the chirp mass M_c, the total mass M, the mass ratio q and the merger time of IMRPhenomD signals injected in real test noise with the held-out library, with three estimators: (i) the best template of a 203-template IMR bank (29 total masses $\times$ 7 mass ratios, non-spinning, face-on), (ii) the grid posterior of the same cube (amplitude marginalised, $\sigma_\rho = 10$, flat priors), and (iii) a 1-D CNN with a heteroscedastic Gaussian head (mean and log-variance of the four standardised targets) trained with fresh injections of SNR 6–20. Two heads are compared: global max pooling (the project CNN, which discards position) and a position-aware flatten head.

quantity	estimator	$\rho = 8$	10	12	16	20
median $\lvert\Delta M_c\rvert / M_c$	bank MLE	0.075	0.073	0.068	0.076	0.067
	posterior mean	0.067	0.073	0.068	0.069	0.063
	CNN (max pool)	0.132	0.120	0.101	0.084	0.088
	CNN (flatten)	0.172	0.150	0.135	0.130	0.107
median $\lvert\Delta q\rvert / q$	posterior mean	0.25	0.25	0.24	0.24	0.21
	CNN (max pool)	0.27	0.28	0.28	0.25	0.28
median $\lvert\Delta t_{\mathrm{merge}}\rvert$ [ms]	posterior mean	1.3	1.0	1.1	1.1	1.1
	CNN (flatten)	37	28	21	19	15
	CNN (max pool)	85	82	73	90	87
68 % coverage of M_c	posterior (grid)	0.32	0.28	0.24	0.18	0.12
	CNN (flatten)	0.75	0.73	0.78	0.78	0.81

Table 23. Accuracy and calibration of the estimators on 300 held-out signals per SNR.

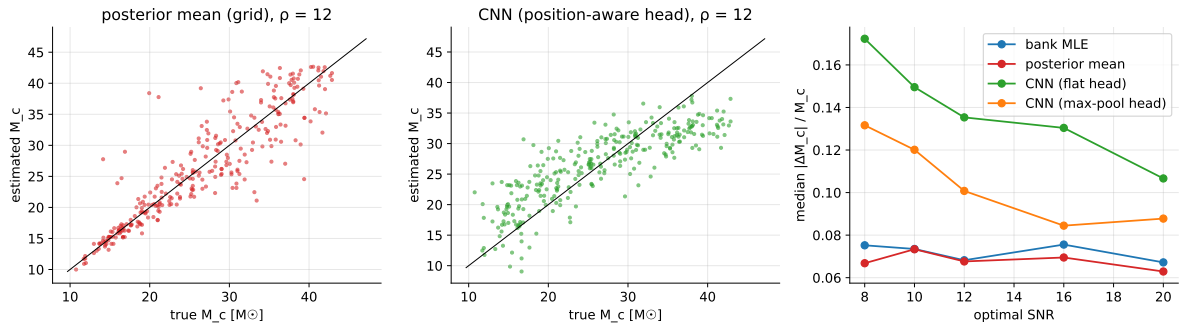


Figure 44. Left, centre: estimated against true chirp mass at $\rho = 12$ for the grid posterior and the CNN. Right: median fractional error of M_c against SNR.

- **The matched-filter estimators win on accuracy**: $\approx 7\,\%$ in M_c and M at every SNR (the error is the mass-ratio and spin degeneracy plus the grid step, not noise: it does not improve with SNR), against 10–17 % for the CNN, which improves slowly with SNR. The merger time is determined to ≈ 1 ms by the bank and to 15–85 ms by the networks (the max-pooling head has no access to position: its error is that of a guess in the window).

- **The grid posterior is badly overconfident**: its 68 % interval contains the truth in only 9–32 % of the cases for M_c and M, and less as the SNR grows. The reason is the model mismatch (non-spinning, face-on templates against aligned spins and random inclination) plus the discrete grid: the likelihood becomes sharp on the wrong point. The interval is a statement about the template model, not about the signal.
- **The CNN is better calibrated** (0.65–0.82 coverage for a nominal 0.68 in M_c, M and the merger time with the flatten head), because it is trained on the true family and learns its own error.
- q is poorly measured by every estimator (21–30 %): at these SNRs it is essentially unconstrained within its prior range.

This is not a full Bayesian parameter estimation (no nested sampling, spins, sky position or two detectors); the grid posterior is the cheap Bayesian counterpart of the bank, and its failure shows what the full analysis (extension 35) would have to repair.

13 UNFOLDING THE MASS DISTRIBUTION OF DETECTED BINARIES

The IMR bank returns a reconstructed total mass $\hat{M}$ (the best of eight template masses), so the observed distribution of $\hat{M}$ is a smeared, sculpted version of the true one (the efficiency depends on mass and SNR). Notebook 19 unfolds it with the methods of the course. The response is simulated with the training IMR library, a uniform mass prior and $p(\rho) \propto \rho^{-4}$ on $[6, 40]$ (30 000 events, selection: IMR-bank statistic above the validation threshold at FAP 10^{-2}, efficiency 0.84–0.91 per mass bin); the pseudo-data (12 000 events from the held-out library, other noise windows) follow a very different population (a peak at $45\,M_\odot$ plus a flat tail), so the truth is known. Seven mass bins between 30 and 100 $M_\odot$ are used for the binned methods; OmniFold is unbinned on $(\hat{M}, \hat{q}, \ln\rho^2, \mathrm{lag}) \to (M, q)$ with gradient-boosted classifiers and 4 iterations.

method	$\sum(\text{unfolded} - \text{true})^2/\text{true}$
naive (observed/$\bar{\varepsilon}$)	1216
matrix inversion	4862
IBU, 1 / 2 / 3 / 5 / 10 iterations	1811 / 521 / 199 / 141 / 268
OmniFold, 4 iterations	468

Table 24. Distance of each unfolded spectrum from the true pseudo-data spectrum (seven bins; the absolute value only ranks the methods).

- **Inversion fails**: the response is nearly singular for neighbouring bins (the template masses are coarse), the solution has negative bins and a relative bootstrap spread of 15 %.
- **IBU** converges in 3–5 iterations (the distance is minimal at 5 and grows again at 10: noise amplification, the usual regularisation by early stopping), relative spread 3 %, mean absolute bias 11 %.
- **OmniFold** recovers the shape without binning but leaves a visible bias (1157 against 673 in the 55–65 $M_\odot$ bin), mean absolute bias 19 %, spread 3 %: with only four reco variables and a coarse $\hat{M}$ it brings no gain over IBU.
- **The bootstrap spread is smaller than the bias**: the truth lies within one standard devia-

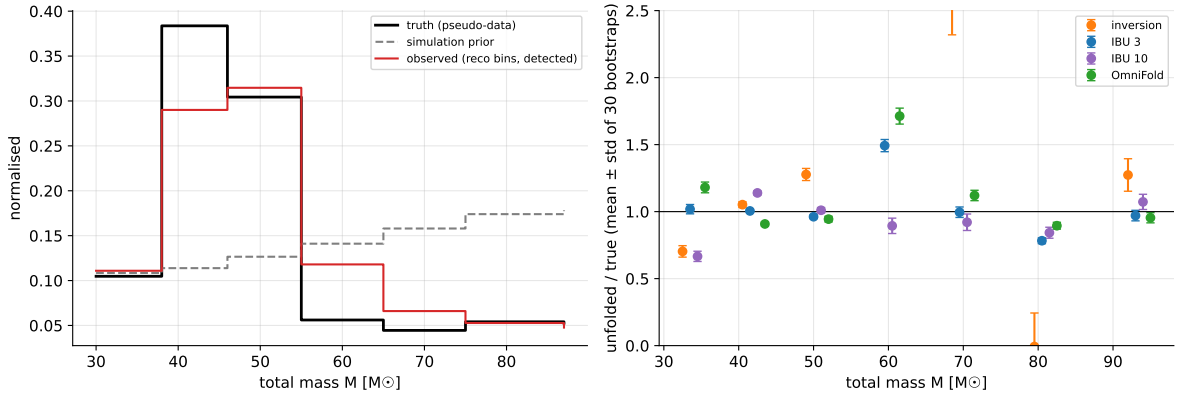


Figure 45. Left: true, prior and observed mass distributions. Right: unfolded over true per bin (mean and standard deviation of 30 bootstraps of the pseudo-data).

tion in only 57 % (IBU) or 0 % (OmniFold) of the bins. The dominant error is the dependence on the simulation prior and the finite resolution, not the statistics of the data; an honest uncertainty needs a prior-variation systematic.

This is the unfolding of a toy population *for this detector and this SNR law*; it is not an estimate of the astrophysical mass function.

Population inference with selection effects. The same pseudo-data can be analysed in the hierarchical way of gravitational-wave population studies (notebook 32): the mass distribution is a mixture of a flat component and a Gaussian of unknown $(f_{\mathrm{flat}}, \mu, \sigma)$, the likelihood of each detected event uses the response matrix of the simulation, and the normalisation divides by the detection efficiency of the population. The posterior means are $\mu = 44.8\,M_\odot$, $\sigma = 4.7\,M_\odot$ and $f_{\mathrm{flat}} = 0.30$, against $(45, 5, 0.30)$ from fitting the true masses (all events, undetected included): selection effects and response are correctly handled. Ignoring them (taking the reconstructed mass as the true one) gives $\mu = 49.6$, $\sigma = 6.4$, $f_{\mathrm{flat}} = 0.36$: a bias of $+4.6\,M_\odot$ in the peak, because the efficiency rises with mass and the template masses are discrete. The hierarchical likelihood is both simpler and more precise than the binned unfolding for a parametric population, and it is the right tool when the question is a parameter rather than a spectrum.

14 Comparison of the three approaches

Figure 46 shows ρ_{50} of every method on one axis and Fig. 47 the pipeline; Table 25 collects what each approach gives on the same data, and what it does *not* give.

14.1 Which background sets the threshold?

Every ρ_{50} depends on the noise sample on which the false-alarm threshold is measured. Table 26 gives the matched-filter bank with the threshold taken from the training, validation, pooled and test noise (same injections, notebook 12). At FAP 10^{-2} the choice moves ρ_{50} by about 0.5 (sine-Gaussian 5.5, 4.9, 4.9 for train, validation, test); at 10^{-3} by several units (10.8

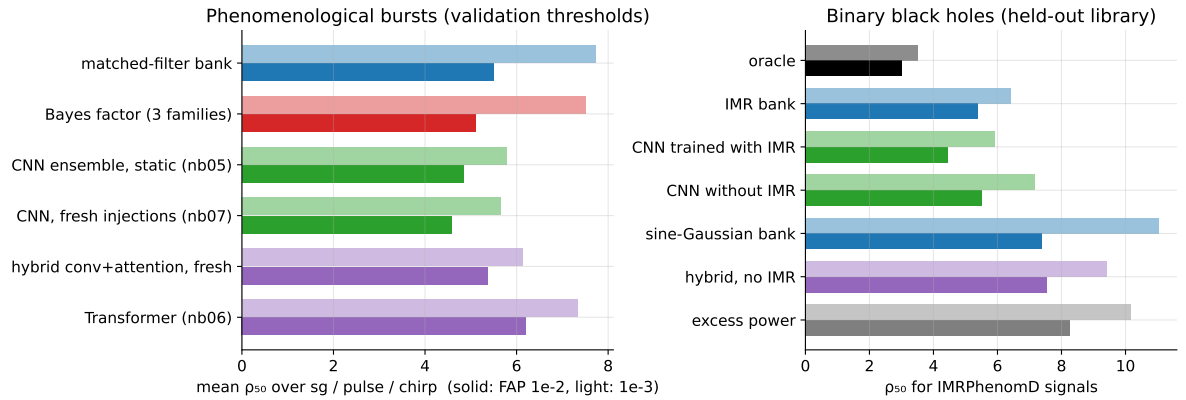


Figure 46. ρ_{50} of the detectors compared in this report, thresholds from the validation background; solid: FAP 10^{-2}, light: 10^{-3}. Left: mean over the three phenomenological morphologies; right: IMRPhenomD signals (held-out library).

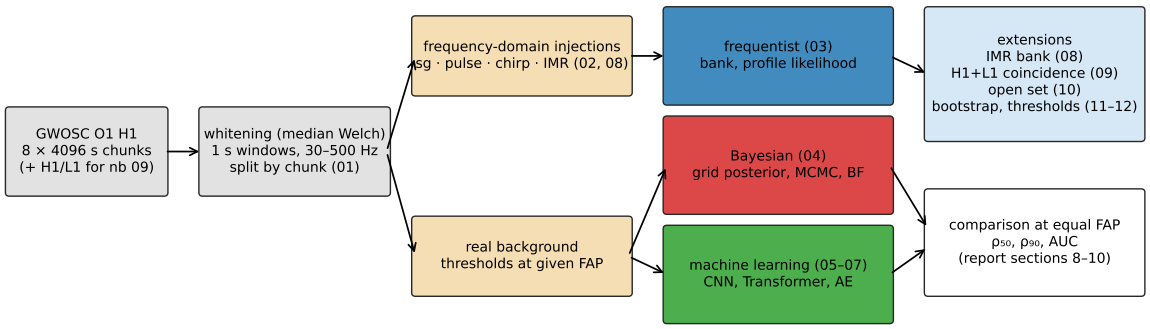


Figure 47. Pipeline of the analysis; numbers refer to the notebooks.

with the training background, 6.8 with the validation one, 14.2 with the test one), because the tail is set by a few glitches of the epochs sampled. *Convention of this report:* all cross-method comparisons (Sections 6–10 and this one) use the validation background; the training-background numbers appear only in the calibration discussion of Section 4; the test noise is never used to set a headline threshold.

14.2 Why does the CNN beat the bank?

Notebook 15 tests three explanations on the three phenomenological morphologies (thresholds from the validation background, mean ρ_{50} over sine-Gaussian, pulse and chirp, bootstrap error ≈ 0.1 at FAP 10^{-2}, ≈ 0.35 at 10^{-3}; Fig. 48).

1. *A richer bank does not close the gap* (5.54 against 5.58): the advantage is not template coverage.
2. *In Gaussian noise the network is not better than the bank*: 5.63 against 5.38 at 10^{-2} (it is better on pulses, worse on sine-Gaussians), where it was never trained. The CNN advantage of ≈ 1.1 (10^{-2}) and ≈ 2.2 (10^{-3}) seen in real noise is therefore largely an advantage on *real* noise, in which the network was trained: the real noise costs the bank 0.2 at 10^{-2} and 1.7 at 10^{-3} and costs the network nothing. Part of it is a domain effect (the network was not retrained in Gaussian noise), so this is evidence that the advantage lives in the tail of real noise, not a proof of its mechanism.

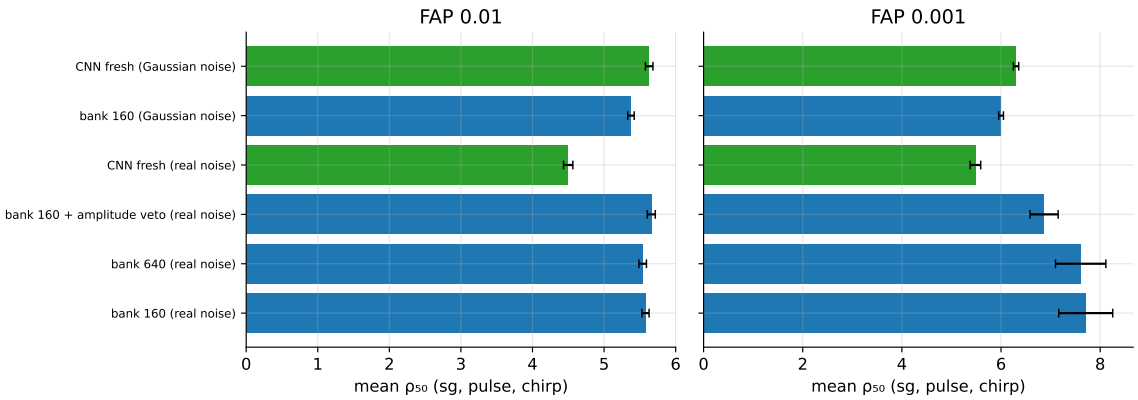


Figure 48. Mean ρ_{50} of the ablation cases of Table 27.

3. *A crude glitch veto recovers part of the gap at* 10^{-3} ($7.71 \rightarrow 6.87$ on the morphologies that still reach 50 %) and nothing at 10^{-2}; signal-consistency vetoes (χ^2) are the natural next step.

14.3 Generalisation to O2 and O3

Notebook 36 applies the unchanged pipeline (own-chunk median-Welch PSD, one-second windows, band 30–500 Hz) to nine new H1 chunks, three each in O2, O3a and O3b ($\geq$ 2000 s from catalogued events). The in-band amplitude spectral density is 0.91 (O2), 0.65 (O3a) and 0.61 (O3b) times that of O1, but the data are much less Gaussian in O3: the excess kurtosis of the whitened samples is 10^5–10^6 and the largest bank statistics reach $7 \cdot 10^5$ (a single window of $\rho \approx 800$), orders of magnitude above O1.

- **The O1 thresholds transfer remarkably well at FAP** 10^{-2}: the observed FAP is 0.9–1.6 % for the bank and 1.4–2.1 % for the CNN, a factor < 2 off the nominal value, and the ρ_{50} with the O1 threshold is the same as in O1 (5.55 for the bank; 3.7–3.9 for the five-network CNN ensemble, against 3.79 in O1, Section 7.4): the whitening with the chunk's own PSD absorbs the change of sensitivity.
- **At FAP** 10^{-3} **the tails decide**: with its own threshold the bank in O3 needs a network SNR above 16 for 50 % efficiency, because three O3 chunks contain windows with a statistic of 10^4–10^6 that push the 99.9 % quantile above any realistic signal; the CNN is far less affected (5.4–6.2). In O3 the data-quality flags of Section 4.11 (not applied here) or time slides are indispensable for a template statistic.
- The CNN trained on O1 only keeps its ranking with the bank in O2 and O3 without retraining (3.7–4.5 against 5.5–5.9 at 10^{-2}): its advantage does not depend on the epoch of the training noise.

What the comparison supports.

1. The frequentist and Bayesian statistics are two views of the same matched-filter cube and detect the same bursts; the Bayesian analysis adds calibrated posteriors and the prior dependence of the evidence, the frequentist analysis adds optimality statements (Neyman–Pearson oracle gap) and guaranteed coverage.
2. A network with no template knowledge reaches matched-filter performance on the family the bank was designed for, and exceeds it on the shapes it has seen; trained with fresh injections at every epoch a single CNN does better than a five-member ensemble trained

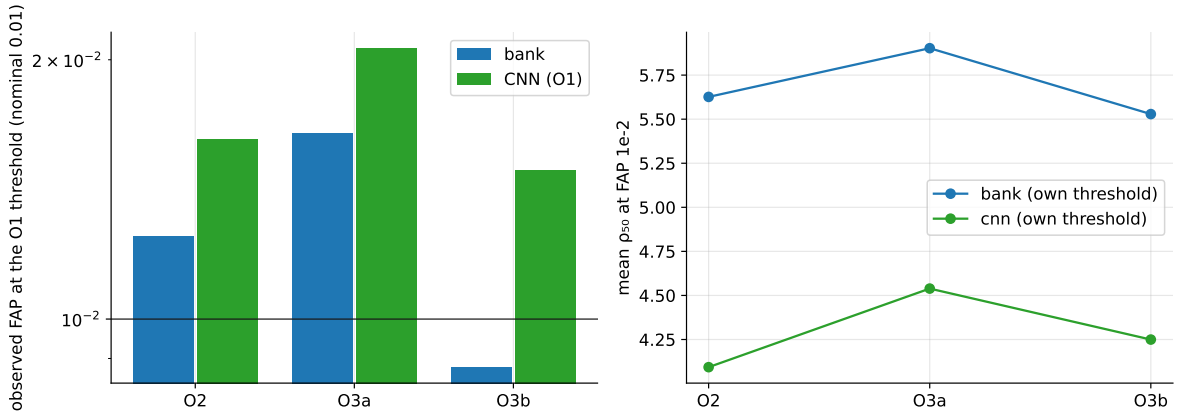


Figure 49. Left: observed FAP of the O1 thresholds in O2/O3 (nominal 0.01). Right: ρ_{50} with the threshold of the own run.

on a fixed set. Its advantage is bounded by its training set: on a morphology it has not seen it falls below the bank, so it is not a replacement for a morphology-independent statistic.

3. **Realistic signals.** On binary-black-hole signals the detectors designed on phenomenological bursts do not transfer for free (the sine-Gaussian bank loses 2.0 in ρ_{50} with respect to an IMR bank, excess power 2.9); the networks trained on the phenomenological morphologies generalise better than the banks of the wrong family, and a network trained with IMR signals beats the IMR bank on waveforms never seen in training, at no cost on the other morphologies.
4. **Two detectors.** Coincidence with time slides moves the false-alarm floor from 10^{-3} (one detector, set by the glitch tail) to 10^{-6} per window at a cost of a few SNR units; the CNN and the matched filter of the right family then both need a network SNR of ≈ 11. The combination statistic must be robust to a glitch in one detector.
5. **Which morphology?** Model comparison with the exact likelihood of each family (Bayes factors) is the best classifier among the networks trained on the static set; a CNN trained with fresh injections ties with it at low SNR and is slightly ahead at high SNR (0.977 against 0.969 at $\rho = 8$). A signal that belongs to none of the families is not flagged by the family decision of any method, but it is by an explicit goodness-of-fit of the matched filter (stochastic bursts) or by the energy score of the network (chirps of the wrong family).
6. **Sequence structure does not help here.** A patch Transformer and a GRU detect worse than the CNN with every lever tried (data, learning rate, patch size, longer training). The patch embedding, not the data or the optimisation, is the limitation: a convolutional stem recovers most of the gap, but the CNN stays ahead on one-second windows with a localised burst.
7. None of the statistics escapes the real noise: the glitch floor dominates the price of a low false-alarm probability, and the calibration of a threshold is a property of the epoch in which it was measured.

What it does not support. These are efficiencies for injections into one-second windows at a given *optimal SNR*; they are not astrophysical sensitivities and not rates. Even the two-detector analysis lacks a coherent network statistic, sky localisation, signal-consistency vetoes and the very low false-alarm rates of real searches.

15 CONCLUSIONS AND LIMITS

Conclusions.

- **The real noise is the main result.** Whitened O1 H1 data are Gaussian in the bulk, but a glitch floor of a few 10^{-3} per second and strong epoch-to-epoch variations decide the price of a false-alarm probability: lowering it from 10^{-2} to 10^{-3} moves the 50 % point of the bank from $\rho \approx 5.5$ to ≈ 11 with the training-background threshold and from 4.9 to 6.8 with the validation one (in Gaussian noise the threshold would move only from ≈ 27 to ≈ 31, i.e. by ≈ 0.4 in SNR), and a threshold calibrated on one set of epochs is off by a factor ≈ 4.5 on others.
- **Frequentist.** The profile-likelihood bank reaches 50 % efficiency at $\rho \approx 5$ (oracle with known parameters: 3.0); the gap is the look-elsewhere penalty, quantified by $N_{\mathrm{eff}} \approx 6800$ independent trials. The MLE saturates the Fisher bound for $\rho \gtrsim 10$; the amplitude estimate is biased and the profile interval under-covers near the boundary, where Neyman/Feldman–Cousins intervals are exact. The dominant analysis systematic is the PSD estimator (mean against median Welch: $+0.025$ on $\varepsilon(\rho = 8)$).
- **Bayesian.** The exact grid posterior is validated against a closed form, MCMC and simulation-based calibration, in Gaussian and in real noise; the Bayes factor ranks windows like $\lambda_{\max}$ but is prior-dependent (Occam factor $-2 \ln \sigma_\rho$) and exceeds 100 on 0.2–0.9 % of pure-noise windows because of glitches. Credible intervals match confidence intervals for $\rho \gtrsim 3$ and differ, depending on the prior, below.
- **Machine learning.** A small CNN ensemble matches the bank on sine-Gaussians and beats it on the pulses and chirps it was trained on by about one SNR unit, but a network trained on one morphology fails on others; an autoencoder trained on noise adds nothing to an energy statistic. Calibration is good (ECE $\lesssim 3$ %), mostly limited by the epoch shift, and ensemble uncertainty identifies the windows where the model should not be trusted.
- **Sequence models and tuning.** A Transformer and a GRU detect worse than the CNN, with the same data and with every lever tried; the training loss of the Transformer stays high (≈ 0.3 against 0.12 for the CNN): the patch embedding cannot build the frequency-selective filters a burst needs. A convolutional stem in front of the Transformer recovers most of the gap but does not overtake the CNN. The largest effect of the whole study is on the CNN itself, which was data-limited: fresh injections at every epoch make a single network better than the five-member ensemble ($\overline{\rho_{50}} = 4.6$ against 4.85 at FAP 10^{-2}).
- **Which morphology.** Bayesian model comparison is the best classifier among the networks trained on the static set (0.97 three-way accuracy at $\rho = 8$); a CNN trained with fresh injections ties with it at low SNR and is slightly ahead at high SNR (0.977), so the ranking of Section 7 was a statement about a data-limited network.
- **Realistic signals.** The detectors designed on phenomenological bursts do not transfer for free to binary-black-hole (IMRPhenomD) signals: the sine-Gaussian bank loses 2.0 in ρ_{50} with respect to an IMR bank (7.4 against 5.4 at FAP 10^{-2}; oracle 3.0); a CNN trained with IMR signals reaches 4.45 on waveforms never seen in training, at no cost on the other morphologies.
- **Two detectors.** Coincidence with time slides moves the false-alarm floor from 10^{-3} to 10^{-6} per window at a cost of 3.7–6.9 in network SNR (CNN and IMR bank: $\rho_{50} \approx 11$ at 10^{-6}); the noises of H1 and L1 are independent (zero-lag against slides of the same chunks: ratio 0.6–1.1), but a combination statistic must be robust to a glitch in one detector.

- **Unknown morphology.** The goodness-of-fit of the matched filter (residual after subtracting the best fit) flags stochastic bursts (AUROC 0.94 at $\rho = 8$, ≥ 0.998 at $\rho \geq 12$) and the energy score of the network flags chirps of the wrong family (0.93 at $\rho = 8$); the Mahalanobis distance of the features fails.
- **What the second round of extensions added.** (i) Errors: the bootstrap error of ρ_{50} is 0.05–0.1 at FAP 10^{-2}, but the seed scatter of the CNN is 0.4 (five seeds), so the ranking CNN $<$ hybrid $<$ Transformer (1–2 units) is robust and differences of tenths between single-seed networks are not. (ii) The advantage of the CNN over the bank lives in the tails of the real noise and in the injection prior: in Gaussian noise the bank is as good, a richer bank does not help, a χ^2 test and the GWOSC CAT2 flags recover most of the gap at FAP 10^{-3} for the bank, and the CNN loses its advantage outside its training ranges (ρ_{50} 8.6 against 5.1 at 400–500 Hz, 0.02 efficiency on loud short pulses) and under a 60 Hz line (FAP 0.4 $\rightarrow$ 17 %). (iii) Physical units: $h_{\mathrm{rss},50} \approx 2.5 \cdot 10^{-23}\,\mathrm{Hz}^{-1/2}$ for single H1, a Euclidean range of 1.3–1.6 Gpc for 30–100 $M_\odot$, and with real antenna patterns the network ρ_{50} at FAP 10^{-6} is ≈ 15, not 11; the coherent statistic tried here brings no gain. (iv) Parameters: ≈ 7 % in M_c and M for the bank, the grid posterior and the exact-model MCMC alike, with calibrated intervals only for the MCMC and the CNN; hierarchical inference with selection effects recovers a mass distribution that unfolding and naive fits do not. (v) Generalisation: the O1 calibration transfers to O2 and O3 to a factor < 2, the CNN trained on O1 keeps its ranking, and the glitches of O3 make the bank unusable at 10^{-3} without data-quality vetoes; a normalising flow or a VAE trained on noise only costs ≈ 2 in ρ_{50} with no prior on the morphology.

Limits.

- **Detectors.** Sections 2–8 use H1 alone, so their false-alarm numbers are per one-second window of a single detector, not rates; Section 10 adds L1 with an incoherent combination (no coherent statistic, no antenna-pattern consistency, no χ^2 vetoes) and six chunks, down to 10^{-6} per window.
- **Injections at a given SNR.** Not converted into distances or energies. The three morphologies are phenomenological; the IMRPhenomD library covers total masses 30–100 $M_\odot$, aligned spins only (no precession, higher modes or eccentricity), and neutron-star binaries are out of reach of one-second windows and a 500 Hz band (Section 8.6: 9 % of the SNR of a $1.4 + 1.4\,M_\odot$ system is retained); precessing, higher-mode and eccentric signals cost at most 0.3–0.9 in ρ_{50} and NSBH signals 5.5 for the IMR bank.
- **Background statistics.** 1.6×10^4 training windows limit the lowest usable FAP to $\sim 10^{-3}$; the quoted 10^{-3} efficiencies carry a large uncertainty from the tail of the background.
- **Eight chunks.** The epoch dependence is measured with two test chunks; a systematic study of its size would need many more.
- **Grid resolution.** The Bayesian grid under-resolves f_0 and t_0 for $\rho \gtrsim 15$ (the SBC coverage of those two parameters, 0.50–0.60, reflects this, not the model).
- **ML numbers** are efficiencies *for the training injection prior*; the supervised detectors are only as good as the injections they have seen, and parameter regression with the CNN (Section 12) is 10–17 % in M_c, worse than the bank.
- **Tuning study.** Single seed for most variants (notebooks 26, 34, 39: the seed scatter of the CNN is 0.4 at FAP 10^{-2}), a budget of 12–30 epochs; contrastive pre-training and overlapping windows for the Transformer not run.
- **Comparison thresholds.** The bank is compared with the ML detectors on the validation

background; the choice of background changes the numbers by up to 0.6 in ρ_{50} (Section 6).

Natural extensions. A full coherent network analysis (sky grid, both polarisations, null stream, χ^2 vetoes) and three-detector time slides; windows of 4–8 s or 16 kHz data for low masses and neutron stars; nested sampling with spins, sky position and several detectors; training the networks with a glitch class, an explicit unknown class and a wider injection prior with no input clipping; more seeds for every variant; contrastive pre-training of the Transformer; the LVK data-quality flags and auxiliary-channel vetoes in the template statistic.

A Reproducibility and origin of the numbers

The project is a pipeline of notebooks in `Code_General/`, run in order with the Jupyter kernel `cern` (environment in `environment.yml`); every random number is generated with a fixed seed and long steps cache their output. `download_data.py` downloads the eight raw chunks (about 0.5 GB in float32; not tracked); `gwosc_tools.py` holds the shared functions (whitening, analytic waveforms, injection, template bank, statistics), `seq_models.py` the models, the dataset and the resumable training loop of the tuning study (it trains the variants in parallel processes), `imr_tools.py` the IMRPhenomD library and the IMR template bank (it needs `lalsuite` only to build the libraries, `python imr_tools.py`), `open_set.py` the family fits and goodness-of-fit scores, and `download_coinc_data.py` the six H1+L1 chunks of Section 10. Run times on a 10-core laptop (on AC power): notebook 01 ≈1 min, 02 ≈1 min, 03 ≈10–15 min (most of it the systematic variants), 04 ≈20 min, 05 ≈30–40 min with training from scratch (a few minutes with the saved models), 06 ≈1–2 h, 07 ≈3 h of training on four processes in total, 08 ≈10 min plus 20 min for the two networks trained with IMR signals, 09 ≈10 min plus the download of the L1 data, 10 ≈10 min (a laptop on battery power runs about three times slower).

The figures of the report are the `figures/fig0N_*.pdf` files written by the notebooks and copied into `Docs/report/figures/`. The report is compiled with `tectonic -X compile main.tex`.

Added tools and tests. `glitch_tools.py`, `pe_tools.py`, `network_tools.py`, `precession_tools.py`, `make_summary_figures.py`; `download_events.py`, `download_coinc_extra.py`, `download_runs.py`, `download_virgo.py`; nineteen automatic tests (`python -m pytest Code_General/tests`), `run_all.sh` and SHA-256 checksums of the raw strain (`results/data_checksums.json`).

References

[1] LIGO Scientific Collaboration and Virgo Collaboration. Observation of gravitational waves from a binary black hole merger. *Phys. Rev. Lett.*, 116:061102, 2016.

[2] LIGO Scientific Collaboration and Virgo Collaboration. Open data from the first and second observing runs of advanced LIGO and advanced Virgo. *SoftwareX*, 13:100658, 2021.

[3] W. G. Anderson, P. R. Brady, J. D. E. Creighton, and É. É. Flanagan. An excess power statistic for detection of burst sources of gravitational radiation. *Phys. Rev. D*, 63:042003, 2001.

[4] S. Klimenko et al. Method for detection and reconstruction of gravitational wave transients with networks of advanced detectors. *Phys. Rev. D*, 93:042004, 2016.

[5] D. M. Macleod et al. GWpy: A python package for gravitational-wave astrophysics, 2021. SoftwareX 13, 100657.

[6] P. D. Welch. The use of fast Fourier transform for the estimation of power spectra. *IEEE Trans. Audio Electroacoust.*, 15:70, 1967.

[7] S. S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *Ann. Math. Statist.*, 9:60, 1938.

[8] R. B. Davies. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 74:33, 1987.

[9] G. J. Feldman and R. D. Cousins. Unified approach to the classical statistical analysis of small signals. *Phys. Rev. D*, 57:3873, 1998.

[10] R. Barlow. Systematic errors: facts and fictions. arXiv:hep-ex/0207026, 2002.

[11] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman. emcee: The MCMC hammer. *Publ. Astron. Soc. Pac.*, 125:306, 2013.

[12] H. Jeffreys. *Theory of Probability*. Oxford University Press, 3 edition, 1961.

[13] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv:1804.06788*, 2018.

[14] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017.

[16] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.

[17] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, 2016.

[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

[19] A. Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[20] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing*, 2014.

[21] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[22] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

[23] LIGO Scientific Collaboration and Virgo Collaboration. Properties of the binary black hole merger GW150914. *Phys. Rev. Lett.*, 116:241102, 2016.

[24] LIGO Scientific Collaboration. LIGO algorithm library – LALSuite. free software (GPL), `https://doi.org/10.7935/GT1W-FZ16`, 2018.

[25] S. Khan, S. Husa, M. Hannam, F. Ohme, M. Pürrer, X. J. Forteza, and A. Bohé. Frequency-domain gravitational waves from nonprecessing black-hole binaries. II. a phenomenological model for the advanced detector era. *Phys. Rev. D*, 93:044007, 2016.

	frequentist (bank)	Bayesian (same likelihood)	machine learning
statistic	$\lambda_{\max} = \max \rho^2$ (profile LR)	$\ln B_{10}$ (prior-weighted mean of the same cube)	CNN logit / ensemble mean
detection, sine-Gaussian (ρ_{50}, FAP 10^{-2})	4.9 (val. bkg)	identical to the bank within 0.01 in efficiency	5.1 (static ensemble); 4.7–4.8 (CNN with fresh injections)
detection, other morphologies	5.3 / 6.3 (pulse / chirp), robust	as the bank	better if trained on them (4.1–4.2 / 4.8–5.0 with fresh injections), worse if not (6.5 / 6.8)
binary black holes (IMRPhenomD)	IMR bank 5.4; sine-Gaussian bank 7.4; oracle 3.0	not computed	CNN trained with IMR 4.45; without IMR 5.5–6.2
two detectors, FAP 10^{-6} (network SNR)	IMR bank 11.0	not computed	CNN 11.0
morphology (3-way accuracy, $\rho = 8$)	0.92 (largest ρ^2)	0.97 (Bayes factors of the families)	0.977 (CNN, fresh), 0.960 (static), 0.87 (Transformer)
unknown morphology	residual of the best fit flags stochastic bursts (AUROC 0.94 at $\rho = 8$)	–	energy score flags chirps of the wrong family (AUROC 0.93)
sequence models	–	–	Transformer and GRU worse than the CNN at detection; a convolutional stem recovers most of the gap, not all
needs	a template family	a template family and a prior	labelled injections; no template
parameters	MLE, saturates the Fisher bound for $\rho \gtrsim 10$	full posterior; calibrated (SBC)	none (not attempted)
uncertainty	profile LR / Neyman / FC intervals; coverage depends on SNR	credible intervals, prior-dependent at $\rho \lesssim 3$	ensemble / MC dropout; reject-uncertain gain
false alarms	calibrated on real background, epoch-dependent	$B_{10} > 100$ in 0.2–0.9 % of noise windows	calibrated on real background, epoch-dependent
cost	seconds	seconds (grid) – minutes (MCMC)	17–80 min training (laptop), ms inference

Table 25. Summary of the three approaches (FAP 10^{-2} per one-second window unless stated; SNR values are optimal SNRs).

FAP	threshold from	sine-Gaussian	pulse	chirp	mean
10^{-2}	train	5.49	5.87	7.05	6.13
	validation	4.94	5.40	6.25	5.53
	train+val	5.27	5.64	6.78	5.90
	test	4.89	5.36	6.18	5.48
10^{-3}	train	10.78	11.38	14.06	12.07
	validation	6.82	7.39	9.08	7.76
	train+val	10.08	10.83	13.33	11.41

Table 26. ρ_{50} of the sine-Gaussian bank as a function of the background used for the threshold. Bootstrap error ≈ 0.1 at 10^{-2} and ≈ 1 at 10^{-3} (the latter set by the $\approx 16\,000$-window backgrounds). With the test noise at 10^{-3} the chirp curve does not reach 50 % within $\rho \leq 16$.

detector, noise	FAP 10^{-2}	FAP 10^{-3}
bank (160 templates), real noise	5.58	7.71
bank (640 templates), real noise	5.54	7.61
bank (160) + amplitude veto, real noise	5.66	6.87
CNN (fresh injections), real noise	4.50	5.48
bank (160), *Gaussian* noise	5.38	6.00
CNN (same network, not retrained), Gaussian noise	5.63	6.30

Table 27. Mean ρ_{50} of the bank and of the CNN in real and in synthetic Gaussian noise, and with a richer bank or a veto. The amplitude veto removes the largest-amplitude windows (0.5 % of the noise) and also loud pulses; at 10^{-3} the pulse curve does not reach 50 % with it and is excluded from the mean.

detector	threshold	O2	O3a	O3b
bank, observed FAP at nominal 10^{-2} (10^{-3})	O1 validation	0.0125 (0.0013)	0.0164 (0.0040)	0.0088 (0.0022)
CNN (O1), observed FAP at nominal 10^{-2} (10^{-3})	O1 validation	0.0162 (0.0020)	0.0206 (0.0033)	0.0149 (0.0012)
bank ρ_{50} at FAP 10^{-2} (10^{-3})	O1 threshold	5.55 (7.75)	5.55 (7.76)	5.58 (7.69)
	own-run threshold	5.63 (8.58)	5.90 (>16)	5.53 (>16)
CNN ρ_{50} at FAP 10^{-2} (10^{-3})	O1 threshold	3.69 (5.23)	3.87 (5.33)	3.86 (5.35)
	own-run threshold	4.09 (5.43)	4.54 (6.24)	4.25 (5.46)

Table 28. Transfer of the O1 calibration and sensitivity to O2 and O3 (mean ρ_{50} over the three morphologies; three chunks per run; a threshold from three chunks at 10^{-3} rests on ≈ 12 windows).

report	notebook	result file
Sec. 2: data, PSD, Gaussianity tests	`01_data_whitening_validation`	`results/01_summary.json`
Sec. 3: waveforms, SNR, injection validation	`02_waveforms_and_injections`	`results/02_summary.json`
Sec. 4: bank, background, efficiency, MLE, toys, systematics	`03_frequentist_analysis`	`results/03_summary.json`, `results/03_systematics_raw.json`
Sec. 5: posterior, MCMC, Bayes factors, SBC, coverage	`04_bayesian_analysis`	`results/04_summary.json`
Sec. 6: features, CNN, autoencoder, calibration, uncertainty	`05_ml_detection`	`results/05_summary.json`
Sec. 7: Transformer, GRU, pretraining, glitches, attention, morphology classification	`06_sequence_models_hypothesis_testing`	`results/06_summary.json`
Sec. 7.3: tuning study (fresh injections, learning rate, patch size, convolutional stem), multiclass with fresh injections	`07_tuning_sequence_models`	`results/07_summary.json`
Sec. 8: binary-black-hole signals, IMR bank, detection and mass dependence	`08_realistic_inspiral_signals`	`results/08_summary.json`
Sec. 10: two detectors, coincidence, time slides	`09_two_detector_coincidence`	`results/09_summary.json`, `results/coinc_chunks.json`
Sec. 11: goodness-of-fit and open-set scores	`10_open_set_goodness_of_fit`	`results/10_summary.json`
Sec. 7.4, 14.1: bootstrap errors, threshold convention	`11_bootstrap_error_bars`, `12_threshold_convention`	`results/11_summary.json`, `results/12_summary.json`
Sec. 8: real events (GW150914, GW151012, GW151226)	`13_real_events`	`results/13_summary.json`, `results/events_chunks.json`
Sec. 10: mini continuous search	`14_continuous_search`	`results/14_summary.json`
Sec. 14.2, 4.8: CNN-versus-bank ablation, χ^2 and ranking statistics	`15_cnn_vs_bank_ablation`, `16_chi2_veto_ranking`	`results/15_summary.json`, `results/16_summary.json`
Sec. 6.5: hard negatives, glitch families	`17_hard_negative_mining`, `18_glitch_families`	`results/17_summary.json`, `results/18_summary.json`
Sec. 13: unfolding and population inference	`19_unfolding`, `32_bayesian_extensions`	`results/19_summary.json`, `results/32_summary.json`
Sec. 6.6: prior mismatch, base rate	`20_prior_mismatch_base_rate`	`results/20_summary.json`
Sec. 5.6, 5.7: labelled glitches, exact-model MCMC, SBC	`21_injected_glitches_robust_bayes`, `32_bayesian_extensions`	`results/21_summary.json`, `results/32_summary.json`
Sec. 12: parameter estimation of binary black holes	`22_imr_parameter_estimation`	`results/22_summary.json`
Sec. 4.9, 4.10, 4.11: background rigour, bank design, data-quality flags	`23_background_rigour`, `24_rare_event_mc`, `27_bank_design_local_whitening_psd`, `35_data_quality_flags`	`results/23_summary.json`, `24`, `27`, `35`
Sec. 6.7, 6.8: robustness, conformal sets, scaling, transfer H1→L1	`25_robustness_conformal_interpretability`, `26_scaling_augmentation_adaptation`	`results/25_summary.json`, `results/26_summary.json`
Sec. 9, 8.5, 8.6: physical units, precession and higher modes, limits	`28_physical_units_range`, `29_precession_higher_modes`, `37_limits_low_mass_bns_nsbh_ecc`	`results/28_summary.json`, `29`, `37`
Sec. 7.4, 7.5, 7.6: seeds, other architectures, Transformer tuning	`30_seeds_paired_tests`, `34_other_architectures`, `39_transformer_tuning_round2`	`results/30_summary.json`, `34`, `39`
Sec. 10.4, 10.5: coherent statistic, physical network injections, Virgo	`31_coherent_network`, `38_virgo_snr_budget`	`results/31_summary.json`, `38`
Sec. 6.9: noise-only flow and VAE	`33_unsupervised_flow_vae`	`results/33_summary.json`
Sec. 14.3: O2 and O3 generalisation	`36_o2_o3_generalisation`	`results/36_summary.json`, `results/runs_chunks.json`
exploratory prototype	`00_waveform_exploration`	–